

ECOLOGY OF FISHES AND INVERTEBRATES INHABITING THE CORAL

POCILLOPORA GRANDIS IN HAWAI'I

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ABSTRACT

The environmental variables and biological interactions that shape biological communities have long been a major focus of the science of ecology. When much of the physical structure of the community is biogenic, such as on coral reefs, these factors affect the living habitat itself. Coral reefs around most of Hawai‘i are characterized by large seasonal waves, low coral cover, and low coral species richness. The antler coral, *Pocillopora grandis*, an uncommon member of the coral community, can grow to more than 1.5 meters in diameter. At this size, these highly branching colonies attract substantial assemblages of resident fishes and invertebrates in an otherwise low-lying reefscape that lacks other substantial sources of vertical relief. Antler coral may therefore provide disproportionately important habitat space relative to its total abundance. Surveys at three sites across the southern shore of O‘ahu revealed that reefscape-scale factors had little effect on resident assemblage structure. Most variation in fish and invertebrate assemblages was attributed to colony volume and amount of living tissue, yet there was evidence of agonistic interactions between species, especially at smaller colony sizes. Given these behavioral observations, I conducted a six-month experimental removal of two common residents, the Blue-eye Damselfish (an interference competitor) and the Arc-eye Hawkfish (a mesopredator). Resident fish and invertebrate abundance and species richness were significantly greater in the absence of these strong interactors. Finally, using photogrammetry, I measured host colony growth rates during this experiment on both manipulated and unmanipulated colonies. In the absence of Blue-eye Damselfish, predation on host colonies by coral-eating fishes greatly increased, resulting in reduced growth rates and sometimes mortality. This pattern provided

evidence for a novel indirect mutualism on coral reefs in which a territorial corallivore indirectly benefits its host coral through defense against transient corallivores, reducing total corallivory and increasing host colony growth. Furthermore, on unmanipulated coral colonies, resident fish identity likely affected the magnitude of augmented growth acquired by the host, with species spending more time in direct contact with the colony providing a greater benefit. Despite limited colonization by other fishes when Blue-eye Damselfish and Arc-eye Hawkfish were already present, these two species provided multiple benefits that increased growth and survival of host *P. grandis* colonies.

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CONTRIBUTION OF AUTHORS

The third chapter of my dissertation was a collaboration between myself, Ryan Jones, Daniela Escontrela, and Mark Hixon. Drs. Hixon and Jones helped with the development of the experimental and statistical design, and everyone aided with fieldwork, data collection, and editing of the manuscript. I executed the experiment, conducted all statistical analyses, and wrote the manuscript.

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CHAPTER 1

General Introduction

Erik G. Brush

Tropical coral reefs are characterized by the highest species richness of any marine ecosystem while covering far less than one percent of the global seafloor (Reaka-Kudla 2005; Fisher et al. 2015). The biogenic habitat created by the corals themselves allows for such high levels of biodiversity in relatively small geographic areas, as it provides the physical habitat space (Bellwood et al. 2017) for the many resident fishes (Coker et al. 2014) and invertebrates (Stella et al. 2011). Abiotic and other biotic factors also shape the assemblage structure of coral reef inhabitants along a broad gradient of spatial scales, from individual coral colonies to entire reefs to islands and coastlines to biogeographic regions.

At the regional scales of islands and coasts, geographic distributions of coral reef species can be influenced by abiotic factors such as wave strength and exposure (Friedlander et al. 2003; Nunes et al. 2013), environmental constancy and variation (Abele 1976; Black and Prince 1983), physical disturbance (Connell 1978), and sedimentation (Mallela et al. 2007; Moustaka et al. 2018), to name a few. Across a local reefscape, metrics of assemblage structure have been linked to components of surrounding reef architecture, especially habitat heterogeneity (Wilson et al. 2007; Palacios and Zapata 2014), architectural complexity (Alvarez-Filip et al. 2011; Ferrari et al. 2018) and coral species richness (Roberts and Ormond 1987; Komyakova et al. 2013). Characteristics of the immediately surrounding reefscape, such as rugosity (Friedlander and

Parrish 1998; Harborne et al. 2012; Darling et al. 2017) and reef size (Acosta and Robertson 2002; Karnauskas et al. 2012) can affect assemblage structure on single coral colonies.

At the microhabitat scale of a single colony, assemblage structure can depend on colony structural characteristics like colony size (Abele and Patton 1976; Pereira and Munday 2016; Counsell et al. 2018), colony shape (Holbrook et al. 2002; Agudo-Adriani et al. 2016; González-Rivero et al. 2017), and coral identity (Dunn et al. 2017; Richardson et al. 2017). Biological interactions such as competition for limited resources (Jones 1988; Bonin et al. 2009; Geange et al. 2013) and predation (Hixon and Carr 1997; Beukers and Jones 1998; Hixon et al. 2012) also greatly shape resident assemblage structure. Further, the order of arrival of organisms to open corals, known as priority effects (Gleason 1927), can greatly influence which species are able to subsequently immigrate (Shulman 1985; Almany 2003, 2004).

The structure of species assemblages on coral reefs can greatly affect the state and resilience of the ecosystem as a whole. For example, at larger spatial scales of islands or reefscapes, total herbivore abundance and functional diversity of herbivore types have been linked to the maintenance of a coral dominated ecosystem (Mumby et al. 2007; Adam et al. 2022). At smaller spatial scales, such as a single coral colony, resident mutualists can promote the growth (Meyer and Schultz 1985; Holbrook et al. 2008) and survival (McKeon et al. 2012; Shantz et al. 2023) of their host coral.

The Hawaiian Islands comprise the most isolated archipelago in the world, which leads many reefs to be characterized by chronically low larval recruitment of both fishes (Walsh 1987; DeMartini and Anderson 2007) and corals (Coles and Brown 2007). Low recruitment can affect local assemblage structure, as resources such as available habitat may be less limiting and density-dependent processes such as competition (Forrester 1990; Holbrook and Schmitt 2002)

or predation (Forrester 1995; Hixon and Carr 1997) less pronounced in the regulation of local adult populations (Hixon et al. 2012). At the extreme, recruitment limited populations are governed entirely by larval input rather than post-recruitment density-dependent processes (Doherty 1983; Doherty and Fowler 1994).

Lacking a continental shelf, the reefs of the Main Hawaiian Islands (MHI) are also characterized by high wave energy. As such, the combination of geographic isolation, low recruitment, and high wave energy has likely led to depauperate coral assemblages on Hawaiian reefs relative to much of the tropical Pacific (Dollar 1982; Grigg 1983; Franklin et al. 2013). Branching corals with a higher skeletal strength, such as those of the genus *Pocillopora* (Rodgers et al. 2003), are the most abundant branching taxa in the MHI (Maragos 1972). However, apart from differences in wave strength and depth (Dollar 1982; Friedlander and Parrish 1998), Counsell et al. (2018) found that most of the variation in resident assemblage structure on *Pocillopora meandrina* colonies was explained by colony characteristics such as size and percent living tissue. Variables associated with the surrounding environment had substantially less effect on the resident assemblages. In general, most studies have shown the extreme importance of pocilloporid corals as habitat for small fishes (Coker et al. 2014) and invertebrates (Stella et al. 2011).

Although not considered a major reef-building coral in Hawai'i (Franklin et al. 2013), *P. grandis* is the only species that grows to a large size, reaching diameters of up to 1.5m (pers. obs.). In areas along the south shore of O'ahu where coral cover is low (Franklin et al. 2013) and numerically dominated by low-lying colonies, large *P. grandis* colonies provide substantial vertical, branching habitat where it is otherwise in short supply (pers. obs.). No prior studies of this coral species have been conducted in Hawai'i, despite potential importance of *P. grandis*

providing branching coral habitat on Hawaiian reefs and its known ability elsewhere in the Pacific to host highly diverse fish assemblages (Coker et al. 2014).

Given the paucity of ecological knowledge of *P. grandis* in Hawai‘i, the second chapter of this dissertation, “Assemblage structure of fishes and invertebrates inhabiting the coral *Pocillopora grandis* in Hawai‘i,” reports observational surveys exploring the reefscape- and colony-scale habitat characteristics potentially driving assemblage structure on individual colonies. Due to the findings of Counsell et al. (2018) for *P. meandrina* and relatively low recruitment typically seen on Hawaiian reefs (Walsh 1987), I hypothesized that most of the variation in assemblage structure would be attributable to colony attributes. Specifically, I sought to address how assemblages differed with variation in site, reefscape, and colony characteristics, and how natural disturbance to the colony affected the structure of these assemblages. Encompassing three sites along the south shore of O‘ahu, this study identified the most common inhabitants of these corals: *Plectroglyphidodon johnstonianus*, *Paracirrhites arcatus*, *Dascyllus albisella*, and five species of guard crabs in the genus *Trapezia*.

The Blue-eye Damselfish, *Pl. johnstonianus* is corallivorous (Gochfeld 1991), highly territorial (Gajdzik et al. 2016), and actively chases other fishes from its host coral, particularly other corallivores (Tricas and Boyle 2014). Territorial behavior in damselfishes can affect local patterns of habitat use via interference competition (Eurich et al. 2018a), interspecific aggression (Bay et al. 2001; Eurich et al. 2018b), territory maintenance (Eurich et al. 2018c), and disruption of foraging (Jones 2005; Catano et al. 2014). Competition can also be the ultimate source of mortality via exposure to predation (Hixon and Jones 2005).

The Arc-eye Hawkfish, *Pa. arcatus* is a common ambush mesopredator of small fishes and invertebrates that is ubiquitous on reefs around the MHI (Randall 1963; DeMartini 1996)

and across the Indo-Pacific (Stier et al. 2013; Leray et al. 2015). This hawkfish strongly associates with branching corals, especially *Pocillopora* spp. with wide inter-branch spacing (Kane et al. 2009; Coker et al. 2015). Previous studies have shown that these predators can directly reduce fish recruitment (Shima et al. 2008; Stier et al. 2013), especially of shoaling damselfishes (Holbrook et al. 2011; Gallagher et al. 2016), as well as invertebrate abundance, including mutualist guard crabs of the genus *Trapezia* (Hobson 1974; Leray et al. 2015).

Given the common presence of both an interference competitor (*Pl. johnstonianus*) and a predator (*Pa. arcatus*), the third chapter, “Habitat characteristics and priority effects of strong interactors shape fish and invertebrate assemblages inhabiting the coral *Pocillopora grandis* in Hawai‘i,” involved an experimental manipulation these two species to test their effects on resident assemblage trajectory and structure. The combined presence of a resident interference competitor and predator has been shown to act synergistically to increase mortality of prey species (Hixon and Jones 2005), and prior residence by such species can greatly alter assemblage trajectory (Almany 2003). Therefore, I hypothesized that the presence of these two species would reduce the abundance and species richness of colonizing fishes and invertebrates to small-to-medium *P. grandis* colonies.

As identified by previous surveys, *P. grandis* resident assemblages commonly include multiple mutualist species. Damselfish of the genus *Dascyllus* have been documented aid their host coral in multiple ways, including augmentation of growth rates via nutrient provisioning (Holbrook et al. 2008, 2011), removal of harmful sediment from tissue surfaces (Chase et al. 2020), and greater resilience to bleaching and thermal stress (Chase et al. 2018; Shantz et al. 2023). Although no studies regarding fish-coral mutualisms have been conducted on the endemic *D. albisella* in Hawai‘i, guard crabs in the genus *Trapezia* have been shown elsewhere in the

Pacific to defend host corals from invertebrate corallivores (McKeon et al. 2012; McKeon and Moore 2014) and alleviate the effects of sedimentation (Stier et al. 2012).

Because local distributions of mutualist species can potentially be affected by the prior presence of *Pl. johnstonianus* and *Pa. arcatus*, chapter 4, “Indirect fish-coral mutualisms and resident fish assemblages enhance host coral growth on Hawaiian coral reefs,” investigated indirect effects of the resident interference competitor and predator on the demographic rates of host *P. grandis* colonies. This study used measurements of host colony growth from both the experimental manipulation described in chapter 3, as well as an additional set of coral colonies with naturally occurring and unmanipulated fish assemblages. Specifically, I hypothesized that the presence of the competitor and predator would either decrease host coral growth via a reduction in mutualist abundance or increase growth via a protection mutualism and reduction in total predation of the host colony.

The research reported in this dissertation employed a series of observational and experimental field studies to determine the ecology of *Pocillopora grandis* colonies within Hawaiian reefscapes. This project demonstrated the interconnectedness of geographic site- and reefscape-scale drivers of local species abundances, biological interactions on single coral colonies, and the resulting effects on host colony growth and survival.

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CHAPTER 2

Assemblage structure of fishes and invertebrates
inhabiting the coral *Pocillopora grandis* in Hawai'i

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Abstract

Coral reefs are highly diverse ecosystems, and the relationship between reef structural complexity and the diversity of resident species has long been of interest. Corals in the genus *Pocillopora*, especially one of the largest species, *P. grandis*, are characterized by highly branching morphology that hosts many fishes and macroinvertebrates. I documented the general environment, surrounding reefscape, and colony-specific variables most correlated with resident assemblage structure on *P. grandis* on the island of O‘ahu, Hawai‘i. In general, surrounding reefscape had little effect on assemblage structure writ large, with most variation attributed to colony size, percent living tissue, and general environmental factors associated with each reef. Resident fish and invertebrate abundance, Shannon-Wiener composite α -diversity, species evenness, and species richness increased with colony size, and fish abundance and species richness increased with greater living colony tissue. However, invertebrate abundance and species richness remained unchanged across equally sized living and dead colonies, although live colonies supported more coral obligate decapods and dead colonies more facultative species. Live colonies detached from the seafloor yet otherwise mostly intact hosted more abundant and diverse residents than fully intact, attached colonies. The capability of *P. grandis* to grow to large sizes, host diverse resident assemblages, and tolerate moderate physical disturbance, after which assemblages may become larger and more diverse due to increases in habitat heterogeneity, makes this coral an ideal candidate for reef restoration efforts in Hawai‘i.

Keywords:

Community ecology, *Pocillopora grandis*, Hawai‘i, habitat effects, coral reef degradation, species diversity

2.1 Introduction

Tropical coral reefs are characterized by the highest species richness of any marine ecosystem, despite covering less than 1% of the global seafloor (Fisher et al. 2015). This diversity is partly due to the complex physical structure provided by corals (Bellwood et al. 2017), which provides refuge for fishes (Holbrook et al. 2002) and mobile invertebrates (Idjadi and Edmunds 2006), thus mediating the effects on resident species of competition (Bonin et al. 2009) and predation (Beukers and Jones 1998). At the reefscape scale, species richness of fishes increases with greater habitat heterogeneity (Wilson et al. 2007) and coral species richness (Komyakova et al. 2013). On an individual coral, resident assemblage structure is affected by colony size and morphology (Abele and Patton 1976; Holbrook et al. 2002), coral identity (Richardson et al. 2017), and surrounding habitat rugosity (Darling et al. 2017).

Coral reefs are also among the most endangered ecosystems in the world (Bellwood et al. 2004), so the tight associations between coral colonies and their resident fishes and invertebrates are under threat from a myriad of global and local stressors. These threats include ocean warming (Hughes et al. 2017), overfishing of important functional groups (MacNeil et al. 2015; Shantz et al. 2020), sedimentation (Fabricius 2005), and eutrophication (Fabricius 2011). While coral reefs in Hawai‘i have not yet been as severely affected by global stressors as other regions (Sully et al. 2019), annual bleaching is projected by mid-century (van Hooidonk et al. 2016) and the reefs are heavily exposed to local stressors that scale with human population density (DeMartini et al. 2008). Reefs around the Main Hawaiian Islands (MHI) experience a gradient of human influence, with those around O‘ahu being the most degraded (Friedlander and DeMartini 2002; Friedlander et al. 2018).

Without continental shelves, corals along the coast of Hawai‘i are naturally and greatly influenced by wave exposure (Dollar 1982; Grigg 1983; Franklin et al. 2013). The southern and western facing shores of O‘ahu were particularly affected by two major hurricanes in 1982 and 1992 that vastly diminished coral cover at many sites, with limited recovery (Coles and Brown 2007; Friedlander et al. 2008). Corals of the genus *Pocillopora* are most commonly found in reef habitats along the west- and south-facing shorelines of O‘ahu. Pocilloporid corals are important habitat for many species of both small fishes (Shima et al. 2008) and mobile invertebrates (Stella et al. 2010), hosting high richness and abundance of the latter due to their highly branching morphology, even at small colony sizes (Stella et al. 2010; Counsell et al. 2018).

Most pocilloporid species in Hawai‘i do not exceed the size of a basketball. However, antler coral (*Pocillopora grandis*) can grow to approximately 1.5m in diameter (pers. obs.), making it the most prominent coral species providing high relief and habitat complexity on exposed O‘ahu reefs. Prior studies elsewhere in the Pacific have investigated *P. grandis* with regards to damselfish recruitment (Schmitt and Holbrook 2000) and habitat use (Sakanoue and Sakai 2019), hawkfish habitat preference (Kane et al. 2009), and a multitude of mutualisms. Mutualist residents increase coral growth rates through fertilization by fish excreta (Holbrook et al. 2008a, 2011), defend against invertebrate corallivores (McKeon and Moore 2014), and bolster resilience to bleaching (Shantz et al. 2023). Coker et al. (2014) found that, despite not being a thicket-forming species, *P. grandis* is among the top 10% of studied corals in terms of resident fish diversity. However, no studies of *P. grandis* have been conducted in Hawai‘i, and the drivers of resident assemblage structure are poorly understood globally.

Given the threats facing coral reefs and ongoing degradation of Hawaiian reefs near large human populations, it is important to understand the factors that structure resident fish and macroinvertebrate assemblages on *P. grandis*. Such knowledge would foster understanding how these assemblages may change under future conditions and whether this species could be leveraged for coral restoration efforts. Therefore, I explored the general environmental, surrounding reefscape, and colony-specific characteristics that potentially drive resident assemblage structure on *P. grandis* colonies. Specific questions addressed were: (1) How does assemblage structure differ relative to variations in a) site-specific environments, b) surrounding reefscape and c) colony-scale characteristics? (2) Is there evidence that the presence of particular species affect assemblage structure? and (3) How does physical disturbance affect resident assemblage structure?

2.2 Materials and Methods

Study sites

Surveys were conducted at three sites along the south and west facing shores of O‘ahu, Hawai‘i (Figure 1). The Hanauma Bay site (21°16'7.35"N, 157°41'34.47"W; ~20ha, 60% hardbottom, 40% sand, 68 surveyed colonies) is located on the southeast side of the island. A fully protected Marine Life Conservation District since 1967, this continuous reef had the highest coral cover (Franklin et al. 2013), greatest coral diversity (pers. obs.), highest abundance of large-bodied fishes (Friedlander et al. 2019), and ranged in depth from 5 to 18m. The Waikīkī site was located approximately 800m offshore of Waikīkī Beach along the Halekulani Sand Channel (21°16'11.00"N, 157°50'18.53"W; ~10ha, 60% hardbottom, 40% sand, 92 surveyed colonies). This spur-and-groove coral reef had a comparable depth to Hanauma Bay, was

dominated by colonies of *Porites lobata* and *Pocillopora meandrina*, had the lowest coral cover of the three sites (Franklin et al. 2013), and lacked larger bodied fishes, likely due to fishing and chronic human influence (Friedlander et al. 2018). The Kahe Point site (21°21'13.37"N, 158°7'54.83"W; ~1ha, 90% hardbottom, 10% sand, 50 surveyed colonies) was slightly shallower than the others (2 – 14m) with coral cover intermediate to Hanauma Bay and Waikīkī, and was dominated by the same coral species as Waikīkī.

Pocillopora grandis surveys

Visual surveys of *P. grandis* colonies (n = 502 surveys, 210 colonies) were conducted on SCUBA from 2017 to 2023 (Appendix A: Table 1). The first set of surveys was completed during the spring and summer of 2017 at all three sites to document fish recruitment to *P. grandis*. A second set was conducted at Hanauma Bay and Waikīkī from spring 2019 to spring 2020 to record a minor bleaching event (Jones et al. 2021). No coral mortality occurred, and resident assemblages did not change (unpub. data). A final set was completed at Waikīkī from summer 2020 to summer 2021 prior to a manipulative experiment focused on ecological interactions driving assemblage structure. Surveys were conducted on colonies ranging from 0 to 100% live tissue and in a variety of colony conditions (Appendix B: Figure 1): 1) fully intact and attached to the seafloor (n = 435 surveys, 183 colonies), 2) detached from the seafloor and leaning but otherwise intact (n = 39 surveys, 23 colonies), 3) detached and split into 2 – 4 large pieces (n = 14 surveys, 9 colonies), and 4) detached and highly fragmented (n = 5 surveys, 5 colonies). All consecutive surveys of the same colony occurred at least 50 days apart unless a major change in condition occurred.

Surveys included a full visual census of all resident fishes and macroinvertebrates. I could not regularly identify hermit crabs to species *in situ*, so these were grouped into a single higher taxonomic level. Live colony tissue was estimated to the nearest 5%. I estimated colony volume for a subset of surveys with measurements to the nearest 1cm of maximum diameter (length), maximum orthogonal diameter (width), and height to calculate the volume of an elliptical cylinder ($volume = \pi \times (\frac{length}{2}) \times (\frac{width}{2}) \times height$). I began measuring inter-branch spacing in 2018 (no measurements occurred at Kahe Point) by averaging the distance to the nearest 0.5cm of five random branch tips to the nearest adjacent branch tip. Finally, I recorded a video of the surrounding reefscape following each survey to estimate the structural complexity and dominant benthos in an approximate 5m radius around each colony. Occurrences of overlapping radii between *P. grandis* colonies was rare, and any fishes that could occupy multiple colonies were not recorded as permanent residents. Variables describing structural complexity included the proportions estimated to the nearest 10% of the surrounding habitat that contained three-dimensional complexity from any source (“Complexity” in figures), *Pocillopora meandrina* (“POME”), mounding *Porites* spp. (“POLO”), *Porites compressa* (“POCO”), hard substrate without living coral (“Hard Non-coral”, hereafter referred to as hardbottom in text), rubble, and sand. Additionally, I characterized the surrounding reefscape as shorter, approximately the same height, or taller than the focal antler coral (“Relief”).

Statistical analysis

Reefscape variation

All statistics were completed within the software *R* (R Core Team 2023). To characterize the reefscape surrounding each colony, I ran a Principal Component Analysis (PCA) on the first

survey of all intact colonies ($n = 180$). A combination of habitat characteristics that loaded most strongly on the first two PCA axes (i.e., “Complexity”, “Hard Non-coral”, and “Sand”; Appendix B: Figure 2a) were used in all following univariate and multivariate analyses (see Appendix A: Table 2). I included *Pocillopora meandrina* coverage (“POME”) because it occupied unique ordination space and is related and structurally similar to *P. grandis*. Because “Relief” was measured in tandem with “Complexity” to describe surrounding structure and isolation, this metric was included. Lastly, depth was included because it can affect species distributions (Counsell et al. 2018).

Assemblage structure on intact colonies

Mixed effects models were used to evaluate the effects of colony and reefscape characteristics on resident assemblage structure. Models included fixed effects for site, depth, colony volume, inter-branch spacing, living tissue, and reefscape characteristics that loaded most strongly or occupied unique ordination space (see above), as well as random effects for colony identity and survey date to account for repeated measures. Surveys were restricted to intact colonies still attached to the seafloor with all measured colony and reefscape characteristics ($n = 263$ surveys, 180 colonies). Response variables included resident fish and macroinvertebrate abundance, species richness (S , number of species), species evenness (J' , Pielou 1966), and Shannon-Wiener composite α -diversity (H'). Abundance and species richness were modeled with generalized linear mixed models (GLMMs) with negative binomial and Poisson error distributions, respectively, in the package *glmmTMB* (Brooks et al. 2017). Species evenness was logit-transformed ($\log(\frac{p}{1-p})$), where p is evenness on a scale of 0 – 1. Both evenness and diversity were modeled with linear mixed models (LMMs) with the *lme4* package (Bates et al.

2015). Colony volume was log-transformed to meet model assumptions, and all continuous predictor variables were scaled and centered. Predictor variable significance was determined by type III ANOVAs with Satterthwaite's method from the package *lmerTest* (Kuznetsova et al. 2017) for LMMs and χ^2 likelihood-ratio tests for GLMMs.

To evaluate the effects of reefscape and colony characteristics on assemblage structure within a multivariate framework, distance-based redundancy analysis (db-RDA, function *capscale*) (Legendre and Anderson 1999) with a Bray-Curtis dissimilarity matrix and PERMANOVA were used, both in the *vegan* package (Oksanen et al. 2022). Most colonies (61.7%) were surveyed only once in this data set. Colonies with multiple surveys (Hanauma Bay: 39, Kahe Point: 0, Waikīkī: 30) had 575 ± 227 days (mean $\pm$ SD; range: 254 – 1537) between surveys, representing *de facto* statistically independent colonies that had grown substantially and hosted different assemblages. Because inter-branch spacing was marginally significant but measured at only two sites, analyses were conducted on two datasets: (1) surveys including these measurements (n = 137, 114 colonies, 2 sites) to test this variable's effect, and (2) all surveys (n = 258, 179 colonies, 3 sites) with this variable excluded. Assemblage data were square-root transformed and Wisconsin standardized, and predictor significance determined through permutation tests.

Species co-occurrence

Assemblage structure was further characterized with a species co-occurrence analysis based on the probabilistic model of Veech (2013) using the package *cooccur* (Griffith et al. 2016) and on the same subset of surveys as the full multivariate analyses. Due to the large variation in individual colony volume, surveys were binned into volume ranges. A scatter plot of all

measured intact colony volumes revealed an inflection point that was designated as the largest size class (“Extra-Large” $> 0.18\text{m}^3$). All surveys of colonies less than this volume were equally divided into three smaller categories: “Small” ($0.002 - 0.022\text{m}^3$), “Medium” ($0.022 - 0.05\text{m}^3$), and “Large” ($0.05 - 0.18\text{m}^3$). I ran combined co-occurrence analyses on small to medium colonies and large to extra-large colonies.

2.3 Results

Assemblage characterization

Across all 502 surveys of 210 colonies at 3 sites, 10 fish and 14 invertebrate species were found in at least 5% of surveys (Table 1). Two families (Pomacentridae and Cirrhitidae) comprised 60% of the fish species, and nearly 80% of the invertebrates were decapods, including five species of guard crabs (*Trapezia* spp.). Nearly 80% of species occurred at all three sites, and all but one (the puffer *Canthigaster jactator*) occurred on the full range of colony sizes. In total, 3,568 individual fish and 4,550 invertebrates were recorded. Of these, four fish species were present in at least 20% of surveys: two damselfishes (*Plectroglyphidodon johnstonianus* and *Dascyllus albisella*), an orbicular velvetfish (*Caracanthus typicus*), and a hawkfish (*Paracirrhites arcatus*). Eight invertebrate species were present in at least 20% of surveys, including all five species of trapezid guard crabs in Hawai‘i (*Trapezia flavopunctata*, *T. intermedia*, *T. digitalis*, *T. bidentata*, *T. tigrina*), the murex snail *Galeropsis monodonta*, the coral shrimp *Harpiliopsis depressa*, and hermit crabs of the Superfamily Paguroidea.

Reefscape-level effects on assemblage structure of intact colonies

Principal Component Analysis revealed reefscape-scale differences among sites in the seafloor habitat surrounding surveyed antler corals (Figure 2), with the first two component axes explaining 58% of the variance (Appendix B: Figure 2a, b). Each site occupied a distinct position in ordination space. Hanauma Bay was most closely associated with greater total coral cover (sum of all species; mean $\pm$ SD; $26.8 \pm 23.0\%$) and complexity ($40.8 \pm 25.9\%$), as well as less hardbottom ($24.8 \pm 19.6\%$) than the other sites. Waikīkī, at the other extreme, had greater cover of hardbottom ($78.2 \pm 23.4\%$), and lower live coral cover ($8.7 \pm 9.3\%$) and complexity ($15.9 \pm 12.4\%$) than the other sites. Kahe Point was intermediate in all measurements (coral cover: $15.1 \pm 12.9\%$; complexity: $38.0 \pm 20.2\%$; hardbottom: $56.6 \pm 23.1\%$). All sites had similar coverage of *Pocillopora meandrina*.

Reefscape level factors had very few significant effects on univariate analyses of assemblage structure. Only relief had a significant negative effect on *P. grandis* resident fish abundance (GLMM: $\chi^2 = 6.69$, $p = 0.035$). However, site was a significant predictor for fish abundance (GLMM: $\chi^2 = 7.85$, $p = 0.020$) and invertebrate diversity (LMM: $F(2, 125.3) = 4.33$, $p = 0.015$), species richness (GLMM: $\chi^2 = 8.66$, $p = 0.013$), and abundance (Figure 3 left, Appendix B: Figure 3 left) (GLMM: $\chi^2 = 12.63$, $p = 0.002$; see supplementary Figure 4 in Appendix B for Figure 3 plotted on a linear scale; see Appendix A: Table 3 and Table 4 for all univariate model fixed and random effect statistics). Colonies at Waikīkī had greater fish abundance (model mean $\pm$ SE; 3.65 ± 0.39 ; Tukey: $z = -2.80$, $p = 0.014$) and invertebrate diversity (1.19 ± 0.08 ; Tukey: $t = -2.75$, $p = 0.018$), species richness (4.16 ± 0.32 ; Tukey: $z = -2.79$, $p = 0.014$), and abundance (8.88 ± 0.88 ; Tukey: $z = -3.41$, $p = 0.002$) per colony compared to colonies at Hanauma Bay (fish abundance: 2.17 ± 0.27 ; invertebrate diversity: 0.82 ± 0.09 ;

species richness: 2.83 ± 0.26 ; abundance: 4.87 ± 0.59). Kahe Point colonies also had greater invertebrate abundance (8.46 ± 1.72 ; Tukey: $z = -2.40$, $p = 0.043$) than those at Hanauma Bay.

Multivariate analysis not including inter-branch space found that few reefscape variables contributed significantly to variation in resident assemblages (Figure 4, Appendix B: Figure 5) (db-RDA: $F = 4.48$, $p = 0.001$). Significant predictors included hardbottom (db-RDA: $F = 2.75$, $p = 0.002$), sand (db-RDA: $F = 2.42$, $p = 0.003$), and vertical relief (db-RDA: $F = 1.67$, $p = 0.006$), but each explained less than 1% of the variation according to PERMANOVA. Site (db-RDA: $F = 3.07$, $p = 0.001$) and depth (db-RDA: $F = 4.46$, $p = 0.001$), however, each explained 2.5% and 1.8%, respectively, by PERMANOVA.

Colony-scale effects on assemblage structure of intact colonies

Pocillopora grandis colony volume decreased as depth increased (Appendix B: Figure 6a) (LMM; $F=5.3$, $p=0.02$), albeit mildly ($R^2 = 0.03$), while colony inter-branch spacing increased substantially with depth (Appendix B: Figure 6c) (LMM; $F=14.7$, $p < 0.001$) and colony volume (Appendix B: Figure 6b) (LMM; $F=62.1$, $p < 0.001$). These predictor variables combined explained 32.7% of the variation in inter-branch spacing.

Except for inter-branch spacing, colony-scale variables had comparatively greater effects on assemblage structure of intact *P. grandis* colonies. The effect of colony inter-branch spacing was non-significant in all univariate models, so this predictor was dropped to increase sample sizes. Colony volume (Figure 3 middle, Appendix B: Figure 3 middle) had significant positive effects on diversity (fishes, LMM: $F(1, 200.6) = 124.8$, $p < 0.001$; invertebrates, LMM: $F(1, 197.0) = 39.2$, $p < 0.001$), species richness (Appendix B: Figure 7b, d) (fish, GLMM: $\chi^2 = 138.2$, $p < 0.001$; invertebrates, GLMM: $\chi^2 = 36.5$, $p < 0.001$), species evenness (fishes,

LMM: $F(1, 198.6) = 26.1$, $p < 0.001$; invertebrates, LMM: $F(1, 250.1) = 9.1$, $p = 0.003$), and abundance (Appendix B: Figure 7a, c) (fishes, GLMM: $\chi^2 = 176.4$, $p < 0.001$; invertebrates, GLMM: $\chi^2 = 82.4$, $p < 0.001$). The amount of living colony tissue (Figure 3 right, Appendix B: Figure 3 right) was significantly correlated only with metrics of fish assemblages (abundance, GLMM: $\chi^2 = 44.3$, $p < 0.001$; diversity, LMM: $F(1, 244.1) = 14.4$, $p < 0.001$; species richness, GLMM: $\chi^2 = 23.0$, $p < 0.001$; species evenness, LMM: $F(1, 245.1) = 10.0$, $p = 0.002$).

To evaluate the effects of colony-scale variables within a multivariate framework, I ran two distance-based redundancy analyses, with and without inter-branch spacing. The model with inter-branch spacing included explained 20.9% of the variation in assemblage structure, albeit across only 2 sites (Appendix B: Figure 8a, b). Inter-branch spacing had a moderate effect (db-RDA: $F = 1.87$, $p = 0.022$) compared to living colony tissue (db-RDA: $F = 8.02$, $p = 0.001$) and volume (db-RDA: $F = 2.23$, $p = 0.011$), and explained less variation (PERMANOVA: inter-branch, $R^2 = 0.014$; percent alive, $R^2 = 0.067$; volume, $R^2 = 0.018$).

When inter-branch spacing was dropped from the model and colonies from all three sites included, the full model explained 16.7% of the variation (Figure 4, Appendix B: Figure 5). Colony volume (db-RDA: $F = 5.24$, $p = 0.001$) and percentage of living tissue (db-RDA: $F = 9.93$, $p = 0.001$) explained 2.4% and 4.6%, respectively, of total variation in assemblage structure by PERMANOVA, equaling 47.9% of total attributable variation.

Species co-occurrence on intact colonies

The co-occurrence analysis showed how species associations changed with increasing colony size. All 24 species were present in at least 1 survey of both colony size ranges (Figure 5; for an analysis with all colonies see Appendix B: Figure 9). On small colonies, 9% of

associations were non-random (Figure 5a) (7 negative, 18 positive), compared to 15.6% on larger colonies (Figure 5b) (6 negative, 37 positive).

Of the negative associations, only that between the two hawkfishes (*Paracirrhites arcatus* and *Cirrhitops fasciatus*) remained constant across coral size ranges. Other negative associations between fishes on small colonies included a damselfish (*Plectroglyphidodon johnstonianus*) with another damselfish (*Dascyllus albisella*) and a diminutive wrasse (*Pseudocheilinus tetrataenia*), yet both became random on larger corals. On large colonies, negative associations were found between *D. albisella* and both a hawkfish (*Amblycirrhitus bimacula*) and scorpionfish (*Sebastapistes coniota*). On small colonies, all four negative associations involving invertebrates included either hermit crabs or the brittle star *Breviturma pica*, which often associated with dead coral.

Of all positive associations, six were found in both coral size classes, and all involved guard crabs of the genus *Trapezia* (*T. flavopunctata* with *T. bidentata*, *T. digitalis*, *P. johnstonianus*, and *Caracanthus typicus*; *T. digitalis* with *C. typicus* and *Galeropsis monodonta*). The guard crab *T. intermedia* had only one significant association with another guard crab. By contrast, all other guard crab species were positively associated with 2-5 congeners on larger colonies.

Colony condition and assemblage structure

Due to limitations in the number of surveys of colonies that had experienced some form of disturbance (i.e., detached from sea floor and leaning, split into 2 – 4 pieces, or highly fragmented), only descriptive analyses were appropriate. Generally, uncommon fishes increased in abundance (Figure 6a) and richness (Figure 6b) on the largest colonies and with moderate

disturbance, where total fish abundance increased from a maximum of 118 (mean $\pm$ SE; 26.3 ± 4.7) on intact colonies to 185 (65.0 ± 23.0) on moderately disturbed colonies. Likewise, fish species richness increased from 13 (6.29 ± 0.59) to 18 (10.1 ± 2.16). However, invertebrate abundance (Figure 6c) (intact: 19.6 ± 2.78 ; disturbed: 17.9 ± 2.90) and species richness (Figure 6d) (intact: 5.88 ± 0.49 ; disturbed: 6.11 ± 0.79) did not drastically increase with moderate disturbance, but like fish assemblages, sharply decreased with high fragmentation. However, local guard crab populations decreased after a lower disturbance threshold. Abundance decreased from a maximum of 22 (9.52 ± 1.04) on intact or leaning colonies to a maximum of 10 (4.67 ± 1.26) on split or fragmented colonies.

2.4 Discussion

Structure of resident assemblages on intact colonies

Prior studies across the tropics have found that the abundance and species richness of resident fishes and invertebrates on coral reefs correlate positively with habitat heterogeneity (Wilson et al. 2007) and rugosity (Darling et al. 2017). However, these patterns are weak if present at all in Hawai'i. Neither my study of *Pocillopora grandis* nor that of *P. meandrina* by Counsell et al. (2018) found significant support for effects of surrounding reefscape characteristics on resident assemblages on individual colonies. These patterns lend credence to the possibility that habitat for fishes and macroinvertebrates may not be limiting on Hawaiian coral reefs (Walsh 1987). Nonetheless, Counsell et al. (2018) did find that site-scale variables positively correlated with the abundance and species richness of residents on *P. meandrina*. In this study, site was a significant predictor of local species distributions, fish abundance, and invertebrate abundance, diversity, and species richness, indicating that unmeasured

environmental variables, possibly including wave energy (Dollar 1982; Grigg 1983; Franklin et al. 2013) and sea-surface chlorophyll-a (Counsell et al. 2018) likely contribute to explaining more of the variation in resident assemblages. Depth was also a significant predictor of local species distributions, similar to previous studies of both invertebrates (Counsell et al. 2018) and fishes (Friedlander and Parrish 1998) in Hawai‘i. The lack of significant effects on coarse metrics of assemblage structure, yet existence of potential species-specific abundance patterns by surrounding reefscape characteristics and depth suggests that some degree of environmental filtering of *Pocillopora* associated residents is occurring.

Because most species occurred on colonies of all sizes, trends in assemblage structure were likely affected by a combination of limited recruitment and priority effects. First, the extremely limited variation attributable to reefscape variables could be caused by recruitment limitation (Doherty 2002), whereby local population densities are limited by low or patchy larval input rather than density-dependent processes. This scenario has been hypothesized previously for Hawaiian coral reefs (Walsh 1987) such that habitat for reef fishes may not be limiting (Jones et al. manuscript). Second, priority effects (Almany 2003) and interactions like competition (Bonin et al. 2015) and predation (Hixon 2015) may also be responsible for obscuring any potential effects of surrounding reefscape. The Blue-eye Damselfish (*Plectroglyphidodon johnstonianus*), a territorial interference competitor (Gajdzik et al. 2016), and the Arc-eye Hawkfish (*Paracirrhites arcatus*), a ubiquitous mesopredator of small fishes and invertebrates (Randall 1963) that prefers branching pocilloporids (Kane et al. 2009), were the two most commonly occurring fish species on *P. grandis* across the entire colony size range. This study found evidence for negative interactions between these and other common species on smaller colonies, indicating that these two species may limit colonization by other species until ample

space opens via colony growth. Limitations on coexistence likely relax on the largest colonies, which supported a high abundance and species diversity of residents, especially fishes.

Similar to reefscape, colony inter-branch spacing affected only species-specific distribution patterns and not overall metric of assemblage structure. This is contrary to other studies of coral morphology elsewhere (Holbrook et al. 2002) but similar to the results of Counsell et al. (2018) in Hawai‘i. Additionally, positive and negative associations within and between species typically found at the base of branches (e.g., the scorpionfish *Sebastapistes coniota*) and branch tips (e.g., the damselfish *Dascyllus albisella*) may be evidence for some unmeasured characteristic of *P. grandis* that affects species habitat preferences and local distributions.

Nonetheless, most variation in assemblage structure was explained by colony volume and living tissue, mirroring the findings of Counsell et al. (2018) for assemblages associated with *P. meandrina* around O‘ahu. Colony size has been linked to resident assemblage structure on corals of this genus previously (Abele and Patton 1976), and I found that all fish and invertebrate assemblage metrics increased with colony volume. Comparatively, the percentage of living tissue had a much greater effect on fish compared to invertebrate assemblages. Similarly, Holbrook et al. (2008b) found marked changes in fish assemblages within experimental plots of small, transplanted *P. grandis* colonies across a gradient of plot living tissue, with obligate species never colonizing completely dead plots. However, contrary to this study, a large decline in fish species richness occurred only with 0% live coral in Holbrook et al. (2008b).

Any relationship between living tissue and invertebrate assemblage metrics was likely affected by the influx of uncommon species on larger colonies with lower levels of live tissue, shifting assemblage composition but leaving total abundance and species richness unchanged. In

a study relating live colony tissue and decapod diversity on *P. meandrina* in Hawai‘i, Coles (1980) found that obligate symbiont decapod diversity was positively correlated with live coral surface area. Dead surface area, however, was associated with greater diversity and number of facultative non-symbiont species. After a crown-of-thorns seastar (COTS) outbreak in Mo‘orea, similar trends were seen in assemblages on *P. grandis*. Reductions in live colony tissue led to greater species alpha-diversity, and large colonies were left as the primary survivors, acting as oases for obligate guard crabs (Leray et al. 2012). The current study found evidence of shifts in invertebrate assemblage structure due to colony mortality, as assemblages became dominated by hermit crabs and brittle stars, as also found in *P. meandrina* (Counsell et al. 2018). Large, live colonies of *P. grandis* may, therefore, act as beacons of desirable habitat for trapezid guard crabs.

Patterns of trapezid guard crab abundance and species richness exhibited similarities and differences to assemblages on *P. meandrina* (Counsell et al. 2018). First, *Trapezia intermedia* exhibited distinct ecological patterns from others, inhabiting smaller and more isolated colonies. This species is likely a pioneer of new habitat that is displaced quickly (Counsell et al. 2018). Second, *T. flavopunctata*, the largest species and least abundant on *P. meandrina*, was the most abundant species on *P. grandis*. Most adult guard crabs occur in single mating pairs on pocilloporid colonies (Preston 1973), but I regularly found more than 10 adults on larger *P. grandis* colonies, indicating the existence of multiple pairs per colony. Third, species depth associations were similar to that found on *P. meandrina*, namely *T. digitalis* in the shallows and *T. tigrina* on deeper corals (Counsell et al. 2018). Further, *T. flavopunctata*, a large species typically found at the branch tips, seemed to associate more with *T. digitalis*, a smaller species found near the base. A similar pairing between *T. tigrina* and *T. bidentata* suggests intra-colony microhabitat partitioning. Across all 502 surveys, only 7 found every species coexisting, and

only 2 surveys of extraordinarily large colonies contained adults from each. Lastly, although observed on colonies with only 5% living tissue in this study, as on *P. meandrina*, guard crab abundance and species richness declined with live colony tissue and size (Coles 1980; Counsell et al. 2018). Except on the largest colonies, facultative species replaced guard crabs once live tissue declined to 25%.

Understanding what shapes patterns of guard crab abundance and species richness is critically important due to host coral – resident mutualisms. Guard crabs alleviate the effects of sedimentation (Stewart et al. 2006; Stier et al. 2012) and defend their host coral from invertebrate corallivores, including COTS, cushion stars, and vermetid snails (Stier et al. 2010; McKeon et al. 2012). Different species also better defend against different predators due to differences in size, so greater crab species richness confers greater protection (McKeon and Moore 2014).

Structure of resident assemblages on broken colonies

Exposed Hawaiian reefs are strongly affected by seasonal large waves (Grigg 1983) that play a role in structuring reef fish assemblages (Friedlander and Parrish 1998; Friedlander et al. 2003). Coral species differ in mechanical strength, so wave strength can greatly affect coral community structure at larger spatial scales (Storlazzi et al. 2005). Rodgers et al. (2003) showed that *Pocillopora meandrina* has a high skeletal strength, which greatly reflects the environment where it typically occurs in Hawai‘i (Franklin et al. 2013). *P. grandis* may have even greater skeletal strength (pers. obs.), but large colonies are supported by a narrow base, so colony dislodgement and fragmentation by waves seems to increase with colony size. Barring unusually large disturbance events, this species grows until it can no longer support its own weight and

collapses, but like *P. meandrina* (Rodgers et al. 2003), these colonies can remain alive in a broken state for extended periods of time (pers. obs.). Only highly fragmented colonies suffered rapid mortality, likely due to predation by cushion stars (Escontrela manuscript), which are very common along the south shore of O‘ahu (Glynn and Krupp 1986). Rodgers et al. (2003) found low mortality of small *P. meandrina* fragments in Kāne‘ohe Bay, Hawai‘i, but cushion stars there are very rare (Glynn and Krupp 1986; D. Escontrela, pers. comm.).

Effects of physical disturbance on resident fish and invertebrate assemblages were most notable on the largest colonies, as these were more likely to experience fragmentation. Colonies exposed to moderate disturbance (i.e., detachment from seafloor but remaining at least relatively intact) acquired expanded fish assemblages that included surgeonfish, squirrelfish, and cardinalfish species not common on intact colonies. In an experimental manipulation of coral disturbance, Syms and Jones (2000) found an influx of uncommon species on the largest disturbed colonies, reflecting the importance of habitat heterogeneity and creation of new microhabitat types that attract more rubble-associated species. However, both fish and invertebrate abundance and species richness drastically declined with full colony fragmentation, matching previous studies that showed more extreme levels of habitat disturbance cause the greatest changes in fish assemblages (Syms and Jones 2000; Yahya et al. 2011).

Potential for coral restoration

Pocillopora grandis occupies a unique role on modern Hawaiian reefs. As part of a healthy reef (e.g., Hanauma Bay) with later successional stage coral communities, they contribute substantially less to the overall structural complexity of the reefscape, attracting only typical resident species. With potentially lower levels of recruitment, resident fish and

invertebrate abundance could be diluted by greater habitat availability. However, on degraded reefs (e.g., Waikīkī), large colonies contribute a relatively large proportion of the structural complexity, creating oases of life in an otherwise flat and relatively barren reefscape. As Hawaiian reefs are naturally depauperate of coral species relative to other Indo-Pacific regions (Grigg 1983), only the thicket-forming *P. damicornis* supports greater reef fish diversity than *P. grandis* among Hawaiian coral species (Coker et al. 2014).

The capability of larger colonies to support concentrated assemblages with mutualist species on degraded reefs makes *P. grandis* ideal for coral restoration. Naturally occurring fish mutualists can increase coral growth rates (Holbrook et al. 2008a, 2011) and resilience to thermal stress and bleaching (Shantz et al. 2023). Outplanted *P. grandis* colonies would, therefore, support greater assemblages soon after transplantation, and better survive thermal stress events, contributing to greater long-term restoration success. Indeed, fragmented branches of *P. grandis* outplanted to artificial structures off Waikīkī have grown substantially faster than transplanted fragments and naturally occurring colonies of other species (pers. obs.).

Additionally, the ability of *P. grandis* to become detached from the seafloor yet live for years and host a greater variety of species makes these corals uniquely suited to the higher average wave energy associated with Hawaiian reefs. Highly fragmented colonies also provide natural sources of fragments ideal for outplanting, especially given (1) their large and morphologically complex branches, (2) ability to keep or quickly acquire guard crabs, and (3) high survival and growth post-transplantation (pers. obs.). Given the vast array of threats facing coral reefs, it is imperative that coral restoration efforts use species that are adaptable for changing conditions and able to host abundant mutualist fishes and invertebrates, both of which should enhance reef resilience.

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2.6 Tables

Table 1. Fishes and invertebrates seen in surveys of *Pocillopora grandis* colonies (n = 502 surveys of 210 colonies), listed in descending order of % occurrence. Potential coral mutualists are denoted with asterisks. Sites include Hanauma Bay (H), Kahe Point (K), and Waikīkī (W). Coral colony volume: S (small); M (medium); L (large); XL (extra-large; see Methods for size designations). Coral physical state: 0 (intact), 1 (intact but detached from seafloor and leaning on side), 2 (detached from seafloor and split into 2 – 4 leaning sections), 3 (detached from seafloor and highly fragmented).

Family	Species	Species code	Sites present	% of surveys present	#/survey when present ($\mu \pm \text{SE}$)	Coral volume range (m^3)	Coral % live range	Coral physical state
FISHES								
Pomacentridae	<i>Plectroglyphidodon johnstonianus</i>	PLJO	H, K, W	59.8	1.9 $\pm$ 0.07	S – XL	5-100	0,1,2,3
Cirrhitidae	<i>Paracirrhites arcatus</i>	PAAR	H, K, W	45.6	1.3 $\pm$ 0.03	S – XL	0-100	0,1,2,3
Caracanthidae	<i>Caracanthus typicus</i>	CATY	H, K, W	45.4	2.1 $\pm$ 0.08	S – XL	40-100	0,1,2,3
Pomacentridae	<i>Dascyllus albisella</i> *	DAAL	H, K, W	23.9	7.9 $\pm$ 0.91	S – XL	0-100	0,1,2,3
Tetraodontidae	<i>Canthigaster jactator</i>	CAJA	H, K, W	13.9	1.6 $\pm$ 0.11	M – XL	0-100	0,1,2,3
Labridae	<i>Pseudocheilinus tetrataenia</i>	PSTE	H, K, W	9.2	1.3 $\pm$ 0.09	S – XL	5-100	0,1,2,3
Cirrhitidae	<i>Amblycirrhites bimaculatus</i>	AMBI	H, K, W	9.0	1.2 $\pm$ 0.05	S – XL	45-100	0,1,2,3
Cirrhitidae	<i>Cirrhitops fasciatus</i>	CIFA	H, K, W	7.4	1.3 $\pm$ 0.08	S – XL	0-100	0,1,2,3
Scorpaenidae	<i>Sebastapistes coniota</i>	SECO	H, K, W	7.2	3.0 $\pm$ 0.53	S – XL	50-100	0,1,2,3
Cirrhitidae	<i>Paracirrhites forsteri</i>	PAFO	K, W	7.0	1.2 $\pm$ 0.07	S – XL	75-100	0,1,2

Table 1. (Continued) Fishes and invertebrates seen in surveys of *Pocillopora grandis* colonies (n = 502 surveys of 210 colonies), listed in descending order of % occurrence. Potential coral mutualists are denoted with asterisks. Sites include Hanauma Bay (H), Kahe Point (K), and Waikīkī (W). Coral colony volume: S (small); M (medium); L (large); XL (extra-large; see Methods for size designations). Coral physical state: 0 (intact), 1 (intact but detached from seafloor and leaning on side), 2 (detached from seafloor and split into 2 – 4 leaning sections), 3 (detached from seafloor and highly fragmented).

Family	Species	Species code	Sites present	% of surveys present	#/survey when present ($\mu \pm \text{SE}$)	Coral volume range (m^3)	Coral % live range	Coral physical state
INVERTEBRATES								
Trapeziidae	<i>Trapezia flavopunctata</i> *	TRFL	H, K, W	64.9	2.8 $\pm$ 0.14	S – XL	40-100	0,1,2,3
Trapeziidae	<i>Trapezia intermedia</i> *	TRIN	H, K, W	48.6	1.7 $\pm$ 0.04	S – XL	5-100	0,1,2,3
Muricidae	<i>Galeropsis monodonta</i>	GAMO	H, K, W	45.2	3.6 $\pm$ 0.25	S – XL	40-100	0,1,2,3
Palaemonidae	<i>Harpiliopsis depressa</i>	HADE	H, K, W	38.6	3.4 $\pm$ 0.32	S – XL	45-100	0,1,2,3
Trapeziidae	<i>Trapezia digitata</i> *	TRDI	H, K, W	36.5	1.8 $\pm$ 0.09	S – XL	40-100	0,1,2,3
Trapeziidae	<i>Trapezia bidentata</i> *	TRBI	H, K, W	34.9	1.5 $\pm$ 0.04	S – XL	50-100	0,1,2,3
Trapeziidae	<i>Trapezia tigrina</i> *	TRTI	H, K, W	24.9	1.5 $\pm$ 0.06	S – XL	60-100	0,1,3
Paguroidea †	unidentified hermit crabs	PAGU	H, K, W	23.7	2.8 $\pm$ 0.40	S – XL	0-100	0,1,2,3
Ophiocomidae	<i>Breviturma pica</i>	BRPI	H, K, W	16.3	1.8 $\pm$ 0.14	S – XL	0-100	0,1,2,3
Alpheidae	<i>Alpheus lottini</i> *	ALLO	H, K, W	10.2	1.1 $\pm$ 0.04	S – XL	40-100	0,1
Portunidae	<i>Goniosupradens hawaiiensis</i>	GOHA	H, W	6.0	1.3 $\pm$ 0.15	S – XL	50-100	0,1,2
Domeciidae	<i>Domecia hispida</i>	DOHI	H, K, W	5.6	1.7 $\pm$ 0.20	S – XL	40-100	0,1,2,3
Stenopodidae	<i>Stenopus hispidus</i>	STHI	W	5.2	1.9 $\pm$ 0.17	S – XL	0-100	0,1,2
Ophiocomidae	<i>Ophiocoma erinaceus</i>	OPER	H, K, W	5.0	1.6 $\pm$ 0.23	S – XL	0-100	0,1,2,3

† superfamily

2.7 Figures

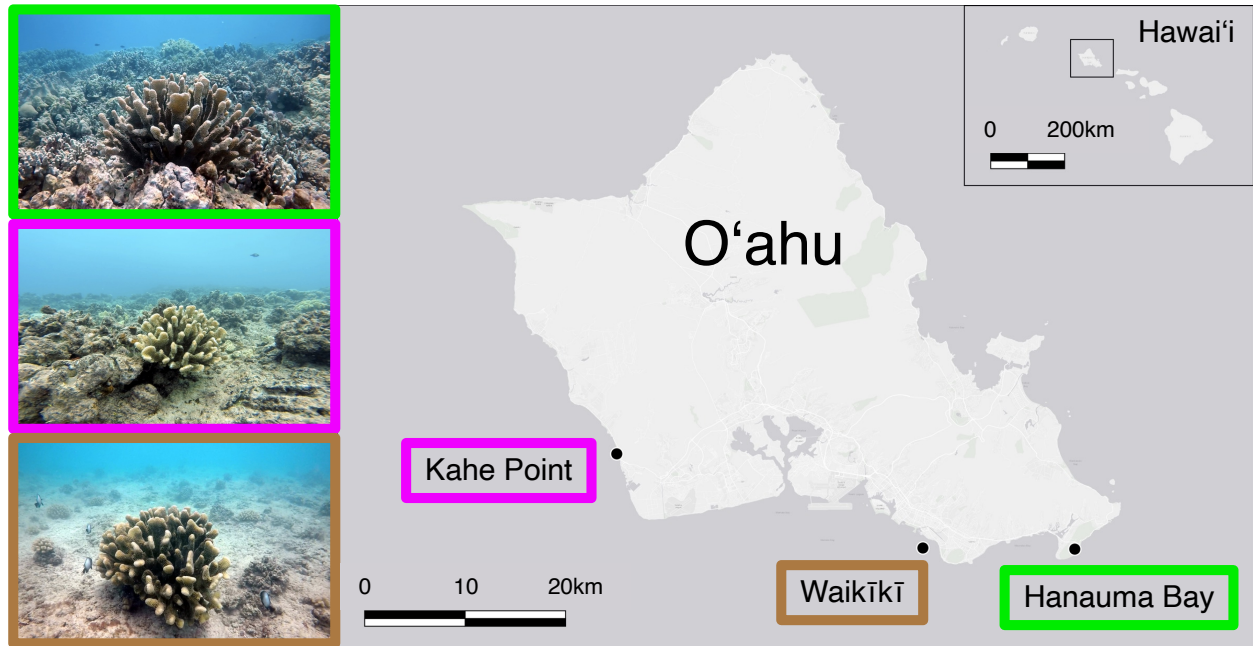


Figure 1. Map of O'ahu, Hawai'i, with the locations of the three study sites. Photos are of *Pocillopora grandis* coral colonies with the typical surrounding reefscape at each site in order of increasing degradation and decreasing habitat complexity (Hanauma Bay: green, top; Kahe Point: magenta, middle; Waikīkī: brown, bottom).

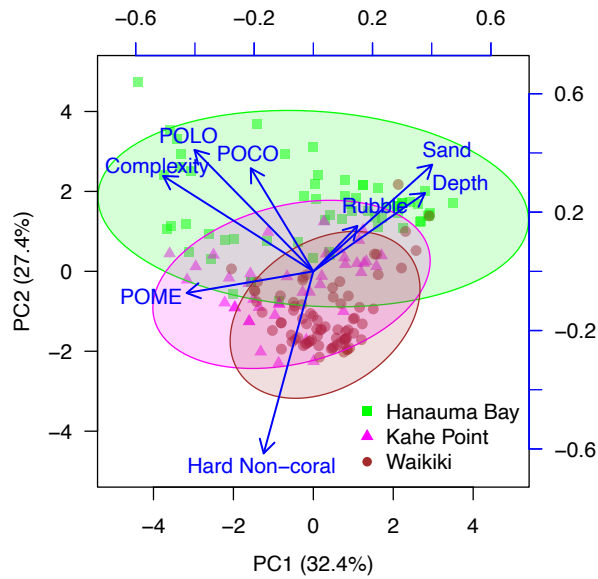


Figure 2. Principal Component Analysis biplot of the seafloor habitat within 5m of 180 intact *Pocillopora grandis* coral colonies (261 surveys; Hanauma Bay: 102 surveys, 62 colonies; Kahe Point: 41 surveys, 41 colonies; Waikīkī: 118 surveys, 77 colonies). Black axes are on the scale of the PC scores (surveys), while blue axes reflect the scale of loadings on each axis (habitat variables). Ellipsoids for each site are standard deviations with confidence limits of 0.95. Coral species within 5m of focal *P. grandis* colonies: POCO = *Porites compressa*; POLO = mounding *Porites* spp.; POME = *Pocillopora meandrina*. Habitat variables are defined in the text and Appendix A: Table 2.

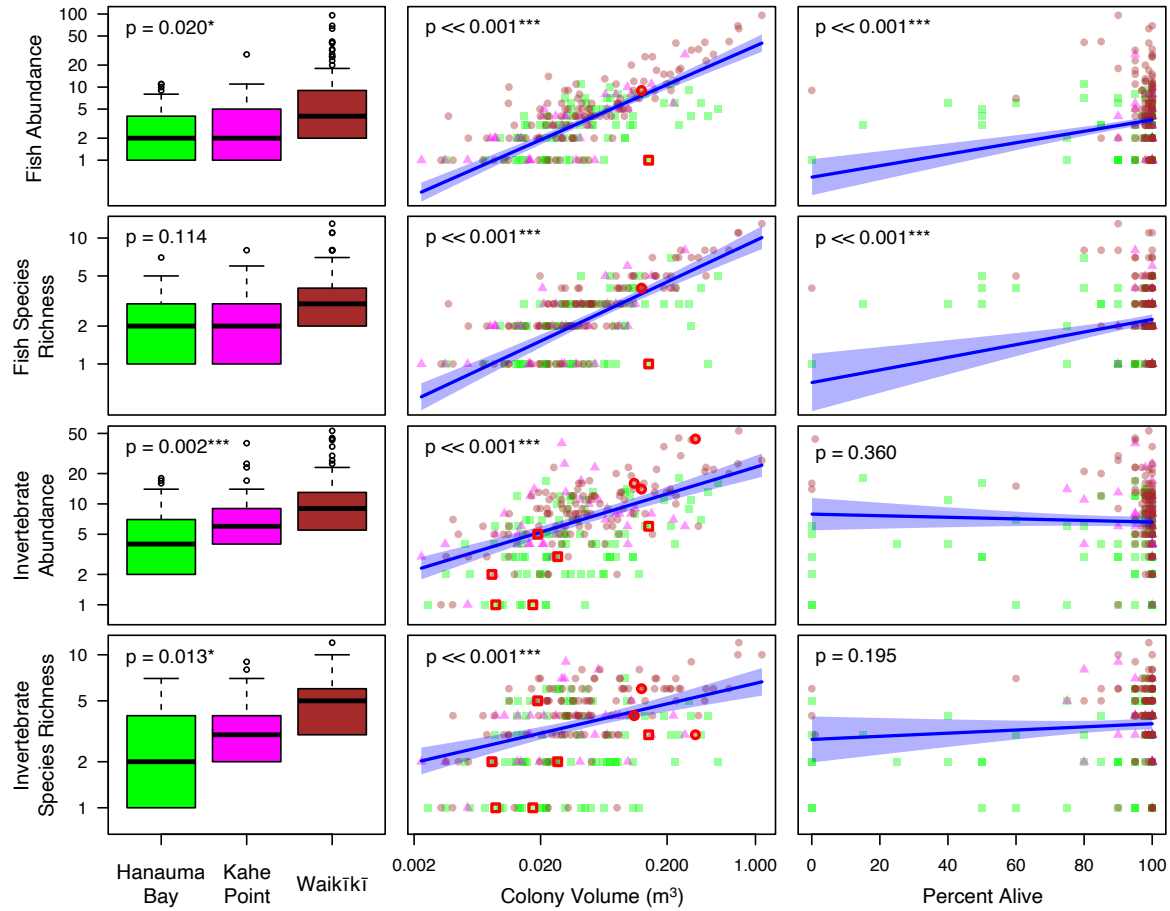


Figure 3. The statistical effects of site (left), colony volume (middle), and percentage of living coral tissue (right) of *Pocillopora grandis* on resident fish abundance (first row), resident fish species richness (second row), invertebrate abundance (third row), and invertebrate species richness (fourth row) in general linear mixed effects models with colony identity and survey date as random effects. Colony volume and all assemblage metrics (responses) are plotted on log-scale axes. The rows with fish and invertebrate metrics are missing 34 and 8 surveys from the raw data, respectively, due to fish and invertebrate absence and inability to plot zeros on the log scale. See Appendix B: Figure 4 for plots with linear axes that show surveys with fishes and invertebrates absent. Points outlined in red are from surveys of colonies with $\leq 5\%$ living tissue, indicating that colony life or death did not clearly affect the general patterns.

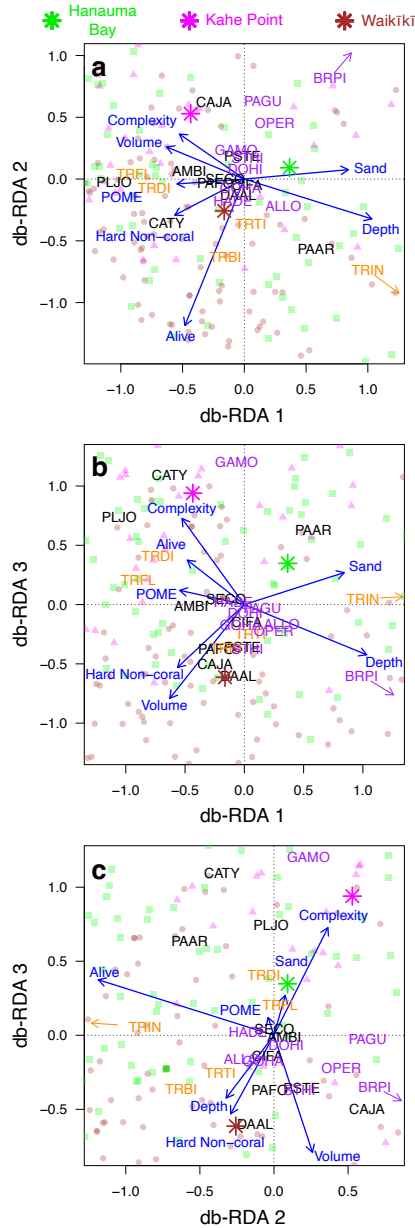


Figure 4. Distance-based redundancy analysis of 258 surveys of 179 intact *Pocillopora grandis* colony resident assemblages: **a** first and second axes, **b** first and third axes, and **c** second and third axes. Analysis is restricted to species listed in Table 1. The effects of colony variables (Volume, Alive) and environmental variables (Depth, Complexity, Relief, Hard Non-coral [hardbottom, dead coral, etc.], Sand, POME = *P. meandrina*) are shown as blue vectors. Values of continuous environmental predictors were scaled by a factor of 1.4 for plotting. Environmental variables were chosen by PCA (see Figure 2) and used in all models. Centroids of each site are denoted by a large asterisk in its respective color (Hanauma Bay: green; Kahe Point: magenta; Waikiki: brown). See Table 1 for color and 4-letter species codes. An arrow associated with a species indicates that point occurred outside this zoomed-in plot; see Appendix B: Figure 2 for the full area of each plot.

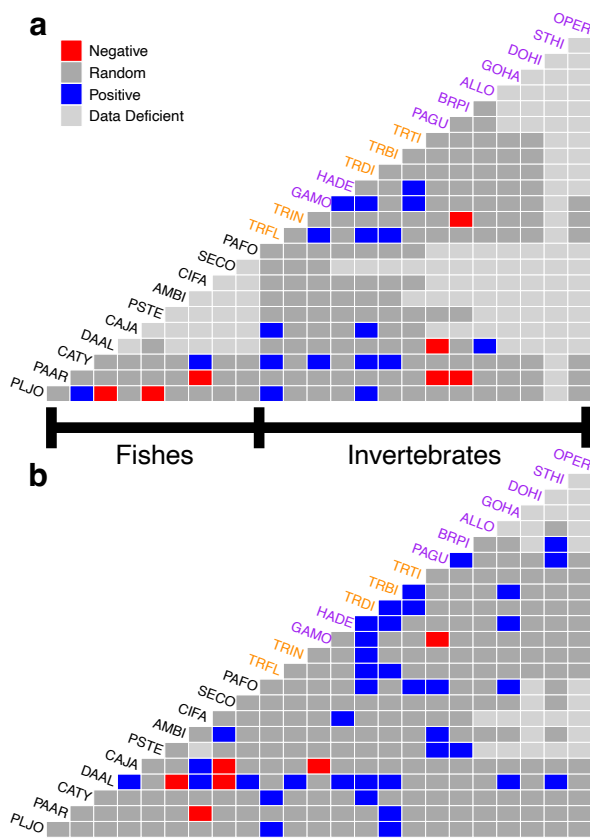


Figure 5. Probabilistic models of species co-occurrence for **a** small *Pocillopora grandis* colonies (n=163 surveys, 117 colonies) and **b** large *P. grandis* colonies (n=95 surveys, 72 colonies; see Table 1 for size categories and species codes). Note that positive associations (i.e., those that happened more often than expected based on random species distributions; blue) and negative associations (less often; red) change as the host coral becomes larger and more habitat space becomes available. See Table 1 for color and 4-letter species codes. See Appendix B: Figure 9 for a probabilistic model of species co-occurrence with all coral size categories included.

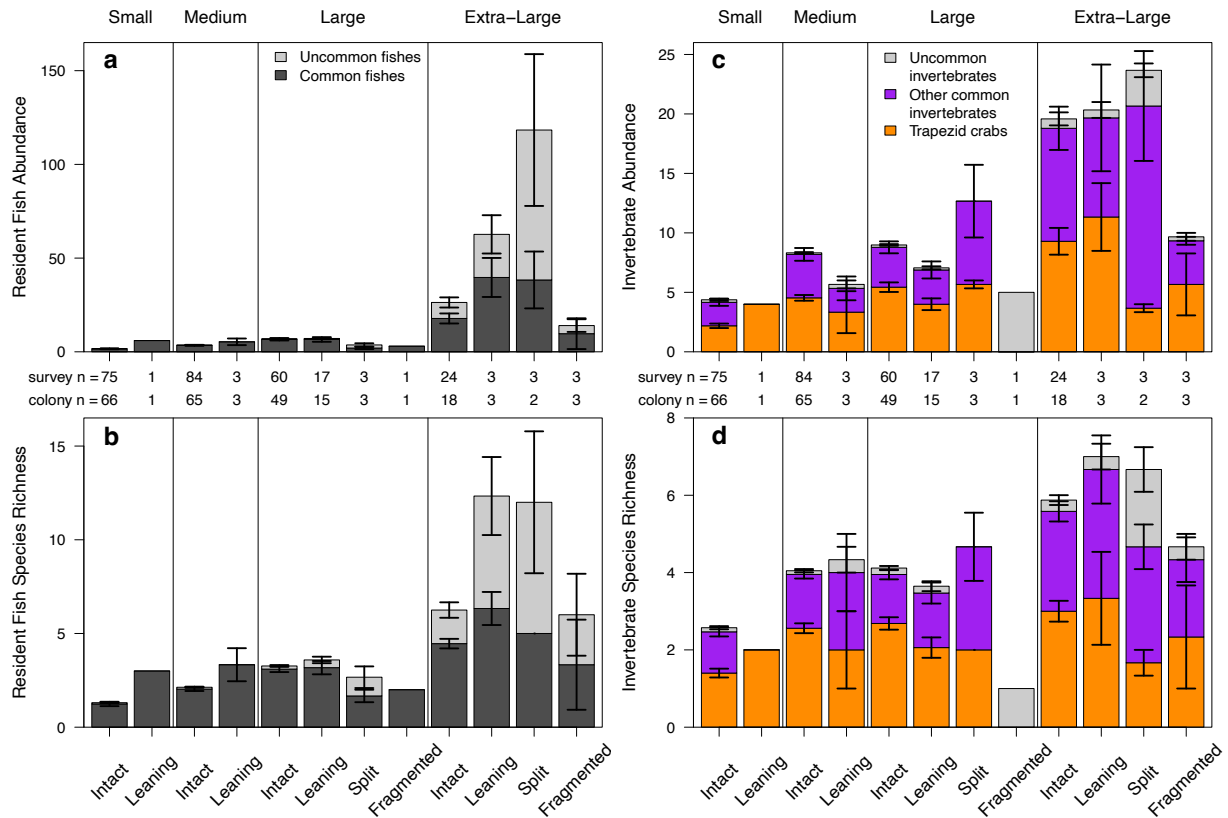


Figure 6. Bar plots of **a** resident fish abundance, **b** resident fish species richness, **c** invertebrate abundance, and **d** invertebrate species richness by *Pocillopora grandis* colony condition and size. Only surveys of colonies with >75% living colony tissue are included. No split or fragmented corals were surveyed in the small or medium colony size classes. Bars are means with standard error of each animal group (grey: uncommon fishes/invertebrates; black: common fishes; purple: other common invertebrates; orange: trapezid guard crabs). See Table 1 for common species and 4-letter species codes.

CHAPTER 3

Habitat characteristics and priority effects of strong interactors shape fish and invertebrate assemblages inhabiting the coral *Pocillopora grandis* in Hawai‘i

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In preparation

Abstract

The structure of ecological communities is driven by an interplay of abiotic and biotic factors, but the history of these factors is less often considered. Priority effects, or how prior residents affect later-arriving species and individuals, can strongly affect the trajectories of local assemblages. On Hawaiian reefs, corals of the genus *Pocillopora* provide the primary source of branching habitat structure for fishes and invertebrates, with *P. grandis*, the largest species, able to host large assemblages at larger colony sizes. At smaller sizes, however, colonies are often dominated by an interference competitor, the Blue-eye Damselfish (*Plectroglyphidodon johnstonianus*), and/or a mesopredator of small fishes and invertebrates, the Arc-eye Hawkfish (*Paracirrhites arcatus*). In a manipulative field experiment of small *P. grandis* colonies off the south shore of O‘ahu, Hawai‘i, we tracked the colonization of fishes and invertebrates in the presence and absence of these two strong interactors to identify potential priority effects. Over 6 months, colonizing fish and invertebrate abundance and species richness were significantly reduced when the competitor and predator were first present. The damselfish *Dascyllus albisella*, a genus which has been shown to benefit host corals through a multitude of mechanisms, was almost entirely prohibited from colonization. However, mutualistic guard crabs of the genus *Trapezia*, while not directly manipulated during the experiment, were minimally affected by the manipulations of strong interactors. Nonetheless, there was potential evidence for priority effects and habitat partitioning among *Trapezia* species. These results indicate that at smaller colony sizes, assemblages of fishes and invertebrates inhabiting *P. grandis* in Hawai‘i are likely affected by priority effects that are less intense on larger colonies as habitat space increases, fostering coexistence and greater local diversity. Understanding the factors that shape community structure

on coral reefs, especially of mutualistic species that promote coral health and ecosystem resilience, is paramount in the face of global and local threats to biodiverse ecosystems.

Keywords: Community ecology, priority effects, *Pocillopora grandis*, Hawai‘i, habitat effects, coral reefs

3.1 Introduction

Identifying the factors in space and time that shape the structure and trajectory of assemblages has been a central tenet of ecology for over a century. Environmental disturbance (Connell 1978), habitat structural complexity (Bracewell et al. 2018), and biological interactions such as predation (Freestone et al. 2011) and competition (Kunte 2008) have all been shown to substantially affect the structure of local assemblages. These forces can also interact across a wide range of spatial and temporal scales (Paine 1980, Connell et al. 1997, Geange and Stier 2010), resulting in alterations to species diversity that can, in turn, affect ecosystem function (Cardinale et al. 2002).

While many studies have examined the effects of abiotic and biotic factors on assemblages, fewer have assessed the role that the order of events can have on assemblage structure and trajectory (see review by Fukami 2015). Evidence suggests that historical contingency can play a large role in shaping assemblages, whether by a previous abiotic disturbance like a storm, or biotic event, such as the arrival of one species to a habitat before another. Such priority effects occur when the arrival order of species to a habitat from the local species pool can affect the colonization of later-arriving individuals, resulting in the possibility of multiple stable assemblages (Diamond 1975). This concept dates back almost a century, when

Gleason (1927) presented an alternative to the dominant paradigm of ecological succession at the time, namely that pre-determined climax communities resulted from predictable species associations. Since then, priority effects have been studied within the context of alternative stable states (May 1977), community assembly theory (Diamond 1975), and more recently, invasive species and ecological restoration (see review by Weidlich et al. 2021).

Empirically, priority effects are inhibitory or facilitative. Inhibitory effects, which have been studied more extensively, limit the colonization ability of later-arriving individuals due to predation, competition for shared resources, or other modifications of available niche space within the local assemblage (Fukami 2015). For example, Kardol et al. (2013) found that an increased time interval between colonization events in a plant community led to greater divergence in assemblage structure, with this effect further augmented in high-nutrient soil that allowed early colonizers to grow faster and shade out latecomers. Modification of available niche space can also cause facilitative effects, as occurred in a study of co-occurring frogs in which the prior arrival of one species led to greater success in both (Alford and Wilbur 1985).

On coral reefs, the majority of priority effects studies have been conducted on the fish and invertebrate residents of individual coral colonies or small patch reefs. Whether priority effects are inhibitory or facilitative have been found to be species-specific. For example, Almany (2003) found that resident predators and territorial damselfishes on patch reefs inhibited recruitment of some species while enhancing that of others. However, most priority effects on coral reefs have been found to be inhibitory, whether by resident competitors (Geange and Stier 2010), predators (Stier et al. 2013), or both (Shulman et al. 1983). Additionally, priority effects have been identified between corals and macroalgae, with each inhibiting the other and leading

to alternative stable states as a result of different disturbance and herbivory regimes (Adam et al. 2022).

In Hawai‘i, most coral reefs are characterized by high wave energy and depauperate coral assemblages relative to much of the tropical Pacific (Grigg 1983). As a result, pocilloporid corals, which have a relatively high skeletal strength (Rodgers et al. 2003), are the most abundant branching taxa in the Main Hawaiian Islands (MHI) (Maragos 1972). However, apart from differentiation by wave strength and depth (Friedlander and Parrish 1998), most of the variation in fish and invertebrate assemblage structure on pocilloporid corals is explained by colony characteristics, with the surrounding reefscape having little effect (Chapter 1, Counsell et al. 2018). It seems likely, then, that biological interactions may play a role in structuring resident assemblages (Counsell and Donahue 2021).

Previous work on *Pocillopora grandis*, the largest species in Hawai‘i and a substantial source of complex vertical relief on many reefs, showed that the two most common resident fish species along the southern shore of O‘ahu were the Blue-eye Damselfish (*Plectroglyphidodon johnstonianus*) and Arc-eye Hawkfish (*Paracirrhites arcatus*) (Chapter 1). The first species is territorial (Gajdzik et al. 2016) and actively chases other damselfishes and corallivorous fishes from its host coral (Tricas and Boyle 2014). The second species, *Pa. arcatus*, is a common ambush mesopredator of small fishes and invertebrates that is ubiquitous on reefs around the MHI (Randall 1963) and Indo-Pacific (Stier et al. 2013, Leray et al. 2015). This species strongly associates with the genus *Pocillopora*, and prefers larger and more open-branched colonies, such as that provided by *P. grandis* (Kane et al. 2009, Coker et al. 2015).

Given the high occurrence of both an interference competitor (*Pl. johnstonianus*) and a mesopredator (*Pa. arcatus*) on *P. grandis* colonies of all sizes, priority effects likely play an

important role in shaping the structure of resident fish and invertebrate assemblages at smaller colony sizes. A previous study of this system found evidence for negative associations between both these focal species and other common inhabitants, especially on smaller colonies, but these restrictions relaxed on larger colonies, where habitat space was more available (Chapter 1). We hypothesized that the absence of these two strong interactors, defined as species that cause pronounced changes in assemblage structure when present (Paine 1980), could affect assemblage colonization trajectories. Specifically, small-to-medium *P. grandis* colonies would host assemblages with a greater abundance and species richness of both resident fishes and invertebrates, be colonized by greater numbers of fish recruits, and be visited more often by a more abundant and diverse array of transient fishes. In particular, colonies without these strong interactors would host more *D. albisella* and a more abundant and diverse guard crab community.

3.2 Materials and Methods

Visual surveys

Visual surveys (see Chapter 1) of assemblages associated with *Pocillopora grandis* were conducted at three sites around O‘ahu, Hawai‘i: 1) Hanauma Bay on the southeastern side of the island (21°16'7.35"N, 157°41'34.47"W), 2) Waikīkī on the southern shore (21°16'11.00"N, 157°50'18.53"W), 3) and Kahe Point on the southwestern side of the island (21°21'13.37"N, 158° 7'54.83"W). Surveyed colonies ranged in depth from 2 to 18m. Compared to the other sites, Hanauma Bay hosted the greatest coral cover (Franklin et al. 2013) and diversity (pers. obs).

To determine the effects of environmental differences at the scale of sites, surrounding reefscape, and colony characteristics on resident assemblages, divers surveyed *P. grandis*

colonies from 2017 to 2023 across virtually their entire size range and from 0 to 100% living colony tissue. The surrounding reefscape within a 5m radius of each *P. grandis* colony was characterized regarding the proportion of surrounding 3D structural complexity and relative vertical relief (average height of the surrounding substrate relative to the *P. grandis* colony), as well as the cover of dominant benthos (the congener *P. meandrina*, sand, and hard substrate lacking living coral, hereafter “hardbottom”) to the nearest 10%. Surveys consisted of a full census of all permanent fish residents, visiting fish that interacted with or inhabited the colony during the survey as part of their larger home range, and resident mobile invertebrates. The amount of living coral tissue was estimated to the nearest 5%. A subset of surveys included measurements of colony volume, which was estimated by measuring the maximum diameter (length), maximum orthogonal diameter (width), and height to the nearest cm, then calculating the volume of an elliptical cylinder ($volume = \pi \times (\frac{length}{2}) \times (\frac{width}{2}) \times height$).

Experimental design

Experimental manipulations were conducted at the Waikīkī site only, which is approximately 800 meters offshore of Waikīkī Beach. The approximately 10ha study site consisted mainly of spur-and-groove coral reef and sandy bottom ranging from 5 to 18m in depth. All surveyed corals were located on hard substrate, which hosted an overall live coral cover of approximately 5% (unpub. data; Franklin et al. 2013) numerically dominated by low lying colonies of *P. meandrina* and *Porites lobata*. Relatively infrequently spaced medium-to-large *P. grandis* colonies provided a significantly greater degree of vertical relief that attracts numerous fish and macroinvertebrate species (Chapter 1).

To test the effects of species interactions on resident assemblage structure, we selected twelve medium-sized *P. grandis*. Colonies ranged from 35 to 70 cm in diameter, and we measured both colony volume and the average distance between five random pairs of branch tips for inter-branch spacing. We also plotted the surrounding 10×10m reef around each colony, and paired plots as best as possible based on *P. grandis* colony characteristics, depth (mean $\pm$ SD; 10.5 ± 3.4 m; range: 5.5 – 14m), location, and surrounding habitat. Each pair ($n = 6$) was randomly divided into a removal plot and a control plot. Beginning in summer 2020, experimental manipulations were conducted on a staggered schedule 1 block (1 removal and 1 control plot) at a time.

Prior to manipulations, we conducted an initial census of all fish and macroinvertebrates associated with each central coral colony as previously described for visual surveys. Due to logistical constraints, only one survey of each colony was conducted prior to the removal experiment. However, a PERMANOVA of repeated visual surveys ($n = 389$) from 2017 to 2023 of 111 colonies showed that the structure of established natural assemblages changed little through time (Appendix C: Table 1). After this pre-removal survey, we conducted a six-month long press removal of *Paracirrhites arcatus* and *Plectroglyphidodon johnstonianus* from the entirety of all six removal plots to provide a buffer zone. A factorial removal of these two species was not possible due to logistic constraints. The Blackside Hawkfish (*Pa. forsteri*) was present in some, but not all plots, as the territories of this larger hawkfish typically include a large prominent coral (often *P. grandis*) (Donaldson 1990). Therefore, the abundance of this resident predator was uneven across plots, so a six-month press removal was also conducted of this species across all plots (removals and controls). *Pa. forsteri* were removed via spearing, and all other species were removed with chopsticks and hand nets (IACUC #19-3255). If absolutely

necessary (and usually only involving *Pl. johnstonianus*), low concentration (<7%) clove oil (a natural fish anesthetic, see review by Priborsky and Velisek 2018) dissolved in seawater was used to aid in fish capture (DAR SAP No. 2019-45). All captured live fish were released over reef at least 1.5km away.

We conducted an initial pulse removal of all non-focal fish species and uncommon invertebrates on each central *P. grandis* colony at the beginning of the experiment. Control colonies were not exposed to clove oil beyond what was necessary for initial pulse removals. We chose to not remove all macroinvertebrates from each colony, including multiple species of guard crabs in the genus *Trapezia*, to avoid subjecting host colonies to an increased risk of predation by invertebrate corallivores. Following this pulse removal, censuses of all resident fishes and invertebrates, as well as visiting fishes on each central colony were conducted every two weeks for six months. Upon conclusion of the press removals, colonies were surveyed twice at 3-month intervals. One of the removal colonies began experiencing substantial mortality six months after the removal period, so this singular survey was excluded from analyses.

Statistical analyses

All statistical analyses were conducted within the software *R* (R Core Team 2023) with univariate analyses conducted with the packages *lme4* (Bates et al. 2015) and *glmmTMB* (Brooks et al. 2017). To analyze species-specific trends of occurrence across visual surveys, data for individual species (*Pl. johnstonianus*, *Pa. arcatus*, *Dascyllus albisella*, and all five *Trapezia* spp.) were analyzed with hurdle models. Species were analyzed first for presence/absence in all surveys (n = 263) in general linear mixed-effects models (GLMMs) with a binomial error distribution, and second for abundance when present with Poisson (resident fishes, included

surveys with no residents, but visiting individuals were recorded) or truncated Poisson (invertebrates) error distributions. All statistical models included predictor variables for colony characteristics (log-transformed volume, percentage of living colony tissue), surrounding reefscape variables (proportional cover of *Pocillopora meandrina*, hardbottom, sand, and structural complexity, and relative vertical relief), depth, and site. We included random effects for colony identity and survey date within models to account for repeated surveys.

To test the effects of the removal experiment on metrics of fish and invertebrate community structure, we used GLMMs with a Poisson error distribution for measures of abundance and species richness, and linear mixed-effects models for composite Shannon-Wiener diversity (H') and species evenness (J' ; Pielou 1966) for each time period (before, during, and after removals). We included only treatment (removals vs. controls) as a fixed effect in models of pre- and post-removal periods. Models of the removal period included fixed effects for treatment, host colony volume (log-transformed), inter-branch spacing, and percentage of living tissue, and the number of surrounding pocilloporid corals (almost exclusively *P. meandrina*) within each plot as a measure of surrounding habitat. We included colony identity as a random effect to account for repeated surveys, as well as survey date, experiment day, and statistical block.

In all statistical models, continuous predictor variables were scaled and centered. For models with Gaussian error distributions, predictor significance was assessed with type III ANOVAs with Satterthwaite's method from the *R* package *lmerTest* (Kuznetsova et al. 2017). All other models used Chi-Square likelihood ratio tests.

3.3 Results

Visual surveys: Fish assemblages

As the two most common fishes in diver surveys of *Pocillopora grandis* colonies, the presence of neither *Plectroglyphidodon johnstonianus* (GLMM: $\chi^2 = 0.59$, $p = 0.744$) nor *Paracirrhites arcatus* (GLMM: $\chi^2 = 2.18$, $p = 0.336$) significantly differed by site or any surrounding reefscape variables (Figure 1, Appendix C: Table 2). However, both strong interactors were more likely to be present (*Pl. johnstonianus* GLMM: $\chi^2 = 68.14$, $p < 0.001$; *Pa. arcatus* GLMM: $\chi^2 = 13.90$, $p < 0.001$) and more abundant on larger colonies (*Pl. johnstonianus* GLMM: $\chi^2 = 52.88$, $p < 0.001$; *Pa. arcatus* GLMM: $\chi^2 = 3.84$, $p = 0.050$) as well as colonies with more live tissue (presence: *Pl. johnstonianus* GLMM: $\chi^2 = 23.49$, $p < 0.001$, *Pa. arcatus* GLMM: $\chi^2 = 8.93$, $p = 0.003$; abundance: *Pl. johnstonianus* GLMM: $\chi^2 = 4.70$, $p = 0.030$) (Figure 1a–d, Appendix C: Table 2). Many smaller colonies with at least 50% living tissue were also often visited by both species even if permanent residents were not present (Figure 1a–d).

Much like the two strong interactors, *Dascyllus albisella* was also more likely to be present (GLMM: $\chi^2 = 17.32$, $p < 0.001$) and more abundant (GLMM: $\chi^2 = 21.59$, $p < 0.001$) on larger coral colonies (Figure 1e–f), albeit in greater numbers at Waikīkī (model mean $\pm$ SE: 3.00 ± 0.85) than Hanauma Bay (0.75 ± 0.33 ; Tukey: $z = -2.56$, $p = 0.028$) (Appendix C: Table 2), and almost non-existent at Kahe Point. Unlike the others, *D. albisella* was affected by reefscape, with presence (GLMM: $\chi^2 = 3.63$, $p = 0.057$) and abundance on *P. grandis* (GLMM: $\chi^2 = 7.24$, $p = 0.027$) declining with greater surrounding complexity and relief, respectively (Appendix C: Table 2). While *D. albisella* co-occurred with the strong interactors on larger

colonies, individuals were more likely to be present and in higher abundances on smaller colonies when *Pl. johnstonianus* was absent (Figure 1e–f).

Visual surveys: Invertebrate assemblages

Across all five species in the genus, *Trapezia* crabs were almost ubiquitous on *P. grandis*, with 94% of colonies with at least 25% living tissue containing at least 1 individual (mean $\pm$ SE: 4.43 ± 0.22 ; range: 0 – 22). Similar to common fish species, guard crabs were much more likely to be present (GLMM: $\chi^2 = 17.71$, $p < 0.001$), in greater abundance (GLMM: $\chi^2 = 72.75$, $p < 0.001$), and more speciose (GLMM: $\chi^2 = 13.35$, $p < 0.001$) (Figure 1g–h, Figure 2a) on larger colonies, as well as colonies with more live tissue (presence GLMM: $\chi^2 = 66.41$, $p < 0.001$; abundance GLMM: $\chi^2 = 8.56$, $p = 0.003$; S GLMM: $\chi^2 = 7.50$, $p = 0.006$) (Appendix C: Table 3). Like *D. albisella*, guard crabs were also more abundant at Waikīkī (5.33 ± 0.42) than at Hanauma Bay (2.86 ± 0.31 ; Tukey: $z = -4.14$, $p < 0.001$) or Kahe Point (2.99 ± 0.51 ; Tukey: $z = -3.00$, $p = 0.008$) (Appendix C: Table 3). At the species level, colony volume was a significant predictor of at least presence (Figure 2b) or abundance (Figure 2c) for all but *T. intermedia* and *T. tigrina*. Living colony tissue was a significant predictor of the local distributions of all 5 species (Appendix C: Table 3). The most common and largest species, *T. flavopunctata*, was more abundant at Waikīkī (2.90 ± 0.30) than Hanauma Bay (1.56 ± 0.29 ; Tukey: $z = -2.60$, $p = 0.025$) and Kahe Point (1.14 ± 0.26 ; Tukey: $t = -3.66$, $p < 0.001$) (Appendix C: Table 3). Most species distributions showed little correlation with surrounding reefscape characteristics, with only *T. digitalis* declining in abundance with increasing cover of *Pocillopora meandrina* (GLMM: $\chi^2 = 12.74$, $p = 0.001$) and *T. flavopunctata* increasing in presence with greater surrounding complexity (GLMM: $\chi^2 = 3.83$, $p = 0.050$). The one exception was *T. intermedia*, which was

more likely to be present with increases in every reefscape variable except general complexity (Appendix C: Table 3).

Removal experiment: Fish recolonization

Resident fish assemblages were not significantly different prior to removals (Appendix C: Table 4), but the experimental removal of *Pl. johnstonianus* and *Pa. arcatus* caused significant increases in resident fish abundance (GLMM: $\chi^2 = 10.08$, $p = 0.001$) (Figure 3a) and species richness (GLMM: $\chi^2 = 8.24$, $p = 0.004$) (Figure 3b). By the end of the removal period, both abundance (mean $\pm$ SE: 7.67 ± 3.78) and species richness (3.00 ± 0.68) on removal colonies were significantly greater than on control colonies (abundance: 0.50 ± 0.22 , species richness: 0.50 ± 0.22). This difference persisted for six months after the termination of the press removal for both abundance (GLMM: $\chi^2 = 10.85$, $p = 0.001$) and species richness (GLMM: $\chi^2 = 14.13$, $p < 0.001$) (Appendix C: Table 4), despite partial recolonization by both strong interactors.

Pomacentrids and cirrhitids were the most numerous fish families to colonize removal colonies as permanent residents, with individuals from each family inhabiting 4 of the 6 removal colonies. Of all the recorded families, only scorpionfishes (Scorpaenidae) were relatively unaffected, occurring in similar numbers on 83% of removal and 67% of control colonies (Appendix C: Table 5).

The most abundant fish species was *Dascyllus albisella*, colonizing twice as many removal colonies as control colonies and in far greater numbers during the removal experiment (Appendix C: Table 5). Similar to total resident fish abundance, *D. albisella* abundance was similar between treatments prior to the experiment, but greater on removal colonies during (GLMM: $\chi^2 = 10.03$, $p = 0.002$) and after the press removal (GLMM: $\chi^2 = 11.58$, $p < 0.001$)

(Figure 3c). Most colonization by *D. albisella*, however, was by the immigration of juveniles and adults, with only 8.6% of *D. albisella* arriving via larval recruitment (34% of total recruitment). Total fish recruitment to *P. grandis* colonies was low in general, but recruit abundance was significantly greater on removal colonies during the experiment (GLMM: $\chi^2 = 16.43$, $p < 0.001$), and nearly absent on control colonies (Figure 3d). As with total resident fish abundance (GLMM: $\chi^2 = 5.18$, $p = 0.023$), fish recruitment decreased on colonies with greater surrounding *P. meandrina* cover (GLMM: $\chi^2 = 7.33$, $p = 0.007$) (Appendix C: Table 4).

Finally, fishes that visited colonies repeatedly throughout surveys were also affected by the two strong interactors. Wrasses (Labridae), surgeonfishes (Acanthuridae), and pufferfishes (Tetraodontidae) were the most common visitors, and all but the latter occurred substantially less often when *Pl. johnstonianus* and *Pa. arcatus* were present (Appendix C: Table 5). Although visiting fish abundance was greater on removal colonies prior to the experiment (GLMM: $\chi^2 = 3.89$, $p = 0.048$), visitor abundance (GLMM: $\chi^2 = 24.44$, $p < 0.001$) and species richness (GLMM: $\chi^2 = 23.48$, $p < 0.001$) were both significantly greater on removal colonies during the experiment (Figure 3e–f). Unlike resident fish, however, visitor abundance (GLMM: $\chi^2 = 2.66$, $p = 0.103$) and species richness (GLMM: $\chi^2 = 0.79$, $p = 0.373$) did not differ between treatments following natural recolonization.

Removal experiment: Invertebrate assemblages

While total invertebrate abundance was greater on removal colonies prior to the experiment (GLMM: $\chi^2 = 6.46$, $p = 0.011$), the difference between treatments increased approximately 4-fold on removal colonies during the experiment (GLMM: $\chi^2 = 13.48$, $p < 0.001$)

and persisted after removals of the strong interactors ceased (GLMM: $\chi^2 = 9.94$, $p = 0.002$) (Figure 4a). Invertebrate species richness also diverged by treatment during the press removal (GLMM: $\chi^2 = 11.73$, $p < 0.001$) and persisted for six months after the experiment (GLMM: $\chi^2 = 6.85$, $p = 0.009$) (Figure 4b, Appendix C: Table 6). The vast majority (91.9%) of invertebrates on all colonies were decapods, with the greatest colonization of removal colonies attributable to the flattened coral shrimp, *Harpiliopsis depressa* (mean $\pm$ SE; beginning: 1.50 ± 0.76 , end: 24.50 ± 5.32). This common member of invertebrate assemblages occurred in 44.3% of visual surveys of colonies at least 50% alive, although at much lower numbers (1.63 ± 0.20) than found on removal colonies in the experiment.

While total invertebrates increased markedly following the removal of *Pa. arcatus* and *Pl. johnstonianus*, guard crabs of the genus *Trapezia* were less affected, as expected because *Trapezia* were not removed at the start of the experiment. Nonetheless, *Trapezia* spp. abundance (Figure 4c) became marginally greater on removal colonies during the experiment (GLMM: $\chi^2 = 3.65$, $p = 0.056$) and remained higher for 6 months (GLMM: $\chi^2 = 4.67$, $p = 0.031$). Species richness (Figure 4d) and recruitment (Appendix C: Table 6) did not change. Abundance (GLMM: $\chi^2 = 9.14$, $p = 0.003$) and species richness (GLMM: $\chi^2 = 7.97$, $p = 0.005$) was greater on colonies with more live tissue during the experiment (Appendix C: Table 6).

3.4 Discussion

Habitat and priority effects on fish assemblages

The two strong interactors, Arc-eye Hawkfish (*Paracirrhites arcatus*) and Blue-eye Damselfish (*Plectroglyphidodon johnstonianus*), occurred on the majority of surveyed *Pocillopora grandis* colonies (Chapter 1) and were more likely to be present and significantly

more abundant as colony size increased. In most habitats, *Pl. johnstonianus* usually occurred as solitary, territorial individuals (Gochfeld 1991, Gajdzik et al. 2016). However, previous studies found that they also live in groups on *P. grandis* colonies in Hawai‘i (Chapter 1, Gochfeld 1991). As an obligate corallivore, *Pl. johnstonianus* also becomes much less abundant when the coral cover of a reef declines (Spalding and Jarvis 2002, Wilson et al. 2008). A reduction in coral cover of reefs around O‘ahu, therefore, could have an effect of concentrating *Pl. johnstonianus* on *P. grandis* colonies, especially the largest colonies observed at the Waikīkī site, where groups of 5 or more were common, with several others occupying surrounding habitat (pers. obs.). *Pa. arcatus* followed the same patterns, but this species is limited to 2 individuals per colony due to its social organization (Donaldson 1990), even on its preferred habitat of larger pocilloporid colonies with wide branch spacing (Kane et al. 2009, Coker et al. 2015). However, with the exception of depth, no other variable was a significant predictor of the distribution of either strong interactor despite differences in common coral species and reefscape characteristics among sites (Chapter 1, Franklin et al. 2013).

Previous work found that resident fish assemblages on *P. grandis* in Hawai‘i were primarily affected by colony characteristics and unknown site-scale differences, with reefscape characteristics and depth explaining only a minor portion of species-specific local distributions (Chapter 1). Smaller colonies typically host small fish assemblages of only a few species that often include *Pl. johnstonianus* and/or *Pa. arcatus*. As expected, the presence of these two species substantially reduced colonization by other fishes. Surrounding *P. meandrina* density was the only significant habitat predictor, negatively affecting resident fish abundance, and species richness was unaffected by all colony and reefscape characteristics. While we did not explicitly test for arrival order, these patterns suggest that priority effects caused by the strong interactors

are more important in governing fish assemblage structure on smaller *P. grandis* colonies. Similarly, Bay et al. (2001) found that among a guild of territorial damselfishes, aggressive interactions were more important than substrate and habitat selection in shaping the local distributions of species involved.

Territorial damselfishes have been widely shown to greatly affect benthic fish communities (see review by Ceccarelli et al. 2001 for herbivorous species), and can restrict colonization, particularly of other damselfishes (Eurich et al. 2018a). This territoriality was reflected in analyses of assemblage structure showing evidence for negative co-occurrence on smaller colonies between *Pl. johnstonianus* with another damselfish, *Dascyllus albisella* (Chapter 1). In the present study, visual surveys indicated that *D. albisella* was more common at the Waikīkī field site, which had the lowest coral cover and structural complexity, and that presence and abundance on *P. grandis* decreased with measures of greater surrounding structural complexity. *D. albisella* was also the most common colonizing species in the absence of the strong interactors. Therefore, the majority of restrictions to adult immigration within the experiment were likely due to the presence of *Pl. johnstonianus*.

While most colonizing fishes were adult immigrants, larval fish recruitment was also greater on colonies with the strong interactors removed. Territorial damselfishes can actively prohibit fish recruitment, but this effect is typically species-specific (Shulman et al. 1983, Almany 2003). A known predator of small fishes (Shima et al. 2008, Stier et al. 2013), *Pa. arcatus* is the most likely cause for this difference between treatments. This difference in fish recruitment also did not appear to decline after the removal period ended, and fewer removal colonies were naturally recolonized by *Pa. arcatus* than *Pl. johnstonianus* at 3- and 6-months post removals. The most common recruit species was the damselfish *D. albisella*, which are

attracted to the presence of previously settled recruits and juveniles (Booth 1992). In Mo'orea, recruit congener damselfishes were vastly reduced by *Pa. arcatus* on pocilloporid corals similar in size to this study, where colonies either hosted large groups of *Dascyllus* recruits or a single hawkfish, and coexistence occurred only on larger colonies (Holbrook et al. 2011). Reduced colonization with greater surrounding structural complexity of both adults and recruit *D. albisella* could reflect either a potential preference for more isolated habitat (Danilowicz 1995) or colonization of *P. grandis* colonies that are less likely to host one or both strong interactors.

Given that resident fish abundance and recruitment on *P. grandis* both decreased with increasing density of surrounding *P. meandrina* colonies, it is possible that habitat may not be limiting (Chapter 1, Walsh 1987), as fish abundance is diluted in more coral-rich areas. This pattern is further highlighted by the low levels of recruitment recorded throughout the entirety of the experiment. Walsh (1987) hypothesized that coral reefs in Hawai'i may be recruitment limited, whereby local populations are constrained by a depauperate or patchy supply of new individuals rather than post-settlement density-dependent processes (Doherty 2002). In a multi-decadal study of *Pa. arcatus* recruitment at multiple sites around the island of Hawai'i, DeMartini et al. (2013) identified sites with consistently low recruitment, indicating that recruitment limitation could be occurring on at least a local scale on Hawaiian reefs. Changes to fish assemblage structure caused by lower recruitment can have indirect effects on host corals, especially regarding *D. albisella*, as this genus provides several key benefits for the host coral (Holbrook et al. 2008, Chase et al. 2018, 2020, Shantz et al. 2023).

Lastly, the presence of visiting fishes that regularly interacted with and whose home range included each *P. grandis*, were also substantially reduced when the two strong interactors were present. However, this result was species specific (Almany 2003), with some species, such

as the chemically protected pufferfish *Canthigaster jactator*, seemingly unaffected and visiting both removal and control colonies in equal numbers. Unlike resident fishes, effects on visitors seemed more transient, with visiting fish abundance and species richness returning to pre-removal levels after natural recolonization was allowed to occur. Territorial damselfishes can alter foraging and habitat use by roving and more transient species (Jones 2005, Eurich et al. 2018b), indicating that *Pl. johnstonianus* presence likely has a larger effect on visiting fishes than *Pa. arcatus*, which usually directs aggression towards conspecifics (Kane et al. 2009). Some of these visiting species were also transient corallivores (e.g., the butterflyfish *Chaetodon multicinctus*), and were recorded almost exclusively on colonies lacking *Pl. johnstonianus*, which actively defends its host colony against corallivorous fishes (Tricas and Boyle 2014).

Habitat and priority effects on invertebrate assemblages

While not initially manipulated like resident fish assemblages, both the abundance and species richness of resident invertebrates increased in the absence of *Pa. arcatus* and *Pl. johnstonianus*, with abundance increasing nearly 4-fold between the beginning and end of the removal period. This increase was most attributable to the commensal shrimp (*Harpiliopsis depressa*) which naturally occurs on *P. grandis* (Chapter 1) and *P. meandrina* (Counsell et al. 2018) colonies around O‘ahu in lower abundances. A factorial manipulation of two predator species on *P. grandis* in Mo‘orea found that the congener *H. spinigera* was less abundant when predators were present (Stier and Leray 2014), but not to the same degree as in this study. Because the diet of *Pa. arcatus* include decapod shrimps (Hobson 1974, Leray et al. 2015), the difference in abundance by treatment was almost certainly due to manipulations of the hawkfish. Additionally, the removal of another invertebrate predator, the Hawaiian Velvetfish (*Caracanthus*

typicus) from all corals where it was present prior to the experiment, may have also contributed to this increase. Congeners of *C. typicus* have been shown to consume (Leray et al. 2015) and reduce the abundance of *Pocillopora*-associated shrimp (Stier and Leray 2014). Therefore, the absence of both predators on removal colonies could have caused the immense increase in *H. depressa*, as a previous factorial manipulation of hawkfish and velvetfish found greater effects when both were present (Stier and Leray 2014). However, any functions or effects this shrimp has on pocilloporid corals are unknown (Leray et al. 2015).

Among invertebrate assemblages, guard crabs of the genus *Trapezia* provide critical benefits to host pocilloporid corals, including removal of sediments (Stier et al. 2012) and defense against invertebrate corallivores (McKeon et al. 2012). Surprisingly, guard crab assemblages during the experiment changed little, despite being known prey of *Pa. arcatus* (Leray et al. 2015), especially in Hawai'i (Hobson 1974). *Trapezia* species richness and recruitment did not differ between treatments during the removal, and there was limited evidence to suggest an increase in abundance on removal colonies. Leray et al. (2015) found that of three predator species of pocilloporid-associated invertebrates in Mo'orea, guard crabs comprised a comparatively smaller proportion of the diet of *Pa. arcatus* than the others, including velvetfish of the genus *Caracanthus*. After initial pulse removals, recolonization by *C. typicus* to removal and control colonies was extremely limited, which could have contributed to the lack of a difference in guard crab assemblages. Given that the most common guard crab species on *P. grandis* colonies in Hawai'i is *T. flavopunctata*, the largest species, it is also possible that *Pa. arcatus* or *C. typicus* presence reduces the abundance of only smaller guard crabs (Stier and Leray 2014). However, new recruits (<0.25cm carapace width), particularly of the smallest species (*T. digitalis*) were regularly seen near the branch tips of both removal and control

colonies. Therefore, it is possible that in the absence of an initial removal of guard crabs, already established assemblages would not change significantly regardless of predator presence.

While effects of the experimental manipulation of either the interference competitor (*Pl. johnstonianus*) or the predator (*Pa. arcatus*) was minimal on guard crab assemblages, there was potential evidence for habitat partitioning between crab species. Previous work showed evidence for possible within-colony microhabitat partitioning, with larger guard crabs occupying the exterior branches and smaller species nearer to the base (Chapter 1, Castro 1976). The differences in occurrences and abundances of each species across colony characteristics, reefscapes, and sites provides further evidence for microhabitat partitioning, as well as potential competition and habitat partitioning at larger scales. Most species were associated with greater colony volume and living tissue, consistent with prior studies (Castro 1976, Gotelli et al. 1985). Species richness also increased with colony volume and living tissue, indicating that microhabitat created by large, healthy colonies likely relaxes the strain of competition, promoting coexistence. Species that were not correlated with colony volume either associated with many reefscape characteristics (*T. intermedia*) or were substantially less common (*T. tigrina*).

Beyond the colony scale, the largest and most common species, *T. flavopunctata*, as well as *T. digitalis*, decreased at greater depths. The latter also decreased in abundance with a greater proportion of surrounding *P. meandrina* colonies. Contrary to this pattern, *T. intermedia* was found to be present more frequently when these predictors and the proportions of surrounding hardbottom and sand increased. This species may be more common on isolated colonies, where displacement by competitively dominant species is less likely (Counsell et al. 2018). Combined with evidence for differences in either presence or abundance by site for 4 of the species, it is

likely that some level of partitioning by reef location (Gotelli et al. 1985), depth, and/or surrounding reefscape is occurring, with potential influence by priority effects on the colony scale.

Conclusions

As foundation species, corals provide the biogenic habitat that can mediate the negative effects of resident species interactions, foster coexistence, and support high local biodiversity. On *Pocillopora grandis*, the microhabitat distributions of resident species are likely shaped by factors that change through the life of the host coral as it increases in size. This study documents the combination that potential priority effects, reefscape, and environmental variables can have on shaping resident assemblage structure through the entire lifespan of a host coral. Prior residence by the two strong interactors – a dominant competitor and a voracious predator – likely restricts resident fish and invertebrate abundance and species richness at smaller colony sizes. However, once living space increases on larger, older colonies, ecological restrictions to coexistence relax and environmental and reefscape variables likely play a larger role. Guard crabs, however, may be governed more by the total space provided by the host coral, intraspecific aggression, and priority effects between similar species at smaller colony sizes.

Understanding the factors that shape community structure on coral reefs across spatial scales is critically important, as coexistence and species interactions shape ecosystem function. Furthermore, mutualistic interactions that benefit species that form the biogenic habit on which so many others rely fosters ecosystem resilience. Pocilloporid corals are common hosts for several known mutualists, each of which is likely affected by different combinations of interactions with other resident species, colony and reefscape characteristics, and environmental

regimes. Understanding the factors that shape resident assemblages and resulting mutualistic interactions with foundation species is critical for expanding our knowledge of the building blocks of ecosystem resilience and function, especially considering current global threats to coral reefs and other highly biodiverse ecosystems.

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3.6 Figures

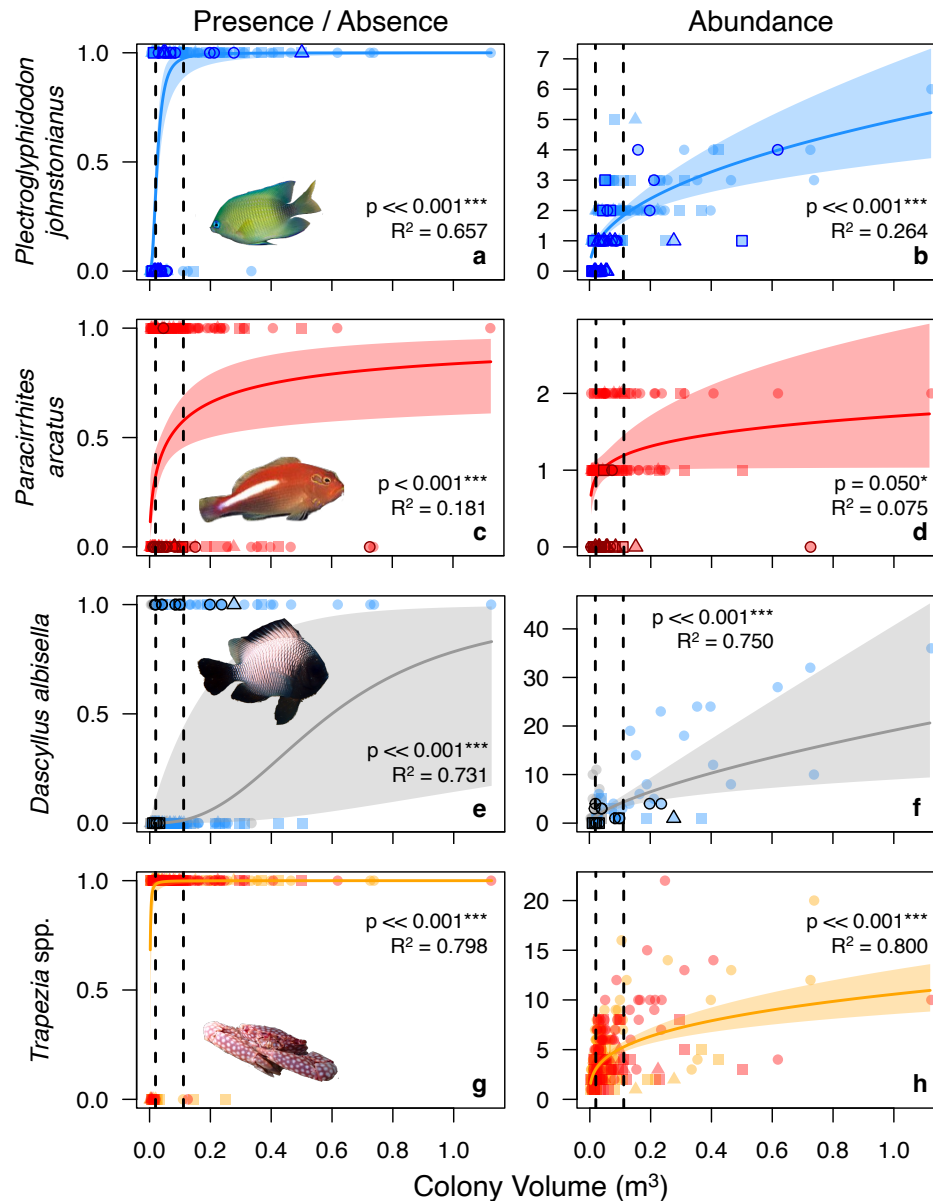


Figure 1. The correlation between *Pocillopora grandis* colony volume and statistical models of presence and abundance of resident *Plectroglyphidodon johnstonianus* (a, b), *Paracirrhites arcatus* (c, d), *Dascyllus albisella* (e, f), and guard crabs of the genus *Trapezia* (g, h). Symbols represent different sites (square: Hanauma Bay, triangle: Kahe Point, circle: Waikīkī). Dashed vertical lines are the range of colony sizes used for the experiment. For *Pl. johnstonianus* and *Pa. arcatus* plots, points outlined in darker blue or red, respectively, are surveys in which visiting individuals were also present. For the *D. albisella* plots, points that are blue indicate surveys in which resident *P. johnstonianus*, an interference competitor, was also present. For the *Trapezia* spp. plots, red points indicate surveys in which resident *Pa. arcatus*, a predator, was present.

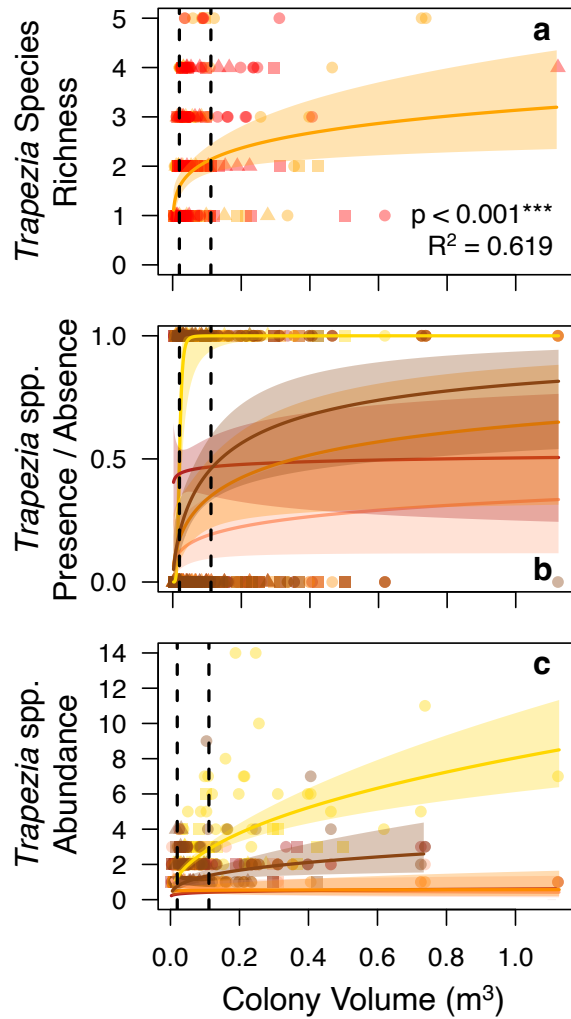


Figure 2. Correlations between *Pocillopora grandis* colony volume and statistical models of *Trapezia* **a** species richness, and **b** presence/absence and **c** abundance of all 5 species. Symbols represent different sites (square: Hanauma Bay, triangle: Kahe Point, circle: Waikīkī). For species richness, red points indicate colonies also inhabited by *Paracirrhites arcatus*, a potential predator, while orange points indicate colonies without this species present. For presence and abundance, each species is represented by a different color (yellow: *T. flavopunctata*; dark red: *T. intermedia*, brown: *T. digitalis*, orange: *T. bidentata*, salmon: *T. tigrina*).

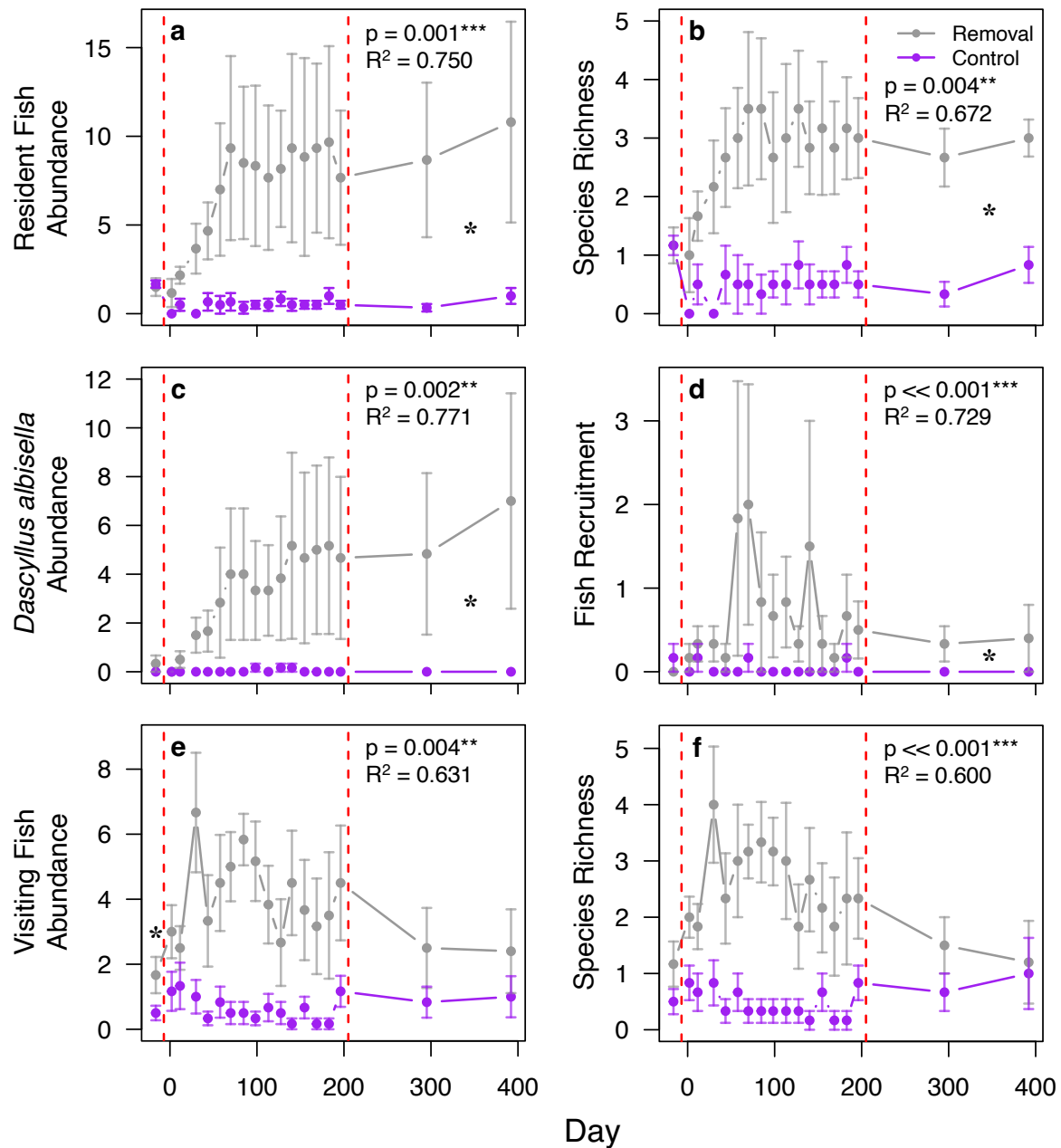


Figure 3. Means and standard errors of **a** resident fish abundance and **b** species richness (S), **c** *Dascyllus albisella* abundance, **d** total resident fish recruitment, and **e** visiting fish abundance and **f** species richness (S) before, during, and after the experimental removal of strong interactors on *Pocillopora grandis*. Colors represent treatment (removals: grey, controls: purple). Dashed red lines indicate the start and end of the press removal. Control colony metrics do not include *Plectroglyphidodon johnstonianus* or *Paracirrhites arcatus*, as they were being actively removed on removal colonies. Asterisks indicate significant differences between treatments during pre- and post-removal surveys. P values correspond to the effect of treatment during the removal period. R^2 values are of the total variation explained by fixed effects during the removal period.

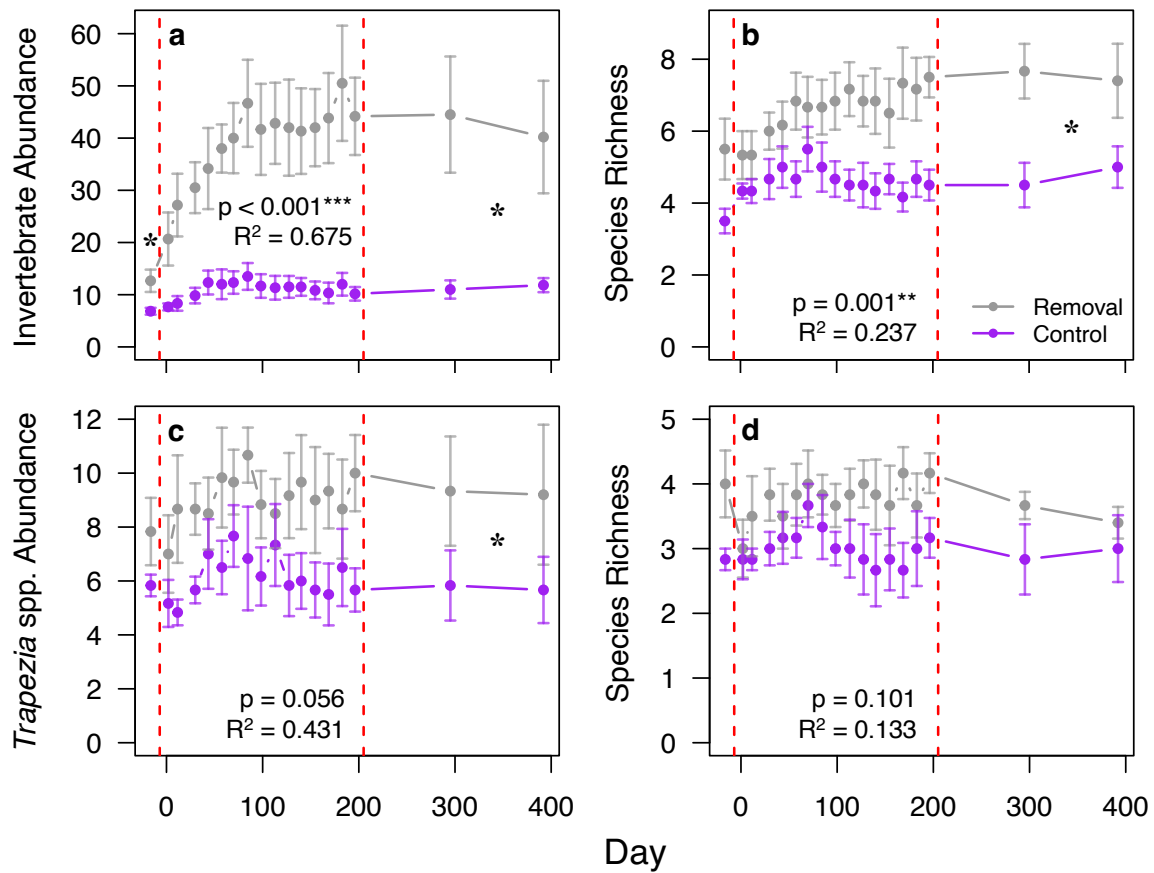


Figure 4. Means and standard errors of **a** total invertebrate abundance and **b** species richness, and **c** guard crab abundance and **d** species richness before, during, and after the experimental removal of strong interactors on *Pocillopora grandis*. Colors represent treatment (removals: grey, controls: purple). Dashed red lines indicate the start and end of the press removal. Asterisks indicate significant differences between treatments during pre- and post-removal surveys. P values correspond to the effect of treatment during the removal period. R² values represent the total variation explained by all fixed effects during the removal period.

CHAPTER 4

Indirect fish-coral mutualisms and resident fish assemblages
enhance host coral growth on Hawaiian coral reefs

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Abstract

Mutualisms are important yet often understudied relationships that may substantially affect the structure of ecological communities. In marine ecosystems, mutualisms often involve foundation species such as corals that provide critical biogenic habitat for other species. In Hawai‘i, the antler coral *Pocillopora grandis* is a large branching species that provides substantial habitat for fishes and invertebrates, including known mutualists. I conducted a press removal of an interference competitor and obligate corallivore (the territorial damselfish *Plectroglyphidodon johnstonianus*) and a mesopredator (the hawkfish *Paracirrhites arcatus*) on colonies of antler coral to investigate whether these species indirectly affected their host coral. I predicted that their presence would alter fish assemblage trajectory on the host coral, thereby either 1) reducing growth due to less fish colonization and overall nutrient provisioning via excreta and/or 2) enhancing growth due to a decrease in total corallivore abundance. In the absence of the territorial damselfish, host colonies suffered substantially reduced growth rates due to predation by transient corallivorous fishes. This result provides evidence for a novel indirect mutualism between a territorial corallivore and its host coral, whereby the fish protects the coral from intense predation from other corallivores and the coral provides food and shelter for the fish. The mesopredator also likely played a role in augmenting coral growth rates through nutrient provisioning via excreta. Examining unmanipulated colonies, coral growth rates increased with greater resident fish abundance and suggested differences in the positive effects acquired by the host coral depending on resident fish identity. All told, mutualisms involving fishes benefit corals in a variety of ways, thereby increasing the amount of habitat space available for the many other species on coral reefs.

Keywords: coral reef, indirect mutualism, corallivory, territoriality, *Pocillopora*, Hawai‘i, assemblage structure

4.1 Introduction

Mutualisms – interactions between species that benefit both – are often understudied compared to other biological interactions, yet these relationships can play critical roles in structuring ecological communities (Stachowicz 2001). Within marine ecosystems, local biodiversity is often reliant upon and greatly affected by mutualistic relationships (review by Hay et al. 2004), particularly when these interactions involve foundation species (Dayton 1972) or other species that have disproportionately strong effects in shaping community structure. Some foundation species provide the biogenic habitat upon which others rely, such as corals, kelps, and seagrasses. Mutualisms involving foundation species have substantial effects on ecosystem resilience (Angelini et al. 2016, de Fouw et al. 2016), habitat provisioning and availability (Holbrook et al. 2008, González-Duarte et al. 2020), species coexistence (Schmitt & Holbrook 2003, Holbrook & Schmitt 2004), and local biodiversity (Stachowicz & Hay 1999, Creed et al. 2021).

On coral reefs, mutualisms can occur across a wide range of spatial gradients, from the suppression of algal biomass by herbivores that promotes coral dominated reefs (review by Hixon 2015) to the symbiotic relationship between dinoflagellates and their host corals (Muller-Parker et al. 2015). Relationships between fishes with foundational species can increase the diversity of both benthic algae (Hixon & Brostoff 1983) and corals (Gochfeld 2010) across a reefscape. Further, fishes can increase primary productivity and coral growth rates via nutrient provisioning over large areas (Meyer et al. 1983, Meyer & Schultz 1985, Shantz et al. 2015).

The directional effect of nutrient provisioning by resident fishes can be dependent upon environmental context (Chase et al. 2014), and species interactions, including indirect effects, can promote or disrupt mutualistic relationships via unpredictable pathways. For example, Cantrell et al. (2015) showed that the attraction of fish to a cleaning shrimp provided nutrients via fish excreta that increased growth rates of the shrimp's host anemone. Conversely, predators have been shown to reduce the abundance of mutualist guard crabs on host corals (Stier & Leray 2014). Yet the mechanisms by which the indirect effects of species interactions can affect mutualisms are poorly understood.

Priority effects occur when the prior colonization of habitat by a species affects community trajectory by either inhibiting or facilitating species that arrive later (Gleason 1927). Previous studies of coral reef fish communities have documented species-specific inhibition or facilitation of other fishes by the prior arrival of territorial damselfishes and resident predators (Shulman et al. 1983, Almany 2003). The alteration to community trajectory could, therefore, affect mutualist abundance (Holbrook et al. 2011) as well as the magnitude of benefits gained by either party. For example, corals and benthic algae compete for limited space on coral reefs, and their order of arrival to new habitat space created by abiotic or biotic disturbance can lead to different ecosystem states (Adam et al. 2022). Depending on the dominant benthos, nutrient provisioning by resident fishes can shift from facilitating coral growth to increasing macroalgal abundance (Burkepile et al. 2013).

At smaller scales, numerous fish and invertebrate species have been identified as coral mutualists that augment the condition of their host colony in a variety of ways. Due to their highly branching morphology that attracts mutualist species, corals of the genus *Pocillopora* have been well-studied with regards to mutualisms between the host coral and its resident fish

assemblage. Fish of the genus *Dascyllus* benefit their host by providing defense against corallivorous butterflyfishes (Chase et al. 2014), alleviating the effects of sedimentation (Chase et al. 2020a), reducing hypoxic conditions between branches with sleep-swimming behavior (Goldshmid et al. 2004), bolstering resistance to bleaching and thermal stress (Chase et al. 2018, Shantz et al. 2023), and augmenting growth rates through nutrient provisioning from excreta (Liberman et al. 1995, Holbrook et al. 2008, 2011). Herbivorous damselfishes can also indirectly benefit corals within their algal mats through associational defense (Gochfeld 2010).

Coral reefs along the south shore of O‘ahu, Hawai‘i, are characterized by low-lying coral habitat with infrequently dispersed colonies of the antler coral *Pocillopora grandis*, which can grow to 1.5m in diameter. This coral is commonly occupied by two strong interactors (sensu Paine 1980): *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus* (Chapter 1, Chapter 2). *Pl. johnstonianus* is an obligate corallivore (Gochfeld 1991) and territorial damselfish (Gajdzik et al. 2016) that aggressively chases other species, particularly other coral predators (Tricas & Boyle 2014). The hawkfish *Pa. arcatus* is a mesopredator of small fishes (Shima et al. 2008), particularly of recruit-sized damselfishes (Holbrook et al. 2011), and occupies the vast majority of *Pocillopora* spp. colonies in Hawai‘i (DeMartini 1996), especially those of wider branching morphology (Kane et al. 2009). Previous work found evidence for potential priority effects, although order of arrival was not specifically tested. The presence of these two strong interactors reduced resident fish abundance and species richness, particularly of damselfish from a known mutualistic genus, *Dascyllus albisella* (Chapter 2).

Because *P. grandis* colonies commonly host 1) an interference competitor and obligate corallivore (*Pl. johnstonianus*) known to restrict the colonization of other fishes, particularly other coral predators and the potential coral mutualist *D. albisella*, and 2) a mesopredator (*Pa.*

arcatus) known to restrict fish recruitment, I hypothesized that the absence of these two strong interactors would indirectly affect coral growth rates by altering the direct and indirect effects of local fish assemblages. I predicted that without these two species, the trajectory of host coral colonization by fishes would result in either 1) reduced growth from less overall fish abundance and nutrient provisioning from excreta, or 2) enhanced growth due to a decrease in total corallivory via protection from other coral predators. To test these predictions experimentally, I conducted a press removal of the two strong interactors on half of the involved colonies and a pulse removal of all other resident fishes on all colonies. I then followed recolonization, comparing coral growth rates between replicate removal and control colonies. I also tested the hypothesis observationally by measuring growth rates of colonies with naturally variable resident fish assemblages.

4.2 Materials and Methods

Study site

I conducted this study off the south shore of O‘ahu, Hawai‘i, specifically offshore of Waikīkī (21°16'11.00"N, 157°50'18.53"W). This site consists of spur-and-groove coral reef with low vertical relief, an average coral cover of 5% (unpub. data; Franklin et al. 2013), and depth ranging from 5 to 18m. Small colonies of *Pocillopora meandrina* and *Porites lobata* make up the bulk of the coral cover across the site, but irregularly spaced, large colonies of *Pocillopora grandis* provide oases of high vertical relief. Larger bodied fishes are not common across the site due to fishing and chronic stressors of the large, nearby human populace of Honolulu (Friedlander et al. 2018).

Experimental design

To test how prior occupancy by the two strong interactors can indirectly affect host colony growth by altering resident assemblage structure and the abundance of important mutualists, I selected twelve *Pocillopora grandis* colonies across the study site, ranging in size from 34 to 65cm in diameter (mean $\pm$ SD; 50.1 ± 9.3 cm) and occurring at depths from 5.5 to 14m (10.5 ± 3.5 m). The 100-square meters of reef surrounding each colony were mapped, and the 10 $\times$ 10m plots were paired based on central colony size and proximity, so that paired colonies experienced similar local conditions. Paired central colonies were randomly divided into removal and control treatments, and initial surveys of each central *P. grandis* colony were conducted prior to experimental manipulations. Surveys consisted of a full census of all permanent fish residents (present for the entire survey period), visiting fish that interacted with or resided on the colony as part of their home range (present for a portion of the survey period), and resident mobile invertebrates.

In the summer of 2020, I conducted experimental manipulations of *P. grandis* fish assemblages on a staggered schedule, starting the experiment on each block (1 removal and 1 control plot) in sequence (ranging from June to August). Prior to monitoring the experiment within each block, I began a 6-month long press removal of *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus* from the entirety of each removal plot to provide a buffer zone for effects on the *P. grandis* assemblage. Logistic constraints made a factorial removal of these two species not achievable. Additionally, the relatively uncommon Blackside Hawkfish (*Pa. forsteri*), was removed from all plots for the entirety of the experiment due to its unequal presence and potential predatory influence across plots. The territories of this species are larger than 100m² and typically include a large coral such as *P. grandis* (Donaldson 1990). Finally, just before the

experiment began, I conducted a pulse removal on each central *P. grandis* colony of all non-focal fish species, as well as invertebrates not commonly present in natural assemblages (common species such as guard crabs of the genus *Trapezia* were not manipulated). Throughout the experiment, I removed Blackside Hawkfish via spearing, and all other species were removed from coral colonies with chopsticks and hand nets (IACUC #19-3255). If needed and typically only for *Pl. johnstonianus*, low concentration (<7%) clove oil (a natural fish anesthetic, see review by Priborsky and Velisek 2018) dissolved in seawater aided in capturing fishes designated for removal (DAR SAP No. 2019-45). I released all live captured fish at least 1.5km away, and there was no evidence of homing. Control colonies were exposed to clove oil only when necessary for initial pulse removals. It is unlikely that clove oil affected coral growth rates during this experiment because previous studies have shown that low concentrations of clove oil, especially lacking ethanol as in this study, has no significant effect on coral health (Frisch et al. 2007) or growth (Boyer et al. 2009).

Observational study design

I selected 15 *P. grandis* colonies (44.7 ± 8.8 cm diameter; range: 27 – 60cm) with different naturally occurring fish assemblages to further explore effects of assemblage variation on host colony growth rates. Colonies were located at depths of 7.6 to 18.3m (10.8 ± 2.6 m). These assemblages were grouped into three variants based on previous findings that *Dascyllus albisella*, a shoaling damselfish, was likely inhibited from colonizing *P. grandis* by the presence of *Plectroglyphidodon johnstonianus*. These assemblage variants were: 1) *Pl. johnstonianus* present and *D. albisella* absent (n = 6), 2) *D. albisella* present and *Pl. johnstonianus* absent (n = 4, but 1 colony died during the observational period), and 3) both species present (n = 4). A

fourth assemblage variant, in which neither *Pl. johnstonianus* nor *D. albisella* were present was planned. However, only 1 colony was found that fit these criteria, and it died during the study. In the assemblage type with both species, *D. albisella* were actively prevented by *Pl. johnstonianus* from entering between the branches of the colony for long periods of time and were thereby relegated to the colony exterior. These individuals often used other surrounding substrate for shelter in addition to the central colony. I began the coral measurement period and initial surveys of each colony as time allowed during the experiment (ranging from November 2020 to May 2021).

Coral growth measurements

I used structure-from-motion photogrammetry to measure growth rates of each central *P. grandis* colony by creating digital models at the beginning and end of the press removal of strong interactors. For each colony model, photos were taken with a TG-6 camera (Olympus America) with dual Sola Dive 800 lumen S/F lights (Light & Motion) following the photograph pattern described in Lange and Perry (2020). I took additional photos in a series of parallel concentric rings to account for the larger coral size and complexity. I also placed a Rubik's cube next to each colony for digital measurements of an object of known size. The photos were imported into *Metashape* v1.7.2 (Agisoft LLC) for model generation. I followed all protocols by Lange and Perry (2020) with several modifications. First, sparse point cloud generation was completed on the "highest" setting. Second, following Over et al. (2021), I generated models with the goal of attaining an SEUW value as close to 1 as possible and a root mean square (RMS) reprojection error less than 0.18 pixels (see Appendix E: Table 1 for model diagnostics). Third, dense point

clouds were generated under the “high” setting. Initial colony volume was also measured within *Metashape*.

I imported each pair of dense point clouds for a colony into *CloudCompare* v2.12.4 (Zephyrus) to measure apical extension (Appendix F: Figure 1). Prior to photography, stakes were driven into the surrounding substrate to serve as ground control points for manual alignment of models. I measured the apical extension of 20 randomly selected branches on each colony and converted all measurements to yearly linear growth. I followed published protocols to evaluate the precision of model construction and accuracy of manual model pair alignment (Figueira et al. 2015, Ferrari et al. 2017) (Appendix E: Table 2 and Table 3). As an additional evaluation of model accuracy, I measured 9 segments of a known length of the Rubik’s cube for each model within *CloudCompare* and calculated the absolute difference of each digital measurement (Appendix E: Table 4). Measured growth rates of *P. grandis* were comparable to other studies that used other methods (alizarin dye, x-ray densitometry, physical measurements; see Appendix E: Table 5). To my knowledge, this is the first study to use photogrammetry to measure growth rates of *P. grandis*, yet Curtis et al. (2023) found that photogrammetry improved measurements of increases in *Pocillopora* spp. volume in Mo‘orea.

Statistical analyses

I conducted all statistical analyses within the software *R* (R Core Team 2023) with the packages *lme4* for all linear mixed effects models (LMMs) (Bates et al. 2015) and *glmmTMB* for all general linear mixed effects models (GLMMs) (Brooks et al. 2017). Fish biomass was calculated using published length-weight relationships for each species, in Hawai‘i when

possible (Appendix E: Table 6). When a relationship for a species could not be found, a phylogenetically and morphologically similar species was chosen as a proxy.

For the experiment, assemblage metrics were evaluated for each time period separately (before, during, and after press removals). Statistical models for the period during press removals contained fixed effects for treatment (removal or control) and initial colony volume, while those for before and after included only treatment. Random effects were included for block (colony pair), survey date, survey day (due to the staggered start dates among the blocks), and colony identity, with the last accounting for repeated measures. For the observational surveys, all models included natural assemblage variant as a categorical variable (variant 1: *Pl. johnstonianus* present, *D. albisella* absent; variant 2: *D. albisella* present, *Pl. johnstonianus* absent; variant 3: both present) and initial colony volume as fixed effects, as well as date, survey day, and colony identity as random effects. Colony growth was assessed with statistical models that included fixed effects for treatment (experiment) or assemblage type (observational study) and initial colony volume, as well as random effects for colony identity and the first month of the growth measurement period to account for seasonal differences in growth measurement periods. Finally, statistical models correlating coral growth with fish assemblage metrics during the experiment (resident fish abundance and biomass, and corallivore abundance) and observational study (resident fish abundance and biomass) included fixed effects for each assemblage metric averaged across surveys during the coral growth period and initial colony volume. These models also included random effects for colony identity and the first month.

In all models, biomass as a response variable and colony volume as a predictor were square root- and log-transformed, respectively. Resident fish biomass and coral colony growth were evaluated with LMMs and predictors assessed with Type III ANOVAs with Satterthwaite's

method. Resident fish and corallivore abundance were assessed with GLMMs with Poisson error distributions and predictor significance determined through chi-square likelihood ratio tests.

4.3 Results

Effects of strong interactor removals on resident fish assemblages

Despite being removed from half of the experimental colonies, the abundance and biomass of the strong interactors (territorial *Plectroglyphidodon johnstonianus* and mesopredatory *Paracirrhites arcatus*) were included in all relevant analyses of resident fishes due to their potential effect on host colony growth and partial recolonization during the experiment. During the removal period, neither resident fish abundance (GLMM: $\chi^2 = 0.08$, $p = 0.772$) (Figure 1a) nor total biomass (LMM: $F = 1.99$, $df = 1, 6.37$, $p = 0.205$) (Figure 1b, Figure 2d) were significantly affected by the removal of the strong interactors. While both the abundance (mean $\pm$ SE; beginning: 1.17 ± 0.79 , end: 7.83 ± 3.73) and total biomass of resident fish (beginning: 9.20 ± 8.04 g, end: 94.26 ± 46.13 g) increased substantially on removal colonies during the press removal, abundance (beginning: 2.17 ± 0.31 , end: 2.17 ± 0.31) and biomass (beginning: 40.54 ± 11.14 g, end: 31.66 ± 8.28 g) on control colonies remained relatively steady (Figure 1a, b). When *Dascyllus albisella*, a species that spends much of its time feeding on plankton above the coral colony and the most common colonizing fish in the experiment by far, is excluded from calculations of total abundance and biomass, the effects on abundance did not change (GLMM: $\chi^2 = 0.81$, $p = 0.368$) (Appendix F: SM2a), yet total biomass (LMM: $F = 13.94$, $df = 1, 8.98$, $p = 0.005$) was significantly less on removal colonies (Appendix F: Figure 2b).

There was nonetheless a significant difference in average individual fish biomass between treatments during the press removal, with larger fish on average residing on control

colonies (model mean $\pm$ SE; 18.71 ± 3.67 g) compared to removal colonies (5.29 ± 1.79 g; LMM: $F = 12.32$, $df = 1,9.00$, $p = 0.007$) during the press removal (Figure 2c, Appendix F: Figure 3). All metrics of resident fish assemblages were also, at least marginally, positively correlated with colony volume (abundance GLMM: $\chi^2 = 5.21$, $p = 0.022$; total biomass LMM: $F = 6.03$, $df = 1,5.06$, $p = 0.057$; mean biomass LMM: $F = 12.32$, $df = 1,9.00$, $p = 0.075$).

Effects of strong interactor removals on host colony growth rates

During the experiment, control colonies containing *Pl. johnstonianus* and *Pa. arcatus* grew significantly more (35.3 ± 3.67 mm yr⁻¹) than colonies where these strong interactors were removed (23.3 ± 3.38 mm yr⁻¹; LMM: $F = 10.05$, $df = 1,9.00$, $p = 0.011$) (Figure 2a; see Appendix F: Figure 4 for individual colony growth rates). Concurrently, coral growth was not correlated with either resident fish abundance (LMM: $F = 0.06$, $df = 1,9.00$, $p = 0.819$) or total biomass (LMM: $F = 0.09$, $df = 1,9.00$, $p = 0.773$) during the press removal. However, corallivores (excluding *Pl. johnstonianus* throughout) were present during surveys of removal colonies (0.43 ± 0.20) significantly more often than control colonies (0.02 ± 0.02 ; GLMM: $\chi^2 = 8.72$, $p = 0.003$) during the experiment (Figure 1c, Figure 2b). Corallivore abundance also increased with host colony size (GLMM: $\chi^2 = 4.70$, $p = 0.030$). This difference in abundance persisted after press removals concluded and natural recolonization of the strong interactors was allowed to continue (GLMM: $\chi^2 = 7.33$, $p = 0.007$), although a decline in corallivore abundance on removal colonies was noticeable during the six months after the press removal (Figure 1c). Coral growth was negatively correlated with corallivore abundance (LMM: $F = 6.09$, $df=1,9.00$, $p = 0.036$) (Figure 3). In all models correlating coral growth with fish assemblage metrics, initial colony volume had no significant effect on coral growth.

Observational study of resident fish assemblages

Due to low sample size and statistical power, there was no significant difference in resident fish abundance (GLMM: $\chi^2 = 5.01$, $p = 0.082$) (Figure 4b) and total biomass (LMM: $F = 4.07$, $df = 2, 8.50$, $p = 0.058$) (Figure 4c) among unmanipulated assemblages. However, these results do suggest a trend of greater abundance (10.05 ± 3.58) and total biomass ($170.5g \pm 54.8g$) on colonies with *D. albisella* present but *Pl. johnstonianus* absent, especially when compared to colonies with *D. albisella* absent but *Pl. johnstonianus* present (abundance: 3.71 ± 0.87 ; abundance Tukey HSD: $z = -2.22$, $p = 0.068$; biomass: 42.0 ± 16.4 ; biomass Tukey HSD: $t = -2.73$, $p = 0.059$). Fish abundance did, however, increase with initial colony volume (GLMM: $\chi^2 = 6.82$, $p = 0.009$), and there was evidence for a potential positive correlation with total biomass (LMM: $F = 3.64$, $df = 1, 8.13$, $p = 0.092$).

Regarding assemblage composition, there was evidence for a difference in the abundance of resident individuals of the hawkfish, *Paracirrhites arcatus* (GLMM: $\chi^2 = 5.39$, $p = 0.068$) between the three assemblage variants: 1) *Pl. johnstonianus* present but *D. albisella* absent (variant 1), 2) *D. albisella* present but *Pl. johnstonianus* absent (variant 2), and 3) both species present (variant 3). *Pa. arcatus* abundance was greatest on variant 2 (1.50 ± 0.64) compared to variant 1 colonies (0.27 ± 0.12 ; Tukey HSD: $z = -2.66$, $p = 0.021$). However, colony volume had a more prominent effect on *Pa. arcatus* abundance patterns (GLMM: $\chi^2 = 4.57$, $p = 0.033$). Additionally, when all *Pa. arcatus* individuals are included (both residents and those fish that were not present for the entirety of a survey), total abundance did not differ by assemblage variant (GLMM: $\chi^2 = 3.96$, $p = 0.138$). Between assemblage variants 1 and 3, there was no significant difference in resident *Pl. johnstonianus* abundance (GLMM: $\chi^2 = 0.95$, $p = 0.331$; variant 1: 1.45 ± 0.28 ; variant 3: 1.93 ± 0.42). Likewise, between assemblage variants 2 and 3,

there was no difference in resident *D. albisella* abundance (GLMM: $\chi^2 = 0.56$, $p = 0.454$; variant 2: 3.23 ± 0.92 ; variant 3: 2.45 ± 0.55).

Coral growth rates during observational study

Colony growth rate was not significantly different among the 3 assemblage variants (LMM: $F = 1.02$, $df = 2, 7.02$, $p = 0.408$) (Figure 4a). However, regardless of assemblage variant, growth was positively correlated with resident fish abundance (LMM: $F = 42.65$, $df = 1, 7.56$, $p < 0.001$, $R^2 = 0.287$) (Figure 5a), but not total biomass (LMM: $F = 3.15$, $df = 1, 8.47$, $p = 0.112$, $R^2 = 0.075$) (Figure 5b). If *D. albisella*, the most prominent colonizing species during the experiment and a planktivore that spends much of its time up in the water column, was excluded from calculations of biomass, coral growth was positively correlated with this metric (LMM: $F = 9.36$, $df = 1, 7.60$, $p = 0.017$, $R^2 = 0.148$) (Figure 5d). Likewise, if *D. albisella* was removed from calculations of average resident fish abundance, coral growth was positively correlated with fish abundance (LMM: $F = 21.65$, $df = 1, 8.41$, $p = 0.001$) and the model still explains 26.4% of the variation (a loss of only 2.3%; Figure 5c). There was also evidence for decreased coral growth with greater initial colony volume in all models except total biomass with *D. albisella* included (biomass excluding *D. albisella* LMM: $F = 4.71$, $df = 1, 7.00$, $p = 0.067$; resident fish abundance with *D. albisella* LMM: $F = 4.99$, $df = 1, 6.87$, $p = 0.061$; and resident fish abundance without *D. albisella* LMM: $F = 5.70$, $df = 1, 6.97$, $p = 0.048$).

4.4 Discussion

Effects on coral growth of removing strong interactors

While there was no difference in total resident fish abundance or biomass between treatments during the experiment, the abundance of corallivores (excluding *Pl. johnstonianus*) was significantly higher on removal colonies where the two strong interactors, the territorial *Pl. johnstonianus* and the mesopredatory *Pa. arcatus*, were absent. One removal colony was colonized by multiple Leopard Blennies (*Exallias brevis*), while another was colonized by a single *E. brevis* and a Fourspot Butterflyfish (*Chaetodon quadrimaculatus*), both species being obligate corallivores (Hobson 1974, Hourigan 1987, Longenecker 2007) and have been documented feeding on *P. grandis* (Carlson 1992, Kayal et al. 2017). Three additional removal colonies were frequently visited by the Multiband Butterflyfish (*C. multicinctus*), another obligate corallivore (Tricas 1989) that was often seen feeding on *P. grandis* during surveys.

Prior studies indicate that *Pl. johnstonianus* will actively defend territories from all three of these corallivores (Hourigan 1987, Carlson 1992, Tricas & Boyle 2014). In this study, these corallivores were rarely seen during surveys prior to experimental manipulations (2 of 12 surveys, 16.7%) or on control colonies during the press removal (1 of 90 surveys, 1.1%). Each of these three occurrences likely happened due to unique circumstances. First, one of the removal colonies lost its resident *Pl. johnstonianus* from an unknown cause shortly before the pre-removal survey (constituting a “natural removal” within this experiment). This disappearance likely resulted in open access to the colony by other corallivores. The second pre-removal survey was of a colony with a small, subadult *Pl. johnstonianus* that, despite aggressive interactions, was unable to deter larger corallivores during the survey. Lastly, the only control colony survey with one of these corallivorous species present occurred at the very end of the press removal on a

colony where the resident *Pl. johnstonianus*, which had been present for the entire experiment, disappeared. This colony was quickly recolonized by a new *Pl. johnstonianus* within a week, but corallivores gained access to the colony during this period of no occupancy. Furthermore, after the conclusion of the press removal when natural recolonization was allowed to resume, corallivore abundance on removal colonies began noticeably declining. It is important to note that *Pl. johnstonianus* recolonized only half of the removal colonies during the 6 months following the press removal, leaving the other half undefended.

Therefore, despite the apparent lack of similarity of corallivore abundance in pre-removal surveys between treatments, it is highly likely that the presence of these species was almost entirely governed by the absence of resident *Pl. johnstonianus*, while the other strong interactor, *Pa. arcatus*, was present in each of the three described surveys. Protection from these corallivorous species almost certainly contributed to the greater growth rates of control *P. grandis* colonies with *Pl. johnstonianus* and *Pa. arcatus* present, corroborating my second prediction that reduced total corallivory due to protection from resident strong interactors would lead to greater host colony benefits.

While *Pl. johnstonianus* is itself an obligate corallivore (Gochfeld 1991), resident individuals fed on *P. grandis* colonies at a substantially lower rate than that of the other corallivore species (pers. obs.). Resident *Pl. johnstonianus* were also frequently seen feeding on surrounding *P. meandrina* corals, rather than their host *P. grandis*. This suggests that the host coral is not a primary food source and *Pl. johnstonianus* provides a net benefit to the host *P. grandis* through a net reduction in corallivory. It is likely, therefore, that intense feeding by corallivores in the absence of *Pl. johnstonianus* significantly reduced coral growth rates. This is further evidenced by the fact that throughout both the experiment and observational study, *E.*

brevis colonized 6 *P. grandis* colonies and half of these colonies died within several months (one from the experiment after its conclusion, two from the observational study that were excluded from coral growth analyses; see methods). The three colonies with *E. brevis* that survived (two from the experiment, one from the observational study) all experienced severely reduced growth. In fact, of all twelve *P. grandis* colonies measured during the experiment, the two with resident *E. brevis* had the lowest growth rates (see Figure 6 for an example of *E. brevis* corallivory on *P. grandis*). Therefore, this reduction in host colony growth in the absence of *Pl. johnstonianus* is evidence for a novel indirect mutualism on coral reefs (Figure 7). By aggressively chasing transient corallivores, the territorial corallivore reduces the total intensity of predation on its host colony, thereby increasing coral growth rates and receiving food and shelter in return.

In addition to corallivory, other factors may have also contributed to the difference in colony growth rates between treatments of the experiment, although likely to a lesser degree. First, of the three removal colonies visited by *C. multicinctus*, one was visited substantially less often, yet its rate of growth was just as poor. This coral was also not colonized by any resident fishes during the experiment and visited only occasionally by non-residents. Uninhabited *P. grandis* colonies in Mo‘orea grew approximately 30% less than colonies with resident fishes (Holbrook et al. 2008), similar to the reduction found in this study and suggesting that a lack of colonization contributed to the poor growth rate of this colony.

Second, the only removal colony with no recorded presence of transient corallivores was also colonized by more fish than any other. While this led to the greatest growth of all removal colonies, it was still less than the average rate of control colony growth. *D. albisella* was the most abundant colonizing species during the experiment (Chapter 2) and was especially abundant on this colony. Similar to conspecifics, this species spends much of its time feeding on

plankton in the water column far above the colony (Chase et al. 2020b). Due to severely depleted populations of roving piscivores around O‘ahu (Friedlander et al. 2018), individuals may be rarely forced to seek shelter within their host colony due to predation risk during the day. In fact, McCormick & Lönnstedt (2013) found that in low-lying, degraded reefscape, such as that found offshore of Waikīkī, damselfishes exhibited higher feeding rates, likely due to easier detection of approaching predators. Further, average individual fish biomass was significantly lower in general on removal colonies, and when *D. albisella* was excluded from analyses, total fish biomass was greater on control colonies. Therefore, it is possible that in the context of this experiment and contrary to previous studies of *Dascyllus* spp., *D. albisella* may have provided reduced nutrient input to their host due to less total time spent in direct contact with the host colony. In comparison, control colonies had larger resident *Pa. arcatus* and *Pl. johnstonianus* that were present for the entirety of the experiment. These strong interactors spend substantially more time in direct contact with their host coral, which could have stimulated better host colony growth via greater nutrient input (Meyer & Schultz 1985b, Holbrook et al. 2008) and aeration (Goldshmid et al. 2004).

Coral growth during observational study

Unlike colonies involved in the fish removal experiment, observational surveys showed that resident assemblages changed little throughout the growth measurement period, removing the effect of ongoing colonization on coral growth. Growth also did not differ between assemblage variants, despite *D. albisella* being present in approximately twice the numbers as *Pl. johnstonianus*. However, host colony growth was highly correlated with total fish abundance regardless of assemblage variant. If *D. albisella* individuals are excluded from analyses, both

resident fish abundance and biomass are significantly correlated with host colony growth. This pattern indicates further evidence that, in this system, *D. albisella* individuals may contribute less to host colony growth than species that spend a greater total percentage of each day either between the branches (*Pl. johnstonianus*) or perched on the colony exterior (*Pa. arcatus*). This result is contrary to the findings of Holbrook et al. (2011), in which the presence of a single *Pa. arcatus* reduced recruitment and survival of *D. flavicaudus*, resulting in reduced coral growth rates when the hawkfish was present. However, being recruits, the *D. flavicaudus* likely maintained much closer contact with coral colonies compared to *D. albisella* in this study, which were mostly adult immigrants (Chapter 2) that commonly venture far from their host colony (Booth 1992). In Hawai‘i, fish recruitment is much lower compared to other geographic locations (Chapter 2, Walsh 1987). Therefore, on *P. grandis* colonies in Hawai‘i, the two strong interactors, which are also the most common members of resident fish assemblages (Chapter 1), may provide greater nutrient provisioning due to closer average proximity to their host coral.

Additionally, corals hosting greater abundances of fishes such as *Pa. arcatus* that remain perched on or between the branches for extended periods of time seemed to grow faster. The fastest growing colonies were either home to a pair of *Pa. arcatus* present during the majority of surveys or hosted resident assemblages with a large number of scorpionfish (*Sebastapistes coniota* on one colony and *Dendrochirus barberi* on the other). These scorpionfishes are likely unaffected by the presence of either strong interactor and spend significant time during each day deep between the branches of their host colony, leaving its confines only when hunting in the immediate surroundings at night (Hobson 1974). These patterns are consistent with the hypothesis that average proximity, and therefore resident species identity, can affect host coral growth rates via nutrient provisioning and other benefits.

Mutualisms on *P. grandis* in Hawai‘i

Liberman et al. (1995) hypothesized that nutrient provisioning and protection from corallivores are the two most important mutualisms provided to host corals by resident fishes. I found evidence that both of these kinds of mutualism contribute to *P. grandis* growth in Hawai‘i. First, similar to studies elsewhere, greater resident fish abundance and biomass was correlated with greater host colony growth, likely due to an increased provisioning of nutrients via excreta, aeration of inter-branch spaces, and removal of sediment (Meyer & Schultz 1985b, Liberman et al. 1995, Goldshmid et al. 2004, Holbrook et al. 2008, Chase et al. 2020a). Secondly and more importantly, while the strong interactors limited colonization by other fishes, including of another mutualist damselfish (Chapter 2), this did not result in reduced colony growth rates that would be characteristic of a disruption to the resident fish-host coral mutualisms (Holbrook et al. 2011). Instead, I found evidence for a novel indirect mutualism between a territorial corallivore and host coral that results in greater colony growth rates due to a net decrease in total corallivory. Most studies on territorial damselfishes, particularly with regard to effects on benthic community structure, pertain to farming herbivorous damselfishes (see review by Ceccarelli et al. 2001). However, this study provides evidence that other territorial damselfishes may also be mutualists on coral reefs able to substantially affect benthic community structure.

Pl. johnstonianus is the most common resident fish species on *P. grandis* in Hawai‘i, and these colonies are colonized by this territorial damselfish from a very small size (Chapter 1). While many fish species are directly inhibited from colonizing (Chapter 2) due to the territoriality of this species and predation by *Pa. arcatus*, the greater growth rates afforded to occupied corals should cause these colonies to reach larger sizes more quickly. *Pl. johnstonianus* coexists with other fish species on larger colonies, including another damselfish, *D. albisella*

(Chapter 1). Therefore, this situation could reflect a positive feedback loop in which greater colony size leads to greater resident fish species richness and abundance as space restrictions on coexistence ease, which in turn leads to greater nutrient provisioning and colony growth. This is similar to the system described by Schmitt & Holbrook (2003), in which a mutualism between a competitively dominant damselfish and its host anemone led to greater host growth rates and coexistence with an inferior competitor. Such mutualisms represent one mechanistic pathway that enriches local fish diversity on coral reefs.

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4.6 Figures

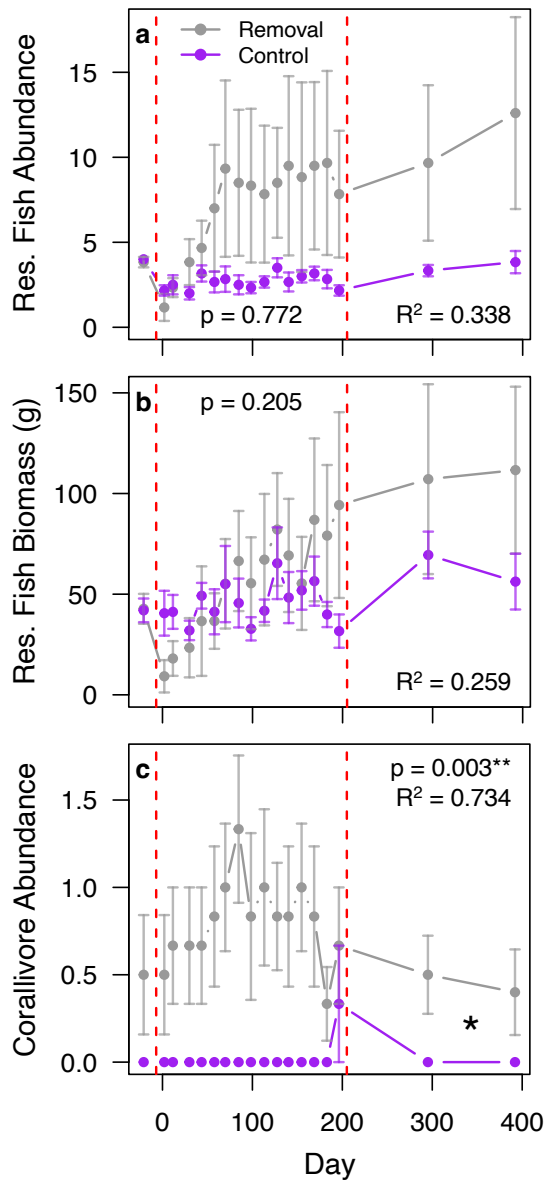


Figure 1. Means and standard errors of **a** total resident fish abundance, **b** total resident fish biomass, and **c** corallivore abundance (excluding *Plectroglyphidodon johnstonianus*) during each survey as a result of the experimental removals of strong interactors *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus* on *Pocillopora grandis*. Colors indicate treatments (removals: grey [n=6], controls: purple [n=6]). The time during which colony growth was measured (start and end of the press removal) is denoted by the two dashed red lines. Abundance and biomass include that of strong interactors, which were present and unmanipulated on control colonies. P and R² values are from statistical models with data from the removal period only. Asterisk indicates significant differences between treatments before or after the removal period.

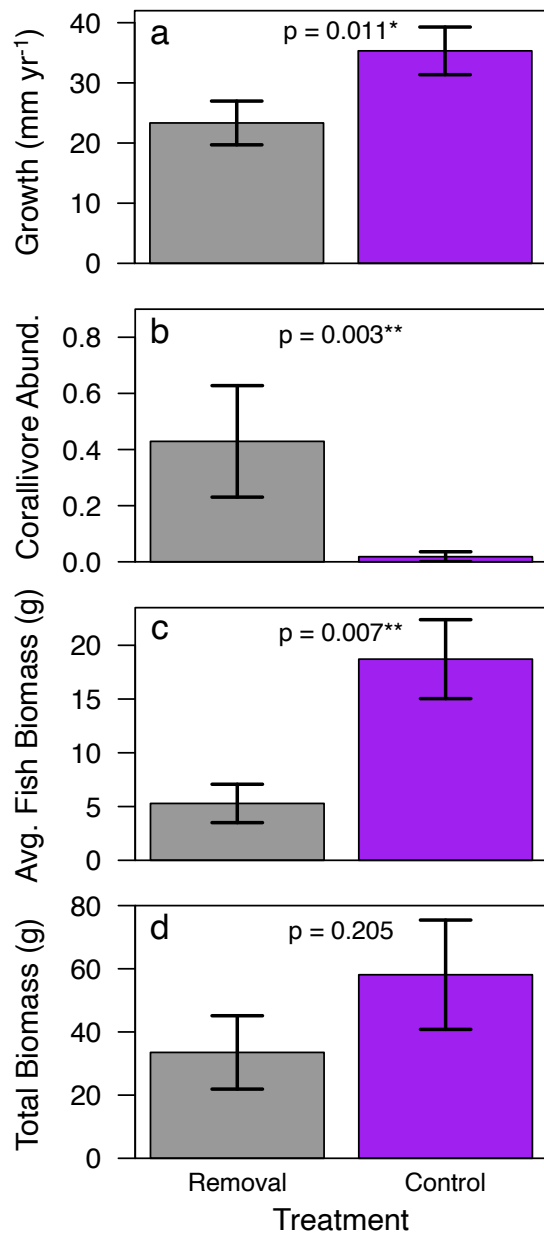


Figure 2. Model means and standard errors of **a** annual apical coral growth (mm yr⁻¹), **b** corallivore abundance (excluding *Plectroglyphidodon johnstonianus*), and **c** average individual fish biomass and **d** total biomass (grams) of all resident fish (including the strong interactors) during the press removal period (between red dashed lines in Figure 1). Colors denote treatment (removals: grey [n=6], controls: purple [n=6]). P values and model-derived means and standard errors are from statistical models using data from the removal period only.

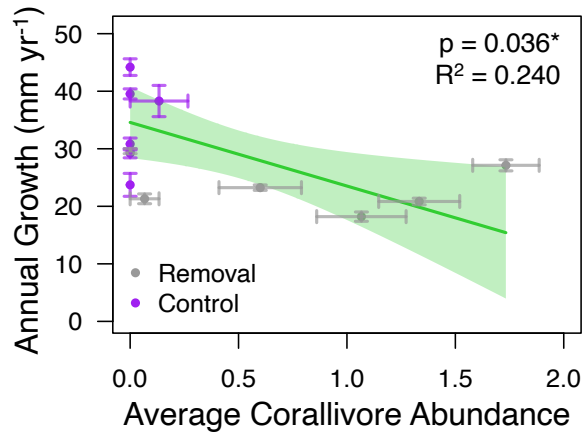


Figure 3. Correlation between annual coral growth (mm yr⁻¹) and the average number of corallivores (excluding *Plectroglyphidodon johnstonianus*) present across 15 surveys during the strong interactor removal period (between dashed red lines in Figure 1). Symbol colors denote treatment, but treatment was not included in the model structure. The p value corresponds to the effects of average corallivore abundance, and R^2 is the total variation explained by all fixed effects.

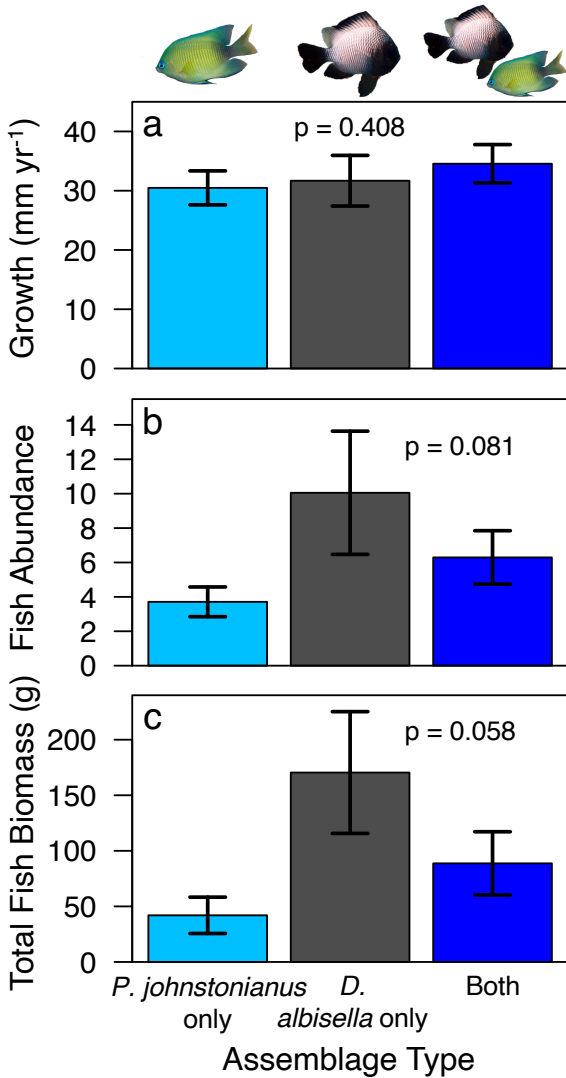


Figure 4. Model means and standard errors of **a** annual coral apical growth (mm yr⁻¹), and **b** abundance and **c** total biomass (grams) of resident fish on unmanipulated colonies during the observational study. Colors denote assemblage variant (*Plectroglyphidodon johnstonianus* present and *Dascyllus albisella* absent [n=6]: light blue, *D. albisella* present and *Pl. johnstonianus* absent [n=3]: black, both species present: dark blue [n=4]). Graphs are of model-derived means and standard errors. P values are for the effect of assemblage variant on each response variable.

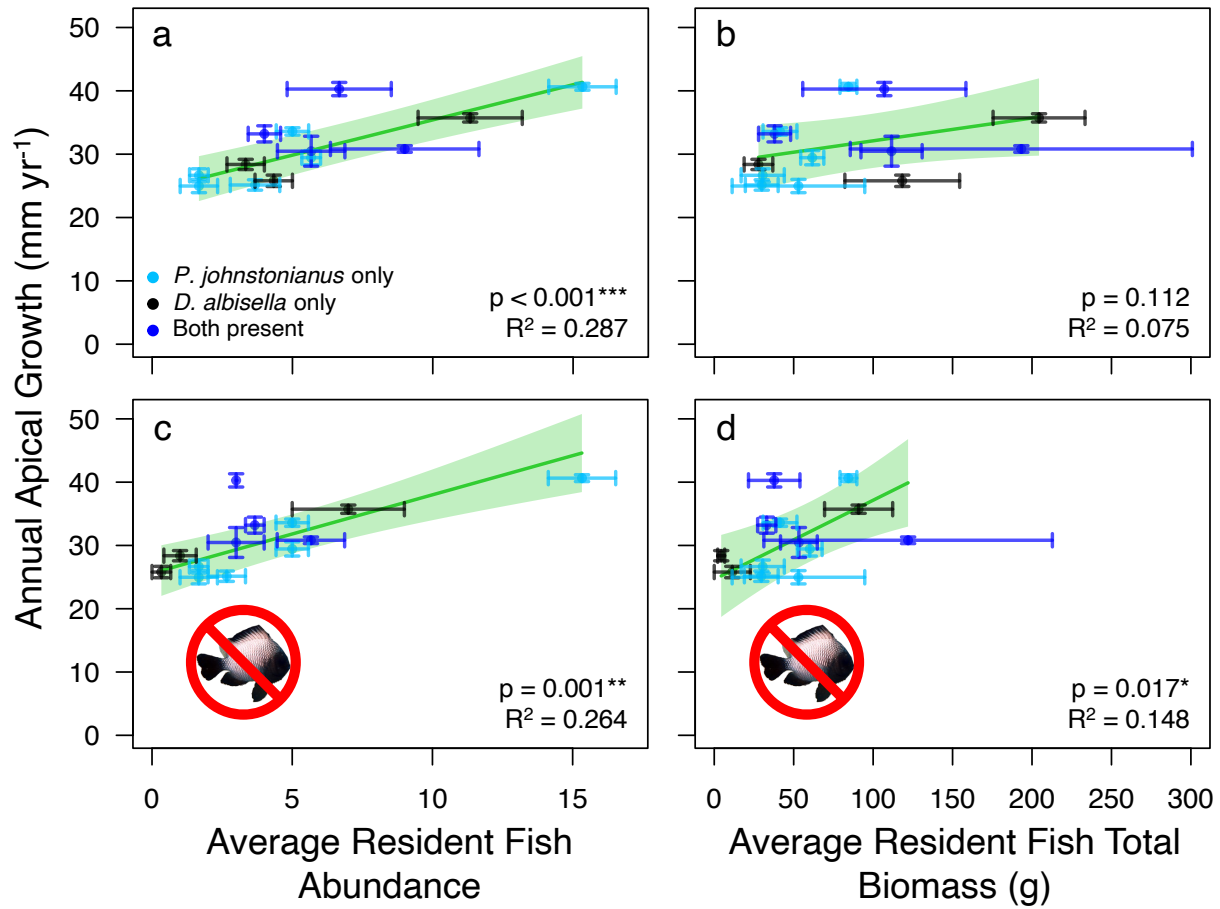


Figure 5. Correlations of annual coral apical growth (mm yr⁻¹) with average resident fish abundance (**a**, **c**) and total biomass (**b**, **d**) across 3 surveys during a 6-month period on unmanipulated colonies during the observational study. Correlations in **a** and **b** include *D. albisella* in the analyses of total abundance and biomass, while **c** and **d** do not. P values correspond to the effects of either abundance or biomass, and R² is the total variation explained by all fixed effects.

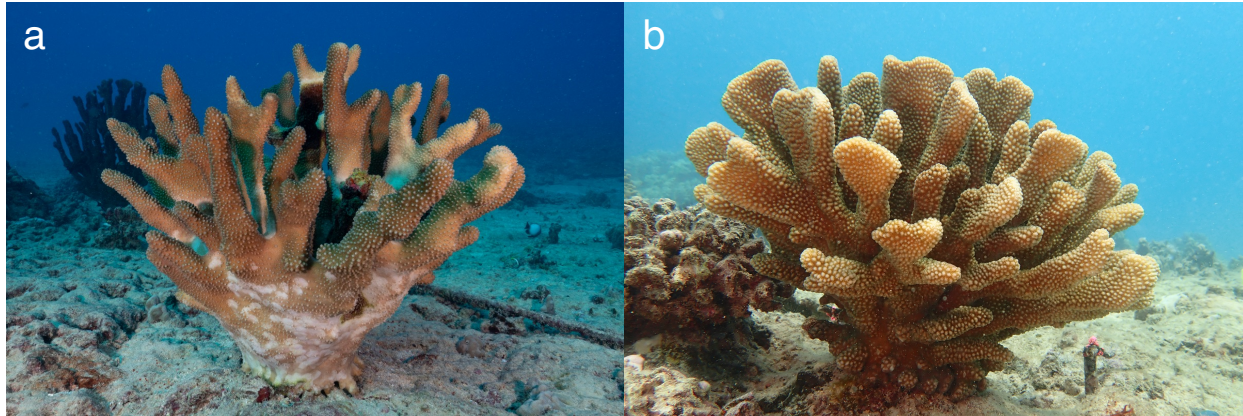


Figure 6. Photographs of *Pocillopora grandis* colonies **a** heavily affected by (photo credit: Bert Weeks) and **b** unaffected by corallivory. The former was bitten extensively by the blenny, *Exallias brevis*.

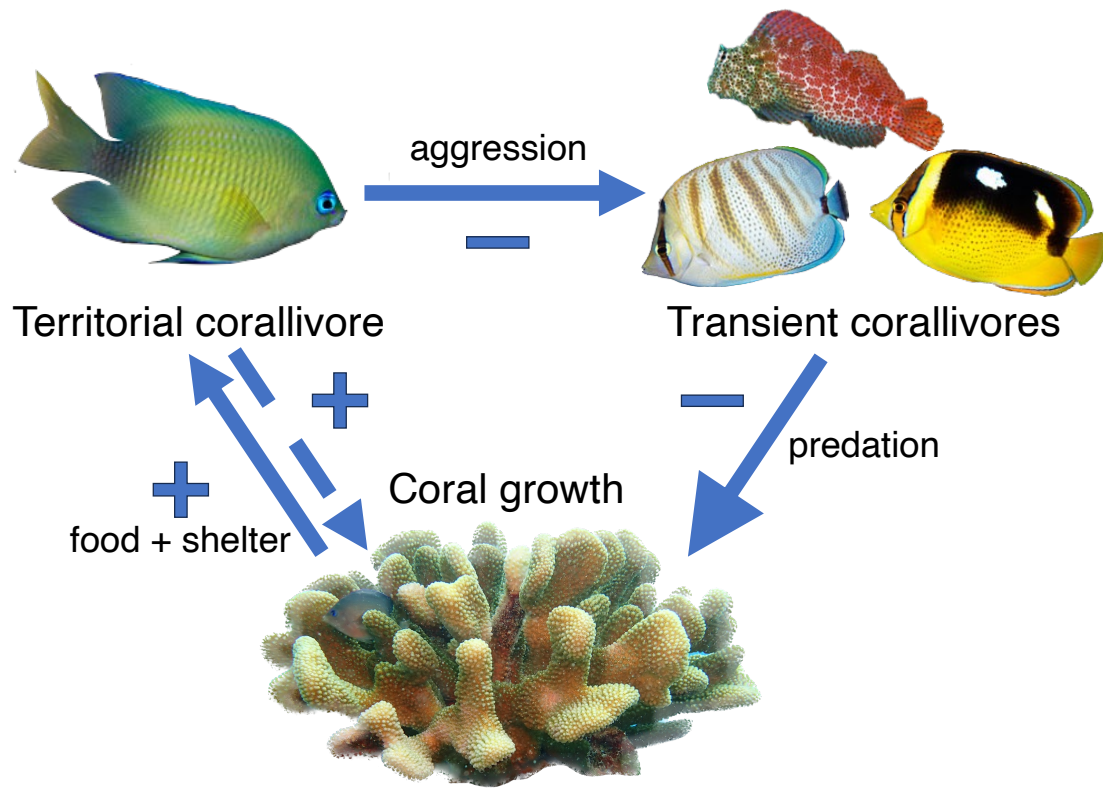


Figure 7. Interaction web illustrating the indirect positive effect of the territorial corallivore *Plectroglyphidodon johnstonianus* on its host coral *Pocillopora grandis*. *Pl. johnstonianus* aggressively restricts access to this resource from transient corallivores (*Exallias brevis*, *Chaetodon multicinctus*, *C. quadrimaculatus*), resulting in increased growth rates of inhabited *P. grandis* colonies. Fish photo credits: Keoki Stender.

CHAPTER 5

General Conclusions

Erik G. Brush

Due to its ability to host large assemblages of fishes (Coker et al. 2014) and provide vertical relief on degraded reefs where complex habitat is limiting, *Pocillopora grandis* is likely an important coral on Hawaiian reefs. However, ecological studies of this large branching coral are lacking, and only in French Polynesia (Schmitt and Holbrook 2000; Kane et al. 2009), Japan (Sakanoue and Sakai 2019), and American Samoa (Smith et al. 2008). Additionally, *P. grandis* can host a multitude of mutualist fish and invertebrate species that provide the host colony with a variety of benefits, including augmented resilience to bleaching (Shantz et al. 2023), defense against corallivores (McKeon and Moore 2014), and increased growth rates (Holbrook et al. 2008a, 2011). Given the ever-increasing human threats to coral reefs and potentially increased resilience granted from resident mutualists, understanding the ecology of *P. grandis* in Hawai‘i could be an important contribution to future management and conservation efforts. This dissertation describes the abiotic and biotic factors that affect the structure of resident fishes and invertebrates on *P. grandis* in Hawai‘i, as well as how these different assemblages can, in turn, benefit their host colony.

The second chapter of this dissertation explored the site-, reefscape-, and colony-level characteristics with the greatest effect on the structure of assemblages inhabiting *P. grandis* along the south shore of O‘ahu, Hawai‘i. Fish assemblages were dominated by pomacentrids and

cirrhitids, common inhabitants of pocilloporid corals (Stella et al. 2011; Coker et al. 2014), while invertebrates were overwhelmingly dominated by decapods. Most variation in resident assemblage structure was attributable to colony-scale variables, with some effect of geographic location and depth, mirroring the results of Counsell et al. (2018) for the smaller *P. meandrina* on O‘ahu. At the scale of the colony, abundance and richness of assemblages were positively associated with colony volume (Abele and Patton 1976; Counsell et al. 2018), but fishes and invertebrates differed in their associations with the percent of living tissue. While fish assemblages were negatively affected by declining amounts of living tissue (cf. Holbrook et al. 2008b), invertebrate abundance and species richness did not change, but rather shifted from obligate coral-dwelling decapods to a suite of facultative species (cf. Coles 1980; Counsell et al. 2018).

There was also some evidence that colony detachment from the substrate and leaning on the reef seafloor, likely caused by high wave energy characteristic of Hawaiian reefs (Grigg 1983), can increase resident fish abundance and species richness due to additional cryptobenthic fishes. Similar to *P. meandrina* (Rodgers et al. 2003), *P. grandis* colonies can remain alive after moderate disturbance for years (pers. obs), allowing these colonies to continue to support these expanded resident assemblages. Similar to the experimental results of Syms and Jones (2000), this expansion in resident fish assemblages was due to an influx of species not commonly found on intact colonies, such as *Pristiapogon kallopterus*, *Sargocentron xantherythrum*, *Dendrochirus barberi*, and *Zebrasoma flavescens* that were likely attracted to the creation of new microhabitats. However, total colony fragmentation, whereby a shattered colony led to individual branches scattered on the seafloor, eliminated these positive effects to assemblage structure (Syms and Jones 2000; Yahya et al. 2011).

The distributions of common resident species on *P. grandis* can be affected by biotic interactions as well as abiotic factors. Positive and negative associations were found between common species, with the some shifting depending on host colony size. Most notably, negative associations were found between two most common residents, the Blue-eye Damselfish (*Plectroglyphidodon johnstonianus*) and Arc-eye Hawkfish (*Paracirrhites arcatus*), which were found on 60% and 45% of colonies, respectively. Territorial *Pl. johnstonianus* (Gajdzik et al. 2016) was also negatively associated with another damselfish, *Dascyllus albisella*, but only on smaller colonies. *Pa. arcatus* was also negatively associated with another hawkfish, the Hawaiian endemic *Cirrhitops fasciatus*, and this was the only negative association that remained constant across small and large colonies. This negative association was previously reported by DeMartini et al. (2013), with between-species competitive dominance potentially determined by individual size, regardless of species. Because the largest *Pa. arcatus* individuals usually inhabit *P. grandis* (Kane et al. 2009; Coker et al. 2015), *C. fasciatus* may be competitively excluded from most colonies.

Given the common occurrence of *Pl. johnstonianus* and *Pa. arcatus* and their potential to restrict access to host *P. grandis* colonies by other species, the third chapter reports an experimental investigation of the effects these two species have on resident assemblages. The distributions of these two strong interactors (Paine 1980) did not differ by geographic site or any reefscape variable, matching previous studies indicating that *P. grandis* may represent preferred habitat for both species (Gochfeld 1991; Kane et al. 2009). Recolonization of *P. grandis* by other fishes was substantially greater on colonies lacking the two strong interactors, as reflected by greater resident fish abundance and species richness, visiting fish abundance and species richness, *D. albisella* abundance, and fish recruitment during this press removal experiment.

Given that *Pl. johnstonianus* is a territorial damselfish (Gajdzik et al. 2016), this species was likely responsible for deterring adult immigration by other fish species. Previous studies have shown that the presence of territorial damselfishes can shape local species distributions (Bay et al. 2001; Almany 2003), particularly of other damselfishes (Eurich et al. 2018), such as *D. albisella* in this study. A known mesopredator of new fish recruits (Shima et al. 2008; Stier et al. 2013), *Pa. arcatus* was almost certainly the cause of significantly reduced fish recruitment to *P. grandis*. These results provide evidence for priority effects, or the effect an already established individual or species can have on newcomers (Gleason 1927), caused by the prior colonization of *P. grandis* by *Pl. johnstonianus* and *Pa. arcatus*, although order of arrival was not explicitly tested.

Fish recruitment to *P. grandis* colonies during the experiment was very low in general, and previous studies have found evidence for low recruitment to reefs in Hawai‘i (Walsh 1987; DeMartini and Anderson 2007). During the experiment, both resident fish abundance and fish recruitment were negatively affected by surrounding coral cover. Furthermore, across the three sites surveyed for chapter 2, total resident fish abundance and *D. albisella* abundance declined on colonies with greater surrounding reefscape complexity, and was greater on colonies at the Waikīkī site, which had the lowest coral cover (Franklin et al. 2013). These patterns lend credence to the possibility that habitat may not be limiting on Hawaiian reefs. On reefs with higher overall coral cover, *P. grandis* colonies may have smaller resident fish assemblages due to a combination of habitat dilution and the common presence of the two strong interactors.

In addition to effects on resident fish assemblages, invertebrate colonization was also substantially affected during the experiment, as abundance and species richness were significantly reduced when *Pa. arcatus* and *Pl. johnstonianus* were present. In fact, by the end of

the experiment, invertebrate abundance was nearly 4-fold greater on colonies with these two species present. As *Pa. arcatus* is a documented mesopredator of decapods (Hobson 1974; Leray et al. 2015), this species was the most likely culprit for this reduction. However, guard crabs of the genus *Trapezia*, which provide important benefits to the host coral, such as defense against corallivores (McKeon et al. 2012; McKeon and Moore 2014) and removal of sediment (Stewart et al. 2006; Stier et al. 2012), seemed largely unaffected by the presence of the strong interactors. While guard crabs were not directly manipulated during the removal experiment, abundance and species richness were only marginally greater on colonies lacking *Pl. johnstonianus* and *Pa. arcatus*, and recruitment was similar, albeit sporadic. Leray et al. (2015) found that guard crabs made up a smaller proportion of the diet of *Pa. arcatus*, indicating that predation by other fishes may be more important.

Additionally, both chapters two and three of this dissertation found potential evidence of habitat partitioning at both the reefscape- and colony-scale between the five species of *Trapezia* spp. in Hawai‘i, indicating that between-species competition may play a substantial role in governing local distributions of guard crab species (Gotelli et al. 1985). Similar to Counsell et al.’s (2018) findings for assemblages inhabiting *P. meandrina*, depth was a significant predictor of individual *Trapezia* species distributions, indicating that some degree of habitat partitioning at the reefscape scale is likely occurring. At the colony scale, microhabitat partitioning seemed apparent, with larger species occupying the exterior branches and smaller species the base (Castro 1976), which likely led to common species associations. It was extremely rare for all five *Trapezia* species to coexist, which occurred only on very large colonies. Lastly, and similar to previous findings for this species on *P. meandrina* colonies (Counsell et al. 2018), *T. intermedia* also commonly associated with smaller colonies and reefscape variables more associated with

colony isolation. This species may be a pioneer that becomes displaced by competitively dominant *Trapezia* spp.

Because the presence of *Pl. johnstonianus* and *Pa. arcatus* was found to greatly affect the local distribution of a potential mutualist, *D. albisella*, the fourth chapter of this dissertation tested the potential indirect effects caused by the presence of these two strong interactors on host coral growth rates. In other geographic locations, *Dascyllus* spp. have been found to confer benefits to host corals, including greater colony growth rates via nutrient provisioning (Holbrook et al. 2008a, 2011), increased resilience to bleaching (Chase et al. 2018) and thermal stress (Shantz et al. 2023), aeration of interbranch spaces (Goldshmid et al. 2004), removal of sediment from the colony's surface (Chase et al. 2020), and protection from corallivores (Chase et al. 2014). Therefore, the strong interactors could negatively affect their host coral through a reduction in *D. albisella* abundance and/or positively benefit the host coral through their own associational defense (Gochfeld 2010) and nutrient provisioning, with the net result being either increased or decreased coral growth.

In the experimental absence of the strong interactors, resident fish abundance did not surpass that of the colonies with *Pl. johnstonianus* and *Pa. arcatus* present. Further, these other fishes were, on average, smaller than the strong interactors, likely resulting in less nutrient provisioning via excreta and reflected by the lower growth rates of host colonies with the strong interactors absent. In an additional group of unmanipulated *P. grandis* colonies, coral growth was positively correlated with resident fish abundance and biomass. However, colonies with *D. albisella* present, and often in numbers greater than other species like *Pl. johnstonianus* and *Pa. arcatus*, saw no additional growth benefit. In fact, the fastest growing colonies often had an abundance of fish species that more closely associate with their host coral, including *Pa. arcatus*

and *Pl. johnstonianus*. Contrary to the results of Holbrook et al. (2011) in Mo'orea, who found that colonies hosting schools of *Dascyllus* recruits grew faster than those with resident *Pa. arcatus*, colonies on Hawaiian reefs may benefit more from other species due to chronically low recruitment (Walsh 1987). Further, most *D. albisella* on *P. grandis* colonies in Hawai'i are adults that spend significant amounts of time feeding in the water column, and depleted piscivore populations around Hawai'i (Friedlander et al. 2018) may necessitate less time sheltering in and fertilizing the host colony.

In addition to and perhaps more important than the potentially greater nutrient provisioning provided to host *P. grandis* colonies by the strong interactors, coral growth in experimentally manipulated colonies was negatively correlated with transient corallivore abundance. Colonies with the strong interactors absent were commonly eaten by corallivore species that are typically chased by *Pl. johnstonianus* (Tricas and Boyle 2014). While *Pl. johnstonianus* is, itself, an obligate corallivore (Gochfeld 1991), careful observation showed that resident individuals rarely fed on their host coral, opting instead to consume coral tissue from the surrounding reefscape. This result, therefore, is evidence for a novel indirect mutualism on coral reefs, in which a territorial corallivore indirectly benefits its host coral via defense against transient corallivores, resulting in a net decrease in total corallivory and greater host colony growth rates. Overall, it appears that the presence of the two strong interactors, which coexist even on smaller colonies, provide multiple net benefits to the host coral.

Colonies of *P. grandis* may play a unique role on Hawaiian reefs. On healthy reefs with abundant coral, these colonies host only the most common members of resident assemblages. However, on degraded reefscales, large *P. grandis* colonies can provide vertical, branching habitat that is otherwise in short supply, attracting large fish and invertebrate assemblages. This

species is also well equipped to survive the high seasonal waves characteristic of Hawai‘i and hosts a diverse community of fish and invertebrate mutualists that confer a wide array of benefits to the host coral. *Pocillopora grandis*, therefore, could represent an excellent candidate for coral restoration efforts in Hawai‘i, especially if outplanted with mutualist communities to potentially boost colony resilience to ever increasing global and local threats to the survival of coral reefs.

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APPENDIX A

Chapter 2 Supplemental Tables

Table 1. Timeline of number of surveys conducted at each site during each quarter of each year.

Site	Season	Number of Surveys						
		2017	2018	2019	2020	2021	2022	2023
Hanauma Bay	January – March	4	1	6	53			
	April – June	45	1	39	6			
	July – September	30		4				
	October – December		5	4		1		
Kahe Point	January – March							
	April – June	39	4					
	July – September	33						
	October – December							
Waikīkī	January – March		1	13	1	11	1	
	April – June	31	8	31	28	18	1	
	July – September	30	2	14	6	15	1	1
	October – December		1		7	6		

Table 2. Descriptions of each *Pocillopora grandis* coral colony and environmental variable, and the number of surveys that include each measurement (n), units, mean, and range of measurements.

Variable	Scale	Measurement	n	Units	Mean	Range
Volume	Colony	Colony size. Measured maximum diameter, maximum orthogonal diameter, and height to nearest 0.01m <i>in situ</i> . Used to calculate the volume of an elliptical cylinder.	304	m ³	0.098	0.00078 – 2.18
Inter-branch ratio	Colony	Colony morphology. Measured 5 random distances between branch tips and 5 associated branch tip widths <i>in situ</i> . Calculated the ratio between the two as a measure of colony openness.	155	ratio	1.27	0.51 – 3.43
Live tissue	Colony	Visually estimated amount of live coral tissue to the nearest 5%.	420	%	88.3	0 – 100
Condition	Colony	Each colony rated based on intactness and attachment to seafloor: 0 = intact and attached, 1 = detached and leaning, 2 = split into 2 – 4 pieces, 3 = highly fragmented.	502	rating	0.17	0 – 3
Depth	Reef	Measured <i>in situ</i> at the base of the coral colony with dive computer.	502	m	10.3	2 – 19.5
Relief	Reef	Average relative height of surrounding reef measured <i>in situ</i> . Habitat classified as 1 = shorter, 2 = same height, or 3 = taller than the focal antler coral.	493	rating	1.66	1 – 3
Complexity	Reef	Based on a video of the surrounding reef, estimated the proportion of the surrounding habitat to nearest 10% of: structurally complex habitat, regardless of source.	491	%	28.2	0 – 100
POME	Reef	<i>Pocillopora meandrina</i> (cauliflower coral)	491	%	3.02	0 – 20
POLO	Reef	<i>Porites lobata</i> (lobe coral)	491	%	11.9	0 – 70
POCO	Reef	<i>Porites compressa</i> (finger coral)	491	%	1.18	0 – 40
Hard Non-coral	Reef	hardbottom habitat and attached dead coral	491	%	51.6	0 – 100
Rubble	Reef	loose rock and dead coral fragments	491	%	5.36	0 – 100
Sand	Reef	sand	491	%	26.9	0 – 100

Table 3. Statistical comparison of univariate models for metrics of fish and invertebrate assemblage structure. Models for abundance and species richness (S) were evaluated with χ^2 likelihood ratio tests (DF = 1). Models for Shannon-Wiener diversity (H') and species evenness (J') were evaluated with F-tests (numerator DF, denominator DF). Significant predictors have bolded p values. Predictors that were not significant in any univariate model are not shown. Corr. = direction of correlation (+: positive; -: negative; W: Waikīkī; H: Hanauma Bay; K: Kahe Point).

	Fishes				Invertebrates			
	χ^2 or F	df	p	Corr.	χ^2 or F	df	p	Corr.
Abundance								
Volume	176.38		<<0.001	+	82.44		<<0.001	+
Live tissue	44.33		<<0.001	+	0.84		0.360	
Site	7.85		0.020	W>H	12.63		0.002	W,K>H
Relief	6.69		0.035	-	2.83		0.243	
Marginal R ²		0.739				0.474		
Conditional R ²		0.852				0.714		
Shannon-Wiener Diversity (H')								
Volume	124.79	1, 200.6	<<0.001	+	39.17	1, 197.0	<<0.001	+
Live tissue	14.36	1, 244.1	<0.001	+	1.70	1, 238.4	0.194	
Site	2.00	2, 105.1	0.140		4.33	2, 125.3	0.015	W>H
Marginal R ²		0.467				0.299		
Conditional R ²		0.725				0.665		
Species Richness (S)								
Volume	138.21		<<0.001	+	36.46		<<0.001	+
Live tissue	23.03		<<0.001	+	1.68		0.195	
Site	4.34		0.114		8.66		0.013	W>H
Marginal R ²		0.489				0.295		
Conditional R ²		0.489				0.354		
Species Evenness (J')								
Volume	26.13	1, 198.6	<<0.001	+	9.08	1, 250.1	0.003	+
Live tissue	9.96	1, 245.1	0.002	+	1.19	1, 251.0	0.276	
Site	2.48	2, 117.0	0.088		0.95	2, 103.3	0.390	
Marginal R ²		0.216				0.084		
Conditional R ²		0.588				0.296		

Table 4. The standard deviation of both random effects (colony ID and survey date) in each statistical model, listed by category (fish or invertebrate) and assemblage metric.

Statistical model	Random effects	
Fishes	Colony ID (n = 180)	Survey Date (n = 90)
Abundance	0.347	0.183
Shannon-Wiener diversity (H')	0.273	0.082
Species evenness (J')	1.761	0.637
Species richness (S)	1.379×10^{-5}	1.386×10^{-5}
Invertebrates		
Abundance	0.238	0.322
Shannon-Wiener diversity (H')	0.223	0.265
Species evenness (J')	2.011×10^{-4}	1.046
Species richness (S)	1.985×10^{-5}	0.151

APPENDIX B

Chapter 2 Supplemental Figures

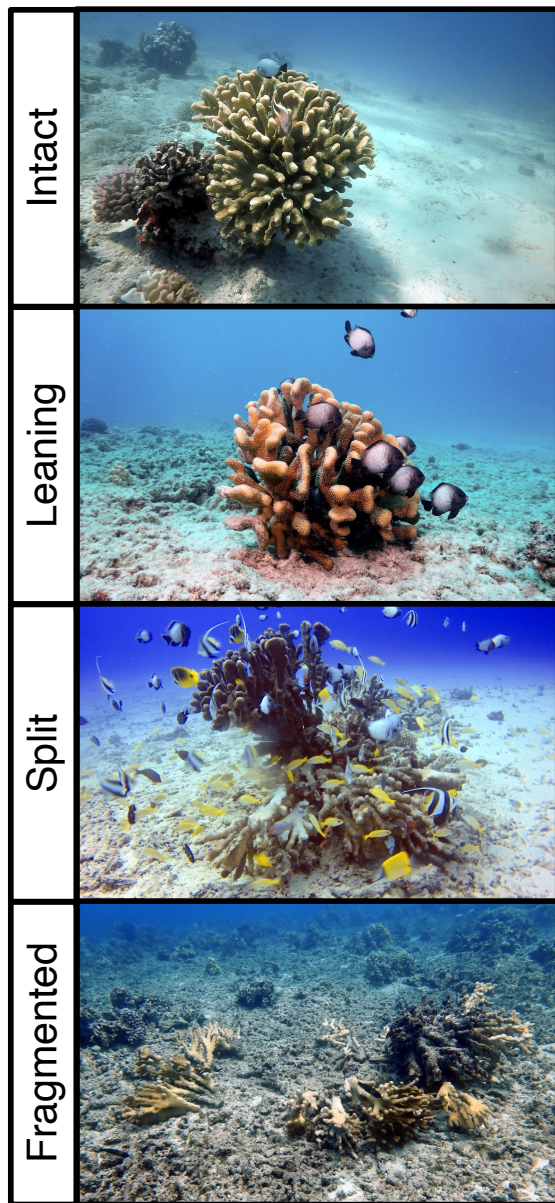


Figure 1. Photographs of four *Pocillopora grandis* colony conditions. Top: intact and attached to the seafloor. Second: Detached from the seafloor and leaning, but intact. Third: Detached from the seafloor, but split into 2 – 4 large pieces. Bottom: detached from the seafloor and highly fragmented.

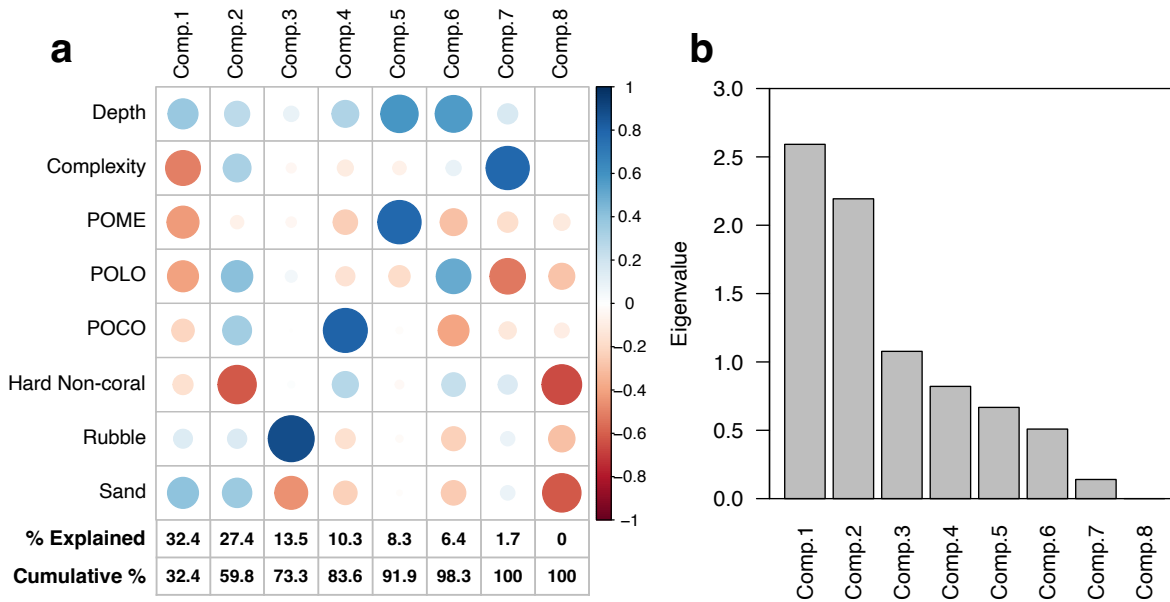


Figure 2. Evaluations of the Principal Component Analysis. **a** Correlation matrix of the 9 environmental variable loadings by principal component axis. The size and color of each circle increases with the contribution of each variable to each component axis (blue=positive, red=negative). “Depth” is depth of the surveyed antler coral. All other variables are proportions of the surrounding substrate estimated to the nearest 10% (Complexity: surrounding reefscape with 3D structure, POME: *Pocillopora meandrina*, POLO: mounding *Porites* spp., POCO: *Porites compressa*, Hard Non-coral: hard substrate lacking living coral). **b** Scree plot of the eigenvalues of each principal component axis. The first two axes explained 58% of the variance in environmental variables.

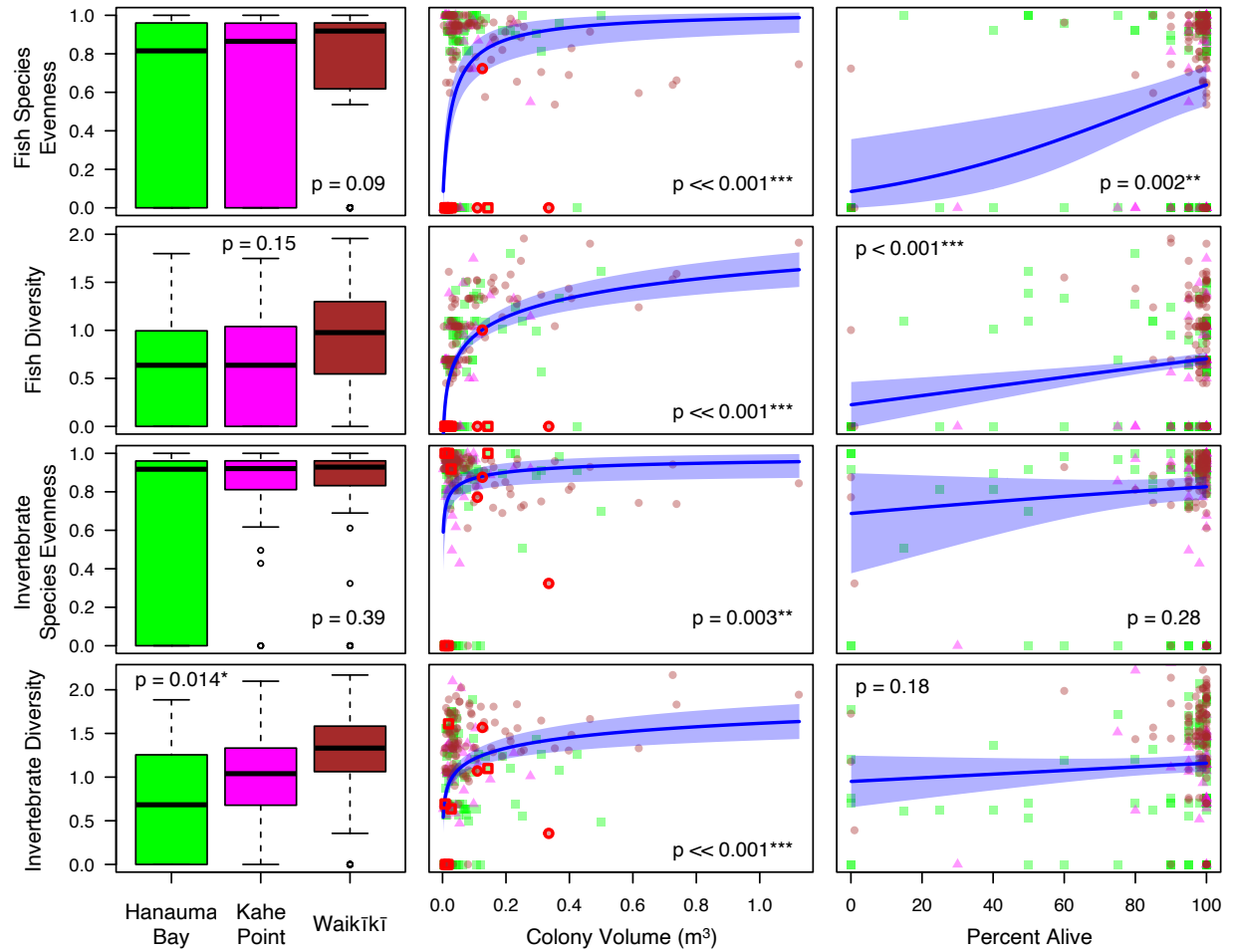


Figure 3. The statistical effects of site (left), colony volume (middle), and percentage of living coral tissue (right) on resident fish species evenness (first row), resident fish Shannon-Wiener diversity (second), invertebrate species evenness (third), and invertebrate Shannon-Wiener diversity (fourth) in linear mixed effects models with colony identity and survey date as random effects. Species evenness was logit-transformed to fit the assumptions of a linear model. All variables are plotted on linear axes. Points outlined in red are from surveys of colonies with $\leq 5\%$ living tissue.

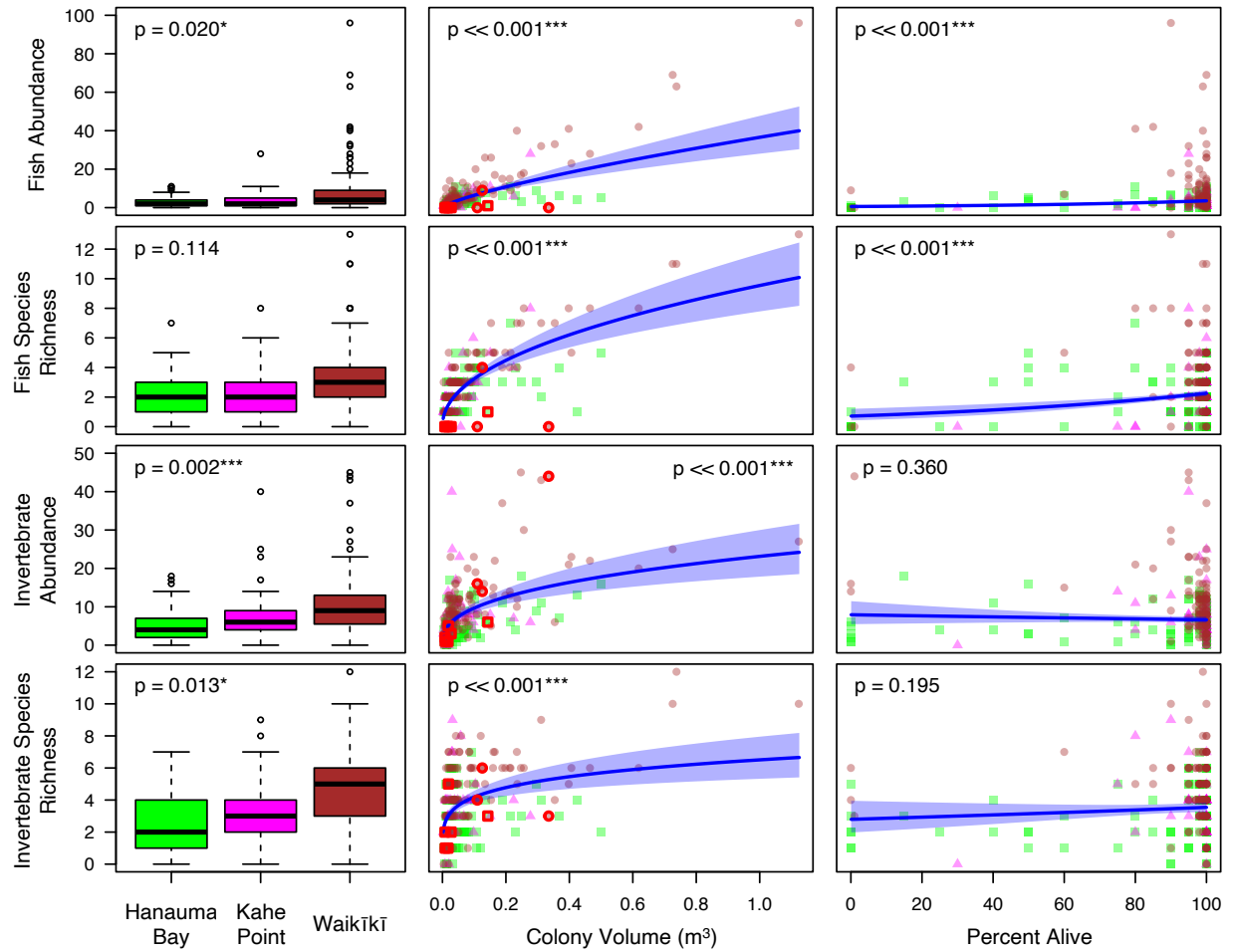


Figure 4. The statistical effects of site (left), colony volume (middle), and percentage of living coral tissue (right) on resident fish abundance (first row), resident fish species richness (second), invertebrate abundance (third), and invertebrate species richness (fourth) in general linear mixed effects models with colony identity and survey date as random effects. All variables are plotted on linear axes. Points outlined in red are from surveys of colonies with $\leq 5\%$ living tissue.

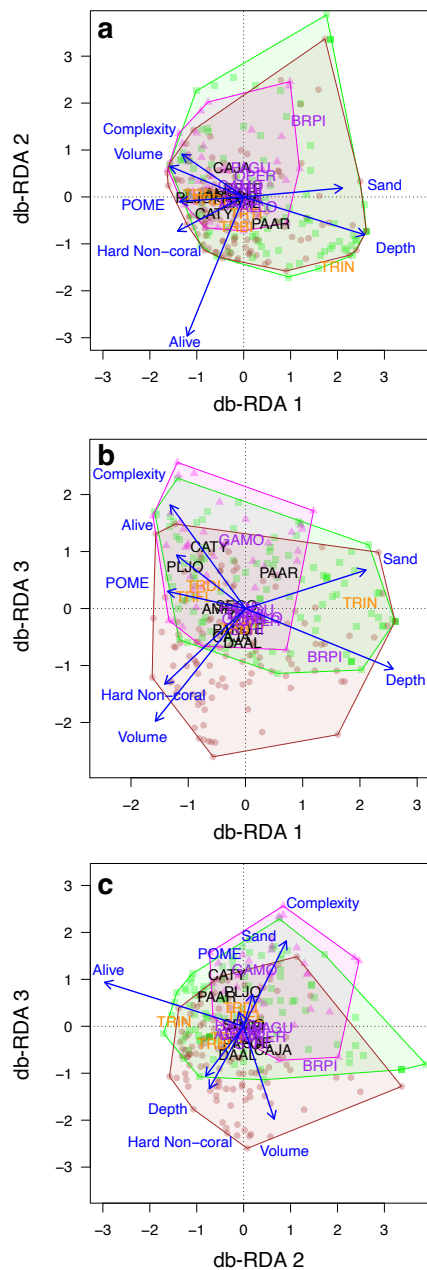


Figure 5. Distance-based redundancy analysis of 258 surveys of 179 intact *Pocillopora grandis* coral colonies: **a** first and second axes, **b** first and third axes, and **c** second and third axes. Analysis is restricted to species listed in Table 1. The effects of colony variables (Volume, Alive) and environmental variables (Depth, Complexity, Relief, Hard Non-coral [hardbottom, dead coral, etc.], Sand, POME = *P. meandrina*) are shown as blue vectors. Values of continuous environmental predictors were scaled by a factor of 3.5 for plotting purposes. Environmental variables were chosen by PCA (see Figure 2) and used in all models. Surveys from each site are represented by their respective colors (Hanauma Bay: green; Kahe Point: magenta; Waikiki: brown). See Table 1 for color and 4-letter species codes.

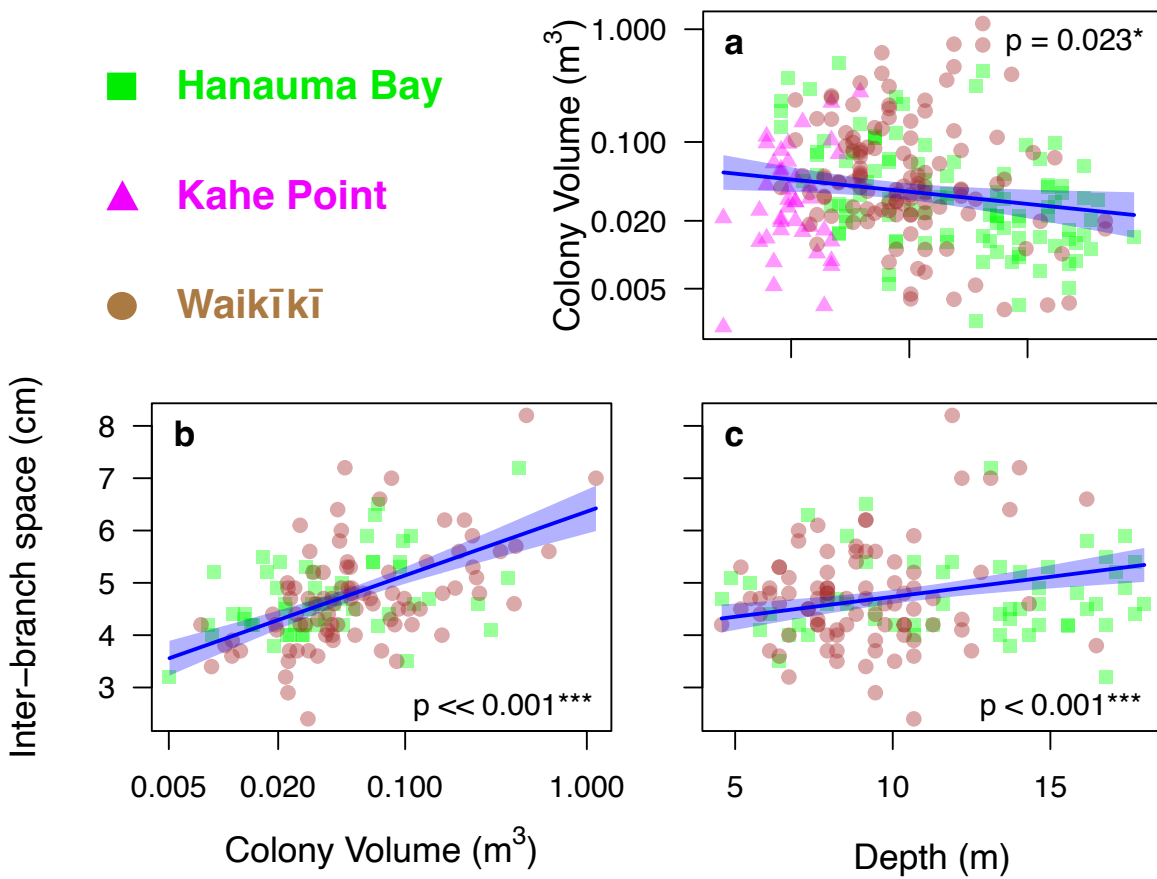


Figure 6. Correlations of *Pocillopora grandis* **a** colony volume by depth ($n=263$) and inter-branch spacing by **b** colony volume and **c** depth ($n=143$) with random effects for colony identity and survey date. Note that **b** and **c** are the results from a single model and contain no data from the Kahe Point site, as no inter-branch spacing measurements were collected during surveys.

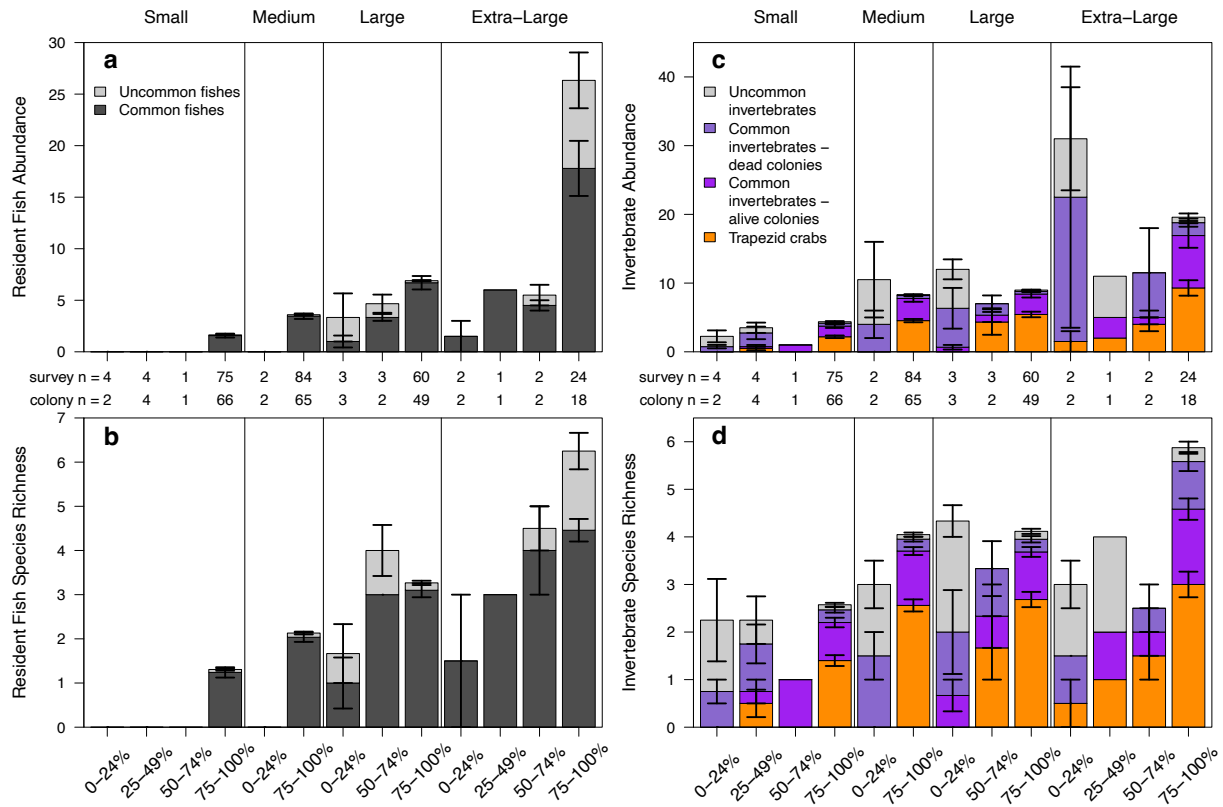


Figure 7. Bar plots of **a** resident fish abundance, **b** resident fish species richness, **c** invertebrate abundance, and **d** invertebrate species richness for surveys of *Pocillopora grandis* colonies that were both structurally intact (no major broken pieces) and attached to the seafloor. Percent living colony tissue has been binned into 4 categories (0-24%, 25-49%, 50-74%, 75-100%). No medium sized colonies with percentages of living tissue between 25 and 74% nor large sized colonies with 25-49% living tissue were surveyed. Bars are means with standard error of each animal group (grey: uncommon fishes/invertebrates; black: common fishes; orange: trapezoid guard crabs; light purple: other common invertebrates common on dead coral [PAGU, BRPI, OPER]; purple: other common invertebrates on live coral). Associations with living and dead colonies determined by Canonical Analysis of Principal Coordinates (Figure 4a-c). See Table 1 for common species and 4-letter species codes.

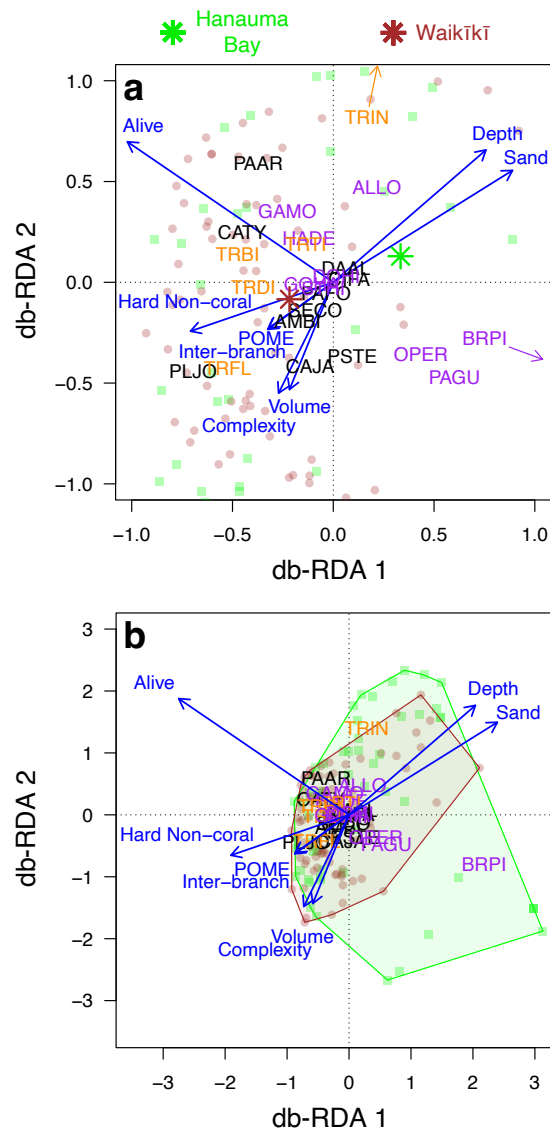


Figure 8. Distance-based redundancy analysis of 137 surveys of 114 intact *Pocillopora grandis* coral colonies that included measurements of inter-branch spacing: **a** zoomed-in view, **b** full plot. Analysis is restricted to species listed in Table 1. Note that no measurements of inter-branch spacing were taken during surveys at the Kahe Point site. The effects of colony variables (Volume, Alive, Inter-branch spacing) and environmental variables (Depth, Complexity, Relief, Hard Non-coral [hardbottom, dead coral, etc.], Sand, POME = *Pocillopora meandrina*) are shown as blue vectors. Values of continuous environmental predictors were scaled by a factor of 1.3 and 3.5, respectively, for plotting purposes. Environmental variables were chosen by PCA (see Figure 2) and used in all models. Centroids of each site are denoted with a large asterisk in its respective color (Hanauma Bay: green; Waikiki: brown). See Table 1 for color and 4-letter species codes. An arrow with a species means that it occurred outside this zoomed-in plot.

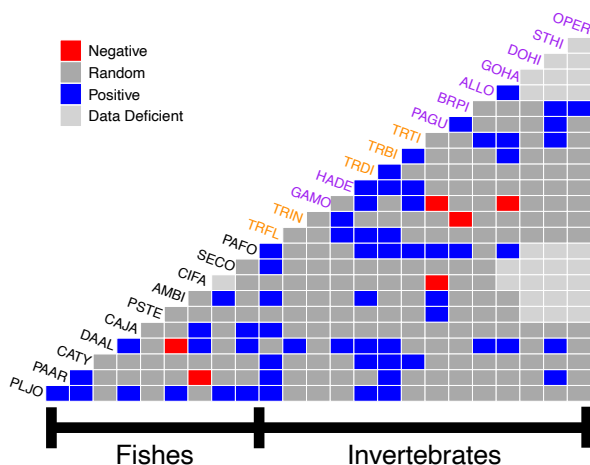


Figure 9. Probabilistic models of species co-occurrence for all *Pocillopora grandis* colonies (n=258 surveys of 179 colonies) encompassing all colony size classes. Positive associations (i.e., those that happened more often than expected based on random species distributions) are blue, negative associations (less often) are red, and random associations are dark grey. See Table 1 for color and 4-letter species codes.

APPENDIX C

Chapter 3: Supplemental Tables

Table 1. Results of a PERMANOVA analyzing the combined effects of colony identity and survey date on resident assemblage structure on *Pocillopora grandis* colonies. Analysis was restricted to visual surveys of 111 colonies that had been surveyed multiple times from 2017 to 2023 (n = 389 total surveys from 3 sites). While both predictor variables were significant, survey date (time) accounted for only 1.4% of the explained variation in assemblage structure. Differences in assemblage structure became noticeable in ordination space only after survey gaps greater than 1 year.

	Colony	Survey Date	Residual	Total
DF	110	1	275	386
Sum of Squares	67.819	0.968	32.705	102.082
F	5.1842	8.1418		
P	0.001	0.001		
R ²	0.66436	0.00949	0.32038	1.00000

Table 2. Correlations of the presence/absence (P/A) and abundance of *Plectroglyphidodon johnstonianus*, *Paracirrhites arcatus*, and *Dascyllus albisella* with colony characteristics, depth, surrounding reefscape, and site from visual surveys of *Pocillopora grandis* colonies. Predictor variable significance was determined with χ^2 likelihood ratio tests. All significant predictors for each statistical model include the direction of the correlation (“Corr.”; +: positive, -: negative, H: Hanauma Bay, K: Kahe Point, W: Waikīkī). Bolded sites indicate a post-hoc test with $p < 0.05$, while nonbolded indicates $0.05 < p < 0.10$. P-value formatting indicates different levels of significance (underlined: $0.05 < p < 0.10$, bolded: $p \leq 0.05$).

		<i>Plectroglyphidodon johnstonianus</i>		<i>Paracirrhites arcatus</i>		<i>Dascyllus albisella</i>	
		P / A	Abundance	P / A	Abundance	P / A	Abundance
Volume (m ³)	χ^2	68.14	52.88	13.90	3.84	17.32	21.59
	p	<<0.001	<<0.001	<0.001	0.050	<<0.001	<<0.001
	Corr.	+	+	+	+	+	+
Live tissue (%)	χ^2	23.49	4.70	8.93	1.06	3.60	0.07
	p	<<0.001	0.030	0.003	0.304	<u>0.058</u>	0.796
	Corr.	+	+	+		+	
Depth (m)	χ^2	4.98	1.91	4.94	0.49	1.01	1.78
	p	0.026	0.167	0.026	0.485	0.315	0.182
	Corr.	-		+			
<i>P. meandrina</i> cover (%)	χ^2	0.004	0.01	0.82	0.56	0.56	0.14
	p	0.949	0.933	0.366	0.455	0.454	0.713
	Corr.						
Complexity (%)	χ^2	0.02	0.44	1.70	0.04	3.63	1.31
	p	0.890	0.507	0.193	0.839	<u>0.057</u>	0.253
	Corr.					-	
Relief (scale)	χ^2	0.63	1.08	0.50	0.15	0.36	7.24
	p	0.73	0.582	0.779	0.927	0.836	0.027
	Corr.						-
Hard Bottom (%)	χ^2	0.19	1.76	0.10	0.44	0.08	0.88
	p	0.666	0.184	0.746	0.508	0.784	0.348
	Corr.						
Sand (%)	χ^2	0.05	0.05	2.38	1.43	3.53	0.37
	p	0.816	0.818	0.123	0.233	<u>0.060</u>	0.543
	Corr.					+	
Site	χ^2	0.59	0.69	2.18	1.74	6.91	6.45
	p	0.744	0.708	0.336	0.418	0.032	0.040
	Corr.						W>H
Marginal R ²		0.657	0.264	0.181	0.075	0.731	0.750
Conditional R ²		0.812	0.264	0.395	0.075	0.946	0.917

Table 3. Correlations of total guard crab presence/absence (P/A), abundance, species richness, and the presence and abundance of each individual species with colony characteristics, depth, surrounding reefscape, and site from visual surveys of *Pocillopora grandis* colonies. Predictor significance was determined with χ^2 likelihood ratio tests. All significant predictors for each statistical model include the direction of the correlation (“Corr.”; +: positive, -: negative, H: Hanauma Bay, K: Kahe Point, W: Waikīkī). Bolded sites indicate a post-hoc test with $p < 0.05$, while nonbolded indicates $0.05 < p < 0.10$. P-value formatting indicates different levels of significance (underlined: $0.05 < p < 0.10$, bolded: $p \leq 0.05$).

		<i>Total Trapezia</i> spp.			<i>T. flavopunctata</i>		<i>T. intermedia</i>	
		P / A	Abundance	S	P / A	Abundance	P / A	Abundance
Volume (m ³)	χ^2	17.71	72.75	13.35	73.70	60.85	0.16	2.81
	p	<< 0.001	<< 0.001	< 0.001	<< 0.001	<< 0.001	0.690	<u>0.094</u>
	Corr.	+	+	+	+	+		+
Alive (%)	χ^2	66.41	8.56	7.50	31.35	1.66	7.45	1.09
	p	<< 0.001	0.003	0.006	<< 0.001	0.198	0.006	0.297
	Corr.	+	+	+	+		+	
Depth (m)	χ^2	3.84	2.53	0.32	6.05	4.09	5.51	0.02
	p	0.050	0.111	0.569	0.014	0.043	0.019	0.882
	Corr.	-			-	-	+	
POME (%)	χ^2	1.03	0.01	0.32	0.001	3.37	3.69	0.16
	p	0.309	0.910	0.572	0.979	<u>0.066</u>	<u>0.055</u>	0.690
	Corr.					+	+	
Comp- lexity (%)	χ^2	2.67	0.14	0.01	3.83	0.0001	1.26	0.30
	p	0.103	0.713	0.928	0.050	0.992	0.262	0.584
	Corr.				+			
Relief (scale)	χ^2	1.08	0.46	1.80	5.11	0.82	6.94	0.30
	p	0.582	0.795	0.406	<u>0.078</u>	0.662	0.031	0.861
	Corr.						+	
Hard bottom (%)	χ^2	1.95	0.61	0.17	1.81	0.23	16.44	0.52
	p	0.162	0.435	0.682	0.179	0.635	<< 0.001	0.471
	Corr.						+	
Sand (%)	χ^2	9.27	2.04	1.53	1.61	0.08	7.44	1.08
	p	0.002	0.153	0.216	0.205	0.773	0.006	0.299
	Corr.	+					+	
Site	χ^2	12.91	17.99	8.29	8.48	14.09	8.49	5.78
	p	0.002	< 0.001	0.016	0.014	< 0.001	0.014	<u>0.055</u>
	Corr.	W>H>K	W>H,K	W>H		W>H,K	H,W>K	
Marginal R ²		0.798	0.800	0.619	0.715	0.735	0.335	0.985
Conditional R ²		0.798	0.909	0.619	0.971	0.735	0.491	0.985

Table 3. (Continued) Correlations of total guard crab presence/absence (P/A), abundance, species richness, and the presence and abundance of each individual species with colony characteristics, depth, surrounding reefscape, and site from visual surveys of *Pocillopora grandis* colonies. Predictor significance was determined with χ^2 likelihood ratio tests. All significant predictors for each statistical model include the direction of the correlation (“Corr.”; +: positive, -: negative, H: Hanauma Bay, K: Kahe Point, W: Waikīkī). Bolded sites indicate a post-hoc test with $p < 0.05$, while nonbolded indicates $0.05 < p < 0.10$. P-value formatting indicates different levels of significance (underlined: $0.05 < p < 0.10$, bolded: $p \leq 0.05$).

		<i>T. digitalis</i>		<i>T. bidentata</i>		<i>T. tigrina</i>	
		P / A	Abundance	P / A	Abundance	P / A	Abundance
Volume (m ³)	χ^2	16.41	8.09	7.22	0.03	2.81	1.24
	p	<<0.001	0.004	0.007	0.856	<u>0.094</u>	0.265
	Corr.	+	+	+		+	
Alive (%)	χ^2	19.73	0.86	8.59	0.87	7.80	4.74
	p	<<0.001	0.355	0.003	0.351	0.005	0.029
	Corr.	+		+		+	-
Depth (m)	χ^2	4.56	6.50	0.14	0.002	1.53	1.65
	p	0.033	0.011	0.710	0.966	0.216	0.199
	Corr.	-	-				
POME (%)	χ^2	0.001	12.74	3.11	3.11	0.54	0.001
	p	0.978	<0.001	<u>0.078</u>	<u>0.078</u>	0.461	0.982
	Corr.		-	+	-		
Complexity (%)	χ^2	0.77	0.25	0.49	0.27	0.24	0.16
	p	0.380	0.617	0.483	0.605	0.626	0.688
	Corr.						
Relief (scale)	χ^2	1.71	2.04	2.59	1.19	4.03	0.59
	p	0.425	0.361	0.274	0.552	0.134	0.745
	Corr.						
Hard bottom (%)	χ^2	0.22	0.11	0.30	0.23	2.51	0.003
	p	0.642	0.740	0.586	0.633	0.113	0.957
	Corr.						
Sand (%)	χ^2	0.003	0.11	0.18	0.42	2.33	1.80
	p	0.957	0.741	0.668	0.519	0.127	0.180
	Corr.						
Site	χ^2	4.95	3.18	7.59	2.45	2.83	1.69
	p	<u>0.084</u>	0.204	0.022	0.294	0.243	0.429
	Corr.	W>H		W>H			
Marginal R ²		0.516	0.500	0.364	0.394	0.270	0.286
Conditional R ²		0.633	0.500	0.615	0.394	0.537	0.286

Table 4. Results of the experimental manipulation of *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus* on fish assemblages. All significant predictors for each statistical model include the direction of the correlation (“Corr.”; +: positive, -: negative, R: removal, C: control). Statistical models for pre- and post-removal period surveys include only Treatment as a predictor. Predictor significance was determined with χ^2 likelihood ratio tests for GLMMs and by type III ANOVAs with Satterthwaite’s method for LMMs. The latter includes numerator and denominator degrees of freedom. P-value formatting indicates different levels of significance (underlined: $0.05 < p < 0.10$, bolded: significant, bolded and underlined: $p < 0.001$).

		Res. Fish Abun.	Res. Fish <i>H'</i>	Res. Fish <i>J'</i>	Res. Fish <i>S</i>	<i>D.</i> <i>albisella</i> Abun.	Vis. Fish Abun.	Vis. fish <i>S</i>	Fish Recruit -ment
Before Experiment									
Treatment	χ^2/F	0.05	0.30 (1,10.0)	0.17 (1,10.0)	0.00	0.39	3.89	1.65	1.27
	p	0.819	0.597	0.685	1.000	0.534	0.048	0.200	0.259
	Corr.	R>C							
Removal Experiment									
Treatment	χ^2/F	10.08	2.92 (1,5.85)	4.21 (1,6.24)	8.24	10.03	24.44	23.48	16.43
	p	0.001	0.140	<u>0.084</u>	0.004	0.002	<u><0.001</u>	<u><0.001</u>	<u><0.001</u>
	Corr.	R>C		R>C	R>C	R>C	R>C	R>C	R>C
<i>P.</i> <i>meandrina</i> density (#)	χ^2/F	5.18	0.40 (1,3.08)	0.06 (1,6.91)	1.70	0.98	1.72	2.03	7.33
	p	0.023	0.572	0.816	0.192	0.323	0.190	0.155	0.007
	Corr.	–							–
Inter- branch space (cm)	χ^2/F	0.53	0.43 (1,5.67)	0.04 (1,6.04)	1.01	1.10	5.69	5.85	5.46
	p	0.468	0.537	0.840	0.314	0.294	0.017	0.016	0.019
	Corr.						+	+	+
Colony volume (m ³)	χ^2/F	2.31	1.57 (1,2.11)	3.61 (1,6.19)	3.00	1.15	2.11	2.80	2.73
	p	0.129	0.331	0.105	<u>0.083</u>	0.284	0.146	<u>0.094</u>	<u>0.098</u>
	Corr.				+			+	–
Colony living tissue (%)	χ^2/F	1.89	0.41 (1,160)	0.22 (1,106)	0.05	0.26	2.25	3.34	4.64
	p	0.169	0.521	0.644	0.823	0.607	0.133	<u>0.068</u>	0.031
	Corr.							+	+
Marginal R ²		0.750	0.395	0.369	0.672	0.771	0.631	0.600	0.729
Conditional R ²		0.960	0.777	0.639	0.835	0.999	0.790	0.632	0.796
After Experiment									
Treatment	χ^2/F	10.85	17.61 (1,7.72)	30.87 (1,3.68)	14.13	11.58	2.66	0.79	5.61
	p	0.001	0.003	0.007	<0.001	<0.001	0.103	0.373	0.018
	Corr.	R>C	R>C	R>C	R>C	R>C			R>C

Table 5. Fish species colonizing *Pocillopora grandis* colonies during the experiment. Families are ordered by recorded number of total resident individuals across the entire study (Res: permanent resident, Vis: occasional resident). Family totals are the number of removal or control colonies occupied by species of that family as either residents or visitors. Multiple species of the same family often occupied the same corals (so this would still be counted as 1 coral). Species totals are the total number of removal or control colonies where a species was present. Note that a species could be represented on the same colony as both a resident and a visitor (still counting as 1 colony), and the maximum possible total is 6 for both totals.

Family	Species	Colonies present						Total # of individuals			
		Removal			Control			Removal		Control	
		Res	Vis	Tot	Res	Vis	Tot	Res	Vis	Res	Vis
Pomacentridae	<i>Dascyllus albisella</i>	4	1	4	1	1	1	295	3	2	1
	<i>Pycnochromis vanderbilti</i>	3	2	3	0	0	0	91	19	0	0
	<i>Plectroglyphidodon imparipennis</i>	0	1	1	0	1	1	0	12	0	1
	TOTAL	4			2			386	34	2	2
Cirrhitidae	<i>Cirrhitop fasciatus</i>	3	1	3	0	0	0	35	2	0	0
	<i>Amblycirrhites bimacula</i>	3	1	3	1	0	1	25	2	1	0
	TOTAL	4			1			60	4	1	0
Labridae	<i>Thalassoma duperrey</i>	4	4	4	0	2	2	15	97	0	8
	<i>Pseudocheilinus tetrataenia</i>	2	2	3	2	1	2	25	4	4	2
	<i>Stethojulis balteata</i>	0	2	2	0	0	0	0	14	0	0
	<i>Oxycheilinus bimaculatus</i>	0	1	1	0	0	0	0	13	0	0
	<i>Pseudojuloides cerasinus</i>	1	1	1	0	0	0	1	3	0	0
	<i>Halichoeres ornatissimus</i>	0	1	1	0	0	0	0	2	0	0
	<i>Macropharyngodon geoffroy</i>	0	1	1	0	0	0	0	1	0	0
	TOTAL	5			4			41	134	4	10
Scorpaenidae	<i>Sebastapistes galactacma</i>	2	0	2	1	0	1	31	0	12	0
	<i>Dendrochirus barberi</i>	1	0	1	2	0	2	3	0	12	0
	<i>Caracanthus typicus</i>	2	0	2	3	0	3	3	0	9	0
	<i>Taenianotus triacanthus</i>	1	0	1	0	0	0	5	0	0	0
	<i>S. coniota</i>	1	0	1	0	0	0	1	0	0	0
	TOTAL	5			4			43	0	33	0
Acanthuridae	<i>Acanthurus nigrofuscus</i>	3	6	6	0	2	2	30	39	0	4
	<i>Zebrasoma flavescens</i>	1	2	2	0	0	0	3	6	0	0
	<i>A. blochii</i>	1	0	1	0	0	0	2	0	0	0
	<i>A. xantherythrum</i>	1	1	1	0	0	0	1	1	0	0
	<i>A. olivaceus</i>	1	2	2	0	0	0	1	3	0	0
	<i>A. nigroris</i>	1	0	1	0	0	0	1	0	0	0
	TOTAL	6			2			38	49	0	4
Blenniidae	<i>Exallias brevis</i>	2	0	2	0	0	0	28	0	0	0
	TOTAL	2			0			28	0	0	0
Tetraodontidae	<i>Canthigaster jactator</i>	3	6	6	3	6	6	14	63	4	34
	<i>C. amboinensis</i>	0	1	1	0	0	0	0	1	0	0
	TOTAL	6			6			14	64	4	34
Chaetodontidae	<i>Chaetodon multicinctus</i>	1	3	3	0	1	1	2	34	0	2
	<i>C. quadrimaculatus</i>	1	1	1	0	0	0	2	6	0	0
	TOTAL	4			1			4	40	0	2

Table 5. (Continued) Fish species colonizing *Pocillopora grandis* colonies during the experiment. Families are ordered by recorded number of total resident individuals across the entire study (Res: permanent resident, Vis: occasional resident). Family totals are the number of removal or control colonies occupied by species of that family as either residents or visitors. Multiple species of the same family often occupied the same corals (so this would still be counted as 1 coral). Species totals are the total number of removal or control colonies where a species was present. Note that a species could be represented on the same colony as both a resident and a visitor (still counting as 1 colony), and the maximum possible total is 6 for both totals.

Family	Species	Colonies present						Total # of individuals			
		Removal			Control			Removal		Control	
		Res	Vis	Tot	Res	Vis	Tot	Res	Vis	Res	Vis
Other	<i>Centropyge potteri</i>	1	1	1	0	0	0	10	1	0	0
	<i>Calotomus carolinus</i>	1	3	3	0	0	0	2	13	0	0
	<i>Parupeneus multifasciatus</i>	0	4	4	0	0	0	0	26	0	0
	<i>Cantherhines sandwichiensis</i>	3	0	3	0	0	0	3	0	0	0
	<i>Pervagor aspricaudus</i>	1	1	1	0	0	0	2	1	0	0
	<i>P. spilosoma</i>	0	0	0	0	1	0	0	0	0	2
	<i>Sufflamen fraenatum</i>	1	0	1	0	0	0	1	0	0	0
	<i>S. bursa</i>	0	1	1	0	0	0	0	2	0	0
	<i>Aulostomus chinensis</i>	1	0	1	0	1	1	1	0	0	1
	<i>Ostracion meleagris</i>	0	2	2	1	1	1	0	2	1	2
	<i>Zanclus cornutus</i>	0	1	1	0	0	0	0	1	0	0
	TOTAL	5			3			19	46	1	5
	GRAND TOTAL							633	371	45	57

Table 6. Results of the experimental manipulation of *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus* on invertebrate assemblages. All significant predictors for each statistical model include the direction of the correlation (“Corr.”; +: positive, -: negative, R: removal, C: control). Statistical models for pre- and post-removal period surveys include only Treatment as a predictor. Predictor significance was determined with χ^2 likelihood ratio tests for GLMMs and by type III ANOVAs with Satterthwaite’s method for LMMs. The latter includes numerator and denominator degrees of freedom. P-value formatting indicates different levels of significance (underlined: $0.05 < p < 0.10$, bolded: significant).

		Invert Abun.	Invert <i>H'</i>	Invert <i>J'</i>	Invert <i>S</i>	<i>Trapezia</i> spp. Abun.	<i>Trapezia</i> spp. <i>S</i>	<i>Trapezia</i> Recruit- ment
Before Experiment								
Treatment	χ^2/F	6.46	1.12 (1,10.0)	7.28 (1,5.00)	2.69	1.76	1.20	6.87
	p	0.011	0.315	0.043	0.101	0.184	0.273	0.009
	Corr.	R>C		C>R				R>C
During Experiment								
Treatment	χ^2/F	13.48	0.92 (1,6.04)	20.94 (1,6.46)	11.73	3.65	2.68	1.57
	p	<0.001	0.375	0.003	<0.001	<u>0.056</u>	0.101	0.210
	Corr.	R>C		C>R	R>C	R>C		
<i>P.</i> <i>meandrina</i> density (#)	χ^2/F	0.67	2.17 (1,4.13)	0.07 (1,7.29)	10.74	0.06	1.03	0.04
	p	0.415	0.212	0.804	0.001	0.805	0.311	0.850
	Corr.				–			
Inter- branch space (cm)	χ^2/F	1.60	0.12 (1,6.40)	0.45 (1,6.35)	3.23	6.73	4.33	2.18
	p	0.205	0.738	0.528	<u>0.072</u>	0.009	0.037	0.140
	Corr.				–	–	–	
Colony volume (m ³)	χ^2/F	3.13	0.62 (1,3.13)	2.72 (1,6.53)	0.002	5.03	1.57	0.80
	p	<u>0.077</u>	0.487	0.146	0.962	0.025	0.211	0.371
	Corr.	+				+		
Colony living tissue (%)	χ^2/F	1.99	1.52 (1,128)	0.69 (1,103)	7.34	9.14	7.97	1.77
	p	0.158	0.220	0.409	0.007	0.003	0.005	0.183
	Corr.				+	+	+	
Marginal R ²		0.675	0.150	0.589	0.237	0.431	0.133	0.215
Conditional R ²		0.938	0.561	0.794	0.237	0.501	0.134	0.492
After Removals								
Treatment	χ^2/F	9.94	0.01 (1,8.30)	8.74 (1,5.08)	6.85	4.67	0.70	2.98
	p	0.002	0.907	0.031	0.009	0.031	0.401	<u>0.084</u>
	Corr.	R>C		C>R	R>C	R>C		R>C

APPENDIX D

Chapter 3: Supplemental Figures

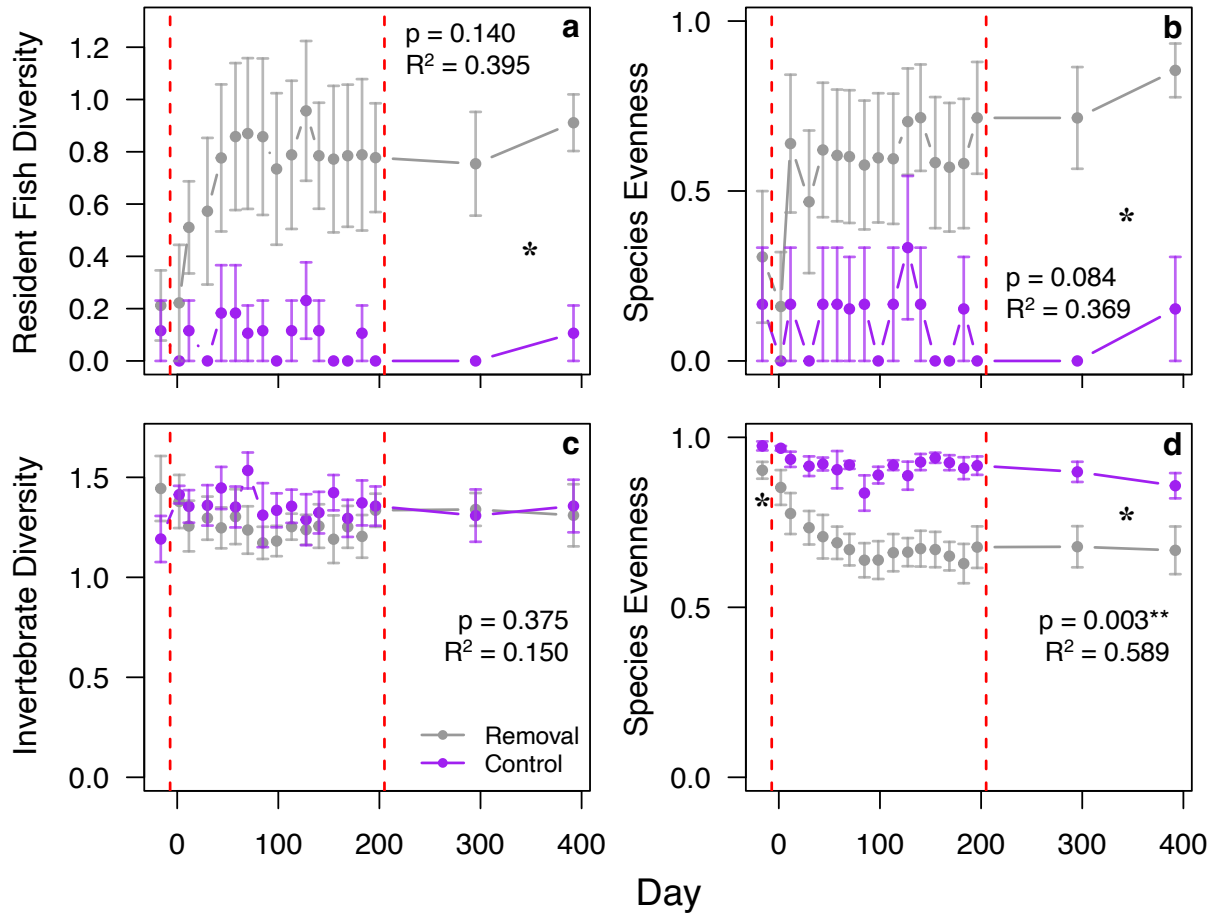


Figure 1. Means and standard errors of **a** resident fish Shannon-Wiener composite diversity (H') and **b** species evenness (J'), and **c** total invertebrate Shannon-Wiener composite diversity (H') and **d** species evenness (J') variables during the experimental manipulation of strong interactors on *Pocillopora grandis*. Colors represent treatment (removals: grey, controls: purple). Dashed red lines indicate the start and end of the press removal. Fish assemblage metrics of control colonies (**a**, **b**) do not include *Plectroglyphidodon johnstonianus* or *Paracirrhites arcatus*, as they were being actively removed on removal colonies. Asterisks indicate significant differences between treatments during pre- and post-removal surveys. P values correspond to the effect of treatment during the removal period. R² values represent the total variation explained by all fixed effects during the removal period.

APPENDIX E

Chapter 4: Supplemental Tables

Table 1. Diagnostics for SfM photogrammetry models created with *Metashape* v1.7.2 (Agisoft, LLC). All values are means $\pm$ standard error. SEUW is a measure of the amount of deviation from the assumed accuracy of the model, or how closely RMS reprojection and estimated errors match. Reprojection error is the distance between the original projection of a 3D point as it was detected on an image and where the point can be projected on the image after model optimization. Average tie point multiplicity is the ratio of the number of projections to tie points for the model. “Dim.” indicates that the measurement is dimensionless and does not contain units.

Metric	Experimental colonies	Observational colonies	Combined
Number of photos	536.75 $\pm$ 19.15	568 $\pm$ 17.67	554.11 $\pm$ 13.05
Tie point accuracy (pixels)	0.12 $\pm$ 0.004	0.12 $\pm$ 0.003	0.12 $\pm$ 0.003
Standard Error of Unit Weight (SEUW)	1.04 $\pm$ 0.009	1.02 $\pm$ 0.005	1.026 $\pm$ 0.005
Target error (pixels)	0.47 $\pm$ 0.015	0.47 $\pm$ 0.013	0.47 $\pm$ 0.010
Scale bar error (mm)	0.24 $\pm$ 0.030	0.36 $\pm$ 0.090	0.31 $\pm$ 0.026
Points left after SPC trimming (%)	13.33 $\pm$ 0.63	12.53 $\pm$ 0.67	12.89 $\pm$ 0.46
RMS reprojection error (dim.)	0.15 $\pm$ 0.003	0.15 $\pm$ 0.003	0.15 $\pm$ 0.002
RMS reprojection error (pixels)	0.40 $\pm$ 0.007	0.41 $\pm$ 0.006	0.41 $\pm$ 0.005
Max reprojection error (dim.)	0.41 $\pm$ 0.003	0.41 $\pm$ 0.002	0.41 $\pm$ 0.002
Max reprojection error (pixels)	2.37 $\pm$ 0.05	2.71 $\pm$ 0.08	2.56 $\pm$ 0.05
Mean key point size (pixels)	2.76 $\pm$ 0.06	2.89 $\pm$ 0.08	2.83 $\pm$ 0.05
Average tie point multiplicity (dim.)	6.00 $\pm$ 0.29	5.85 $\pm$ 0.30	5.92 $\pm$ 0.21

Table 2. To evaluate the photogrammetry model precision, one coral colony was selected at random and six additional models created, each with set of 10% of photos removed from the data set. Each of these models was aligned to the original model in *CloudCompare* v2.12.4 (Zephyrus) to calculate the average distance between point clouds (mean $\pm$ SD; $0.538 \pm 0.0385\text{mm}$). When the original model with all photos was aligned with the model of the same colony from the other time point, branches grew an average of 29.2 mm. The average realignment between the same model is 1.8% of the average growth measurement. Averaged across all six realignments, >95% of points were less than 0.126 ± 0.003 mm away from the original model used for growth measurements. This represents well over an order of magnitude less than the distances measured for colony growth rates.

	Average distance between point clouds (mm)	Distance that >95% of points are less than (mm)
Precision model 1	0.505	0.12496
Precision model 2	0.521	0.12497
Precision model 3	0.578	0.13067
Precision model 4	0.489	0.12117
Precision model 5	0.579	0.12886
Precision model 6	0.557	0.12791
Average	0.538	0.12642
Standard Deviation	0.0385	0.00341

Table 3. To evaluate the accuracy of *Pocillopora grandis* colony growth measurements, manual alignment of same colony model in SM2 in Appendix E was conducted with its paired model (model at month 0 and model at month 6) six additional times, and measurements of growth were taken from the same branches. The coefficient of variation is the ratio of the average growth measured in realigned models to the standard deviation. The standard deviation of average measures of linear extension across models (n=6) was 0.75 mm yr⁻¹, or 20 ± 2.2% (range: 18 – 23%) of the average within colony standard deviation. This also represents 2.57 ± 0.06% (range: 2.44 – 2.62%) of average measured annual linear extension across realigned models.

	Average annual linear extension (mm yr ⁻¹)	Within colony standard deviation (mm yr ⁻¹)
Original	29.22	3.58
Realignment 1	30.63	3.22
Realignment 2	28.51	4.14
Realignment 3	28.95	4.08
Realignment 4	28.96	3.54
Realignment 5	28.80	4.10
Realignment 6	29.05	4.14
Realignment Average	29.15	3.87
Standard Deviation	0.75	0.39
Coefficient of Variation	0.13	

Table 4. Measurements of a Rubik's cube located in every coral colony photogrammetry model. The same measurements were taken within the software *CloudCompare* v2.12.4 (Zephyrus) of the Rubik's cube within each model, three in on each axis (z, x, and y). Values reported are means $\pm$ standard error, with differences in millimeters and error as absolute percent difference from each known measurement. Across all models, there was an average absolute difference (mean $\pm$ standard error) of 0.25 ± 0.0087 mm, resulting in an average difference from true values of $1.38 \pm 0.048\%$.

Measurement	Experimental colonies	Observational colonies	Combined
Z1 difference (mm)	0.27 ± 0.04	0.20 ± 0.03	0.23 ± 0.03
Z1 error (%)	1.51 ± 0.23	1.11 ± 0.16	1.29 ± 0.14
Z2 difference (mm)	0.22 ± 0.04	0.22 ± 0.02	0.22 ± 0.02
Z2 error (%)	1.16 ± 0.19	1.15 ± 0.11	1.16 ± 0.11
Z3 difference (mm)	0.33 ± 0.05	0.29 ± 0.05	0.31 ± 0.03
Z3 error (%)	1.84 ± 0.28	1.59 ± 0.27	1.71 ± 0.19
X1 difference (mm)	0.17 ± 0.03	0.21 ± 0.02	0.19 ± 0.02
X1 error (%)	0.94 ± 0.14	1.15 ± 0.13	1.06 ± 0.10
X2 difference (mm)	0.16 ± 0.03	0.31 ± 0.04	0.24 ± 0.03
X2 error (%)	0.85 ± 0.14	1.65 ± 0.23	1.29 ± 0.15
X3 difference (mm)	0.23 ± 0.03	0.19 ± 0.02	0.21 ± 0.02
X3 error (%)	1.30 ± 0.17	1.04 ± 0.13	1.16 ± 0.10
Y1 difference (mm)	0.23 ± 0.04	0.26 ± 0.04	0.25 ± 0.03
Y1 error (%)	1.27 ± 0.20	1.45 ± 0.20	1.37 ± 0.14
Y2 difference (mm)	0.46 ± 0.06	0.35 ± 0.04	0.40 ± 0.04
Y2 error (%)	2.40 ± 0.33	1.84 ± 0.21	2.09 ± 0.19
Y3 difference (mm)	0.24 ± 0.04	0.24 ± 0.03	0.24 ± 0.02
Y3 error (%)	1.36 ± 0.22	1.34 ± 0.15	1.35 ± 0.13
Average Z difference (mm)	0.27 ± 0.03	0.24 ± 0.02	0.25 ± 0.02
Average Z error (%)	1.50 ± 0.17	1.29 ± 0.11	1.39 ± 0.10
Average X difference (mm)	0.19 ± 0.02	0.24 ± 0.02	0.21 ± 0.01
Average X error (%)	1.03 ± 0.09	1.28 ± 0.11	1.17 ± 0.07
Average Y difference (mm)	0.31 ± 0.03	0.28 ± 0.02	0.30 ± 0.02
Average Y error (%)	1.67 ± 0.15	1.54 ± 0.09	1.60 ± 0.08
AVERAGE DIFFERENCE (mm)	0.26 ± 0.01	0.25 ± 0.01	0.25 ± 0.01
AVERAGE ERROR (%)	1.40 ± 0.08	1.37 ± 0.06	1.38 ± 0.05

Table 5. Previous studies that measured growth as linear extension (mm yr^{-1}) of *Pocillopora grandis* colonies. All measurements are reported as mean $\pm$ standard deviation, range in parentheses, or mean alone.

Growth rate (mm yr^{-1})	n	Location	Method	Source
(14.3 – 24.9)	14	American Samoa	Alizarin dye	Smith et al. (2008)
25.2	3	Samoa	Mean colony diameter	Mayor (1924)
30.0 ± 6.8 (18.2 – 44.2)	25	O‘ahu, Hawai‘i	SfM photogrammetry	This study
30.8 ± 4.8 (21 – 39)	13	Costa Rica	Alizarin dye	Jiménez and Cortés (2003)
33.5 ± 4.3 (28 – 42)		Hawai‘i	Height measurement	Jokiel and Tyler (1992)
47.6 ± 20.5	20	O‘ahu, Hawai‘i	Max diameter	Kolinski (2007)
~50	1	Enewetak	X-ray density bands	Buddemeier et al. (1974)
50.4 ± 13.4 (29 – 68)	16	Johnston Atoll	Height measurement	Jokiel and Tyler (1992)

Table 6. Published length-weight relationships used to calculate biomass for each recorded fish species. Studies conducted in Hawai‘i were given preference. If a study from Hawai‘i could not be found, the next closest location was chosen. If no published relationships for a species were found, a proxy species was chosen that was phylogenetically and morphologically similar.

Family	Species	Location	Source
Angelfishes	<i>Centropyge potteri</i>	Hawai‘i	Friedlander et al. (2013)
Blennies	<i>Exallias brevis</i>	New Caledonia	Kulbicki et al. (2005)
Boxfishes	<i>Ostracion meleagris</i>	New Caledonia	Kulbicki et al. (2005)
Butterflyfishes	<i>Chaetodon multicinctus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>C. quadrimaculatus</i>	New Caledonia	Kulbicki et al. (2005)
	<i>C. fremblui</i>	Hawai‘i	Friedlander et al. (2013)
Cardinalfishes	<i>Pristiapogon kallopterus</i>	Hawai‘i	Friedlander et al. (2013)
Damsel fishes	<i>Dascyllus albisella</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Pycnochromis vanderbilti</i>	Hawai‘i	Longenecker unpub.
	<i>Plectroglyphidodon imparipennis</i>	Hawai‘i	Longenecker unpub.
	<i>P. johnstonianus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Py. hanui</i>	New Caledonia	Kulbicki et al. (2005)
Filefishes	<i>Cantherhines sandwichiensis</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Pervagor aspricaudus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>P. spilasma</i>	Hawai‘i	Friedlander et al. (2013)
Frogfishes	<i>Antennarius commerson</i>	Hawai‘i	Friedlander et al. (2013)
	<i>A. tuberosus</i>	Hawai‘i	Friedlander et al. (2013)
Goatfishes	<i>Parupeneus multifasciatus</i>	Hawai‘i	Friedlander et al. (2013)
Hawkfishes	<i>Cirrhitops fasciatus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Amblycirrhites bimacula</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Paracirrhites arcatus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>P. forsteri</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Zanclus cornutus</i>	Hawai‘i	Friedlander et al. (2013)
Moorish Idol			
Moray Eels	<i>Gymnothorax undulatus</i>	New Caledonia	Kulbicki et al. (2005)
	<i>G. flavimarginatus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>G. meleagris</i>	New Caledonia	Kulbicki et al. (2005)
Parrotfish	<i>Calotomus carolinus</i>	Guam	Kamikawa et al. (2015)
Pufferfishes	<i>Canthigaster jactator</i>	Hawai‘i	Friedlander et al. (2013)
	<i>C. amboinensis</i>	Hawai‘i	Friedlander et al. (2013)

Table 6. (Continued) Published length-weight relationships used to calculate biomass for each recorded fish species. Studies conducted in Hawai‘i were given preference. If a study from Hawai‘i could not be found, the next closest location was chosen. If no published relationships for a species were found, a proxy species was chosen that was phylogenetically and morphologically similar.

Family	Species	Location	Source
Scorpionfishes	<i>Dendrochirus barberi</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Sebastapistes galactacma</i>	Hawai‘i	Longenecker unpub.
	<i>S. coniota</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Taenianotus triacanthus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Scorpaenopsis diabolus</i>	Hawai‘i	Friedlander et al. (2013)
Surgeonfishes	<i>Acanthurus nigrofuscus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>A. olivaceus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Zebrasoma flavescens</i>	Hawai‘i	Friedlander et al. (2013)
	<i>A. nigroris</i>	Hawai‘i	Friedlander et al. (2013)
	<i>A. blochii</i>	Guam	Kamikawa et al. (2015)
	<i>A. xanthopterus</i>	Guam	Kamikawa et al. (2015)
	<i>Naso literatus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>N. unicornis</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Sufflamen bursa</i>	Hawai‘i	Friedlander et al. (2013)
	<i>S. fraenatum</i>	New Caledonia	Kulbicki et al. (2005)
Trumpetfishes	<i>Aulostomus chinensis</i>	Hawai‘i	Friedlander et al. (2013)
Velvetfishes	<i>Caracanthus typicus</i>	Hawai‘i	Friedlander et al. (2013)
Wrasses	<i>Thalassoma duperrey</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Stethojulis balteata</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Halichoeres ornatissimus</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Pseudojuloides cerasinus</i>	Hawai‘i	Longenecker unpub.
	<i>Pseudocheilinus tetrataenia</i>	Unknown	Froese (1998)
	<i>Pseudocheilinus evanidus</i>	Unknown	Froese (1998)
	<i>Pseudocheilinus octotaenia</i>	Unknown	Froese (1998)
	<i>Gomphosus varius</i>	Hawai‘i	Friedlander et al. (2013)
	<i>Oxycheilinus bimaculatus</i>	New Caledonia	Kulbicki et al. (2005)
	<i>Macropharyngodon geoffroy</i>	Hawai‘i	Friedlander et al. (2013)

APPENDIX F

Chapter 4: Supplemental Figures

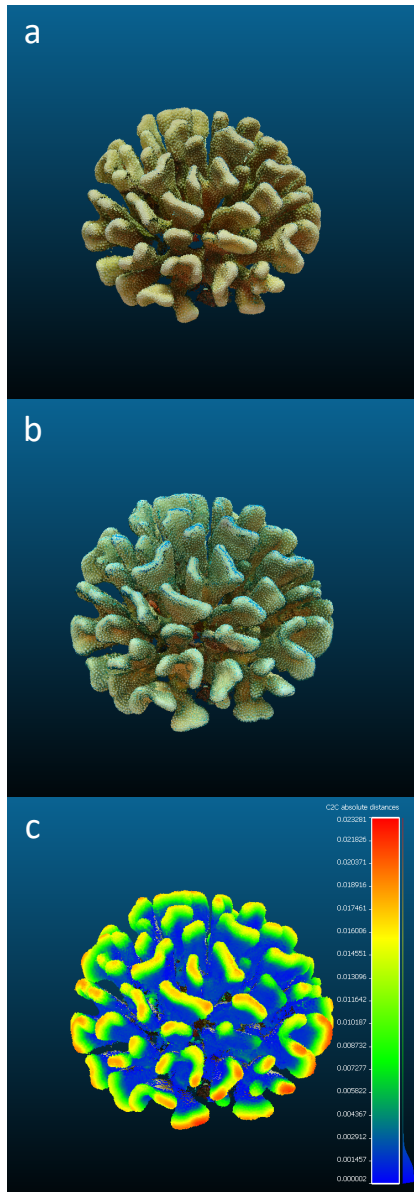


Figure 1. The workflow of manually aligning photogrammetry models for measurements of *Pocillopora grandis* colony growth rates in the software *CloudCompare* (Zephyrus). The initial colony model is shown in **a**, the colony model after 6 months of growth in **b**, and the two models aligned with calculated average cloud to cloud distance in **c**.

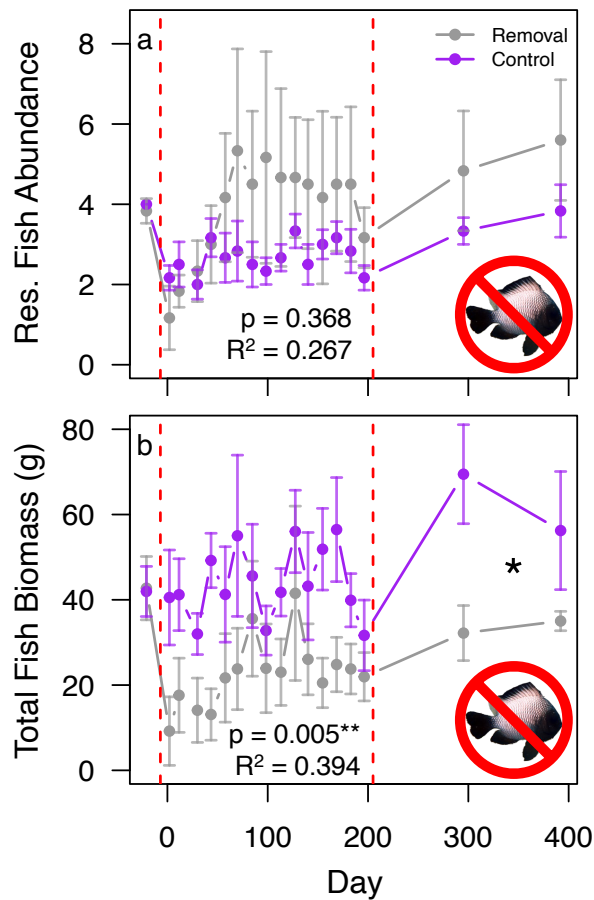


Figure 2. Means and standard errors of total resident fish **a** abundance and **b** biomass not including *Dascyllus albisella* during each survey as a result of the experimental manipulation of fish assemblages on *Pocillopora grandis*. Colors are indicative of treatment (removals: grey [n=6], controls: purple [n=6]). The time during which colony growth was measured (start and end of the press removal) is denoted by the two dashed red lines. Abundance and biomass include that of *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus*, which were present and unmanipulated on control colonies. P and R^2 values are from statistical models with data from the removal period only. Asterisks indicate significant differences between treatments before or after the removal period.

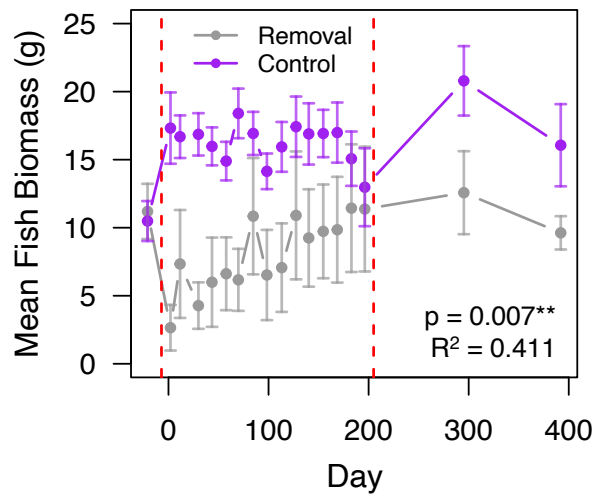


Figure 3. Means and standard errors of average individual fish biomass during each survey as a result of the experimental manipulation of fish assemblages on *Pocillopora grandis*. Colors are indicative of treatment (removals: grey [n=6], controls: purple [n=6]). The time during which colony growth was measured (start and end of the press removal) is denoted by the two dashed red lines. Average individual fish includes that of *Plectroglyphidodon johnstonianus* and *Paracirrhites arcatus*, which were present and unmanipulated on control colonies. P and R^2 values are from statistical models with data from the removal period only.

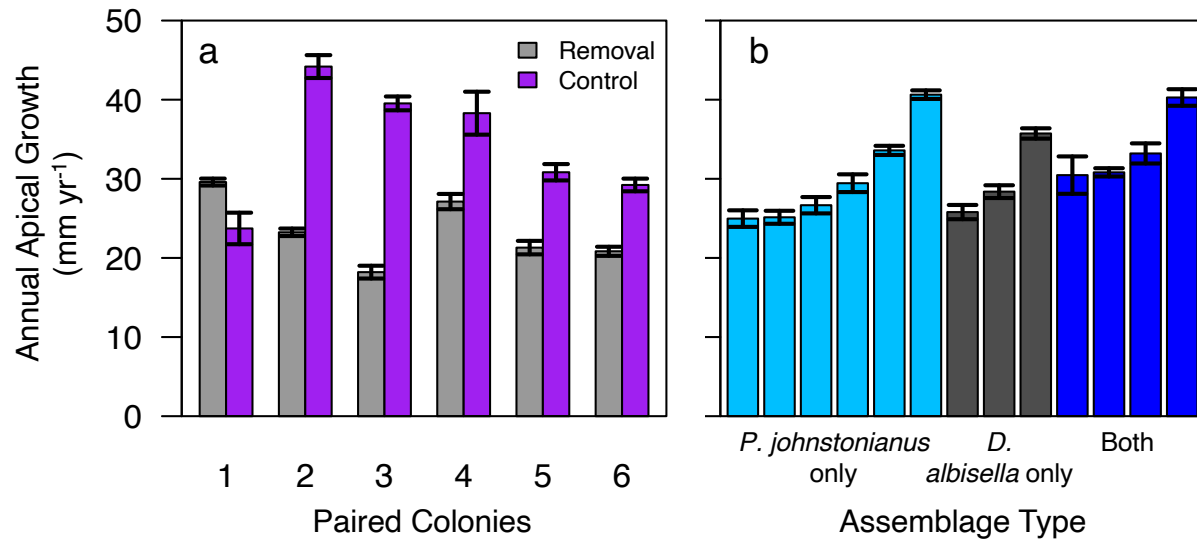


Figure 4. Means and standard errors of annual apical growth (mm yr⁻¹) for each individual *Pocillopora grandis* colony in **a** the removal experiment and **b** the observational study. Colors denote experimental treatment (removals: grey [n=6], controls: purple [n=6]) or naturally occurring assemblage variant (*Pl. johnstonianus* only: light blue [n=6], *D. albisella* only: black [n=4, but 1 colony died], both species present: dark blue [n=4]). Twenty random branches from each coral were measured for apical extension.

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